

<u>Budget Category</u>	<u>US\$ Budget</u>	<u>Calendar Year 2003 Work Program</u>
Drilling & Workover:	600,000.00	Pre-Ops expenses to drill 1 exploration well in April 2004
Geological & Geophysical (G&G):	834,000.00	Seismic data processing, interpretation & prospect justification
Exploration Administration:	144,000.00	Field Offices & Administration
Production:	0.00	
General & Administration (G&A):	522,000.00	Office, Admin, Logistics, Allocated Overheads
Total Budget	2,100,000.00	

Yapen Budget Comparison to PSC Commitment:

This 2003 budget corresponds generally to Yapen PSC contract year four which ends 26-September 2003. The firm work commitment promised Pertamina pursuant to the Yapen PSC for the first four contract years is US\$ 6,000,000. Actual expenditures during the first four contract years are expected to total approximately US\$ 3,100,000 or 20% of the commitment. The full cost of the exploratory well referred to in line one above is expected to exceed US\$ 6,000,000 and meet the PSC Commitment.

The 2003 Yapen PSC Work Program

The objectives of the 2003 annual work program for Yapen Block are briefly described as follows:

1. **Data Processing:** Process 1,250 line kilometers of new HiRes-2D seismic shot in the northern half of the Yapen Block during mid 2001.
2. **Geological Evaluation:** Geological evaluation of the block will continue through 2003. The purpose of this study is to evaluate the petroleum potential of the block and provide a sound geological model for integration with the seismic interpretation for later use in planning more detailed exploration including additional seismic acquisition and drilling prospect identification.
3. **Seismic Interpretation:** The purpose of the 2003 seismic interpretation study is to integrate the newly acquired 1,250 line kilometers of HiRes-2D shot in mid 2001 with vintage data in the area. Minor reprocessing or attribute reprocessing of some intervals may be performed depending upon results. The purpose of the mapping is to interpret all available two dimensional seismic data covering the Yapen PSC area, integrate it with local well control, and produce a reliable suite of maps on all geological horizons of interest. The interpretation should include identification of areas of existing seismic which may benefit from reprocessing and select areas which may benefit from additional seismic acquisition as well as identifying possible drilling targets.
4. **Field Geology:** Little is known about the detailed geology at the formation level of the Yapen Block. Reconnaissance field investigations of the surface geology of Yapen Island may be conducted as necessary during 2003. Field visits would be planned to collect geological samples and make geological measurements located by GPS at selected outcrop areas on the mainland and islands in the vicinity of the Yapen Block. These samples would be described and analyzed to assist geological evaluation.
5. **Reprocessing:** Preliminary seismic data reprocessing tests already conducted have indicated there is little quality improvement to be gained by reprocessing existing, vintage seismic data on the Yapen Block. However, given that the seismic processing of the new data acquired in 2001 may reveal additional information or point to specific features which could be complemented by selective re-processing of 1970's vintage seismic then such reprocessing shall be done to assist the final seismic interpretation prior to drilling.
6. **Orthorectification Survey:** Due to some serious mis-ties of seismic programs and available base maps in the Yapen Block area attributable to differing vintages of seismic and maps not on a common ellipsoid, datum or map projection we plan a small field GPS ground truthing and remote sensing survey to orthorectify available base maps into a reliable, contiguous data unit.
7. **Drilling:** Expenditures for one exploratory well are budgeted to commence in 4th quarter 2003. Due to the remote location of the drilling site and need for specialized drilling equipment considerable advance planning of this well is required once the actual location is determined based on 2003 seismic interpretation plans. It is unlikely that a well location will be ready in time for having a drill ship on location before the end of the weather window for drilling which is April through August. Therefore the actual drilling is expected to occur in this weather window in 2004. However, advance preparation, site investigation and capital materials purchases are expected to commence in 4th quarter 2003 and are budgeted accordingly.

Funding the 2003 Yapen PSC Work Program

The Company's 60% share of Continental Energy Yapen Ltd.'s 10% joint venture share of the 2003 Yapen PSC work program is fully funded by PT Exspan Yapen as per the Yapen Farm Out above described in the sub-section entitled "Yapen Farm Out" in this Item-9.

Yapen Farm Out

Pursuant to a farm out agreement and press release both dated November 5, 2002 the Company made and announced the farm out to PT Exspan Yapen of a 90% share of the "Yapen Block" production sharing contract ("PSC") area offshore Irian Jaya, Indonesia. The Company's 60% owned subsidiary, Continental Energy Yapen Ltd., will retain a fully carried 10% stake in the Yapen Block PSC. PT Exspan Yapen is a

wholly owned subsidiary of "PT Medco Energi Internasional Tbk." (the "Medco Group"). The Medco Group is the largest non-government owned Indonesian oil and gas company. It currently produces approximately 85,000 barrels per day of oil and 100 million cubic feet per day of natural gas. Major shareholders of the Medco Group include the prominent Indonesian Panigoro family, the state oil company of Thailand, and Credit Suisse First Boston. In accordance with the farm out agreement PT Exspan Yapen has paid approximately US\$ 600,000 to discharge Continental Energy Yapen Ltd. accrued liabilities up to the farm out date and has agreed to:

1. Pay for all exploration work commitments necessary to maintain the Yapen PSC in good standing with the Indonesian government; including an obligation to drill at least one exploration well on the Yapen PSC;
2. Pay 100%, and thereby "Carry" the share of all Yapen PSC exploration and drilling costs and expenses attributable to the 10% Yapen PSC working interest retained by Continental Energy Yapen Ltd. from the farm out until any "Plan of Development" for the exploitation of the first of any commercial oil or gas discovery is approved in writing by Indonesian authorities; and
3. Assume the roll of Operator for the Yapen PSC.

The assignment of Yapen PSC interest is subject to approval of Indonesian national oil and gas regulatory authorities which although not yet received at the Record Date of this filing is expected in due course.

MATERIAL COMMITMENTS

A summary of the current status of all material contracts, commitments and expenditure obligations of the Company as of the date of this ANNUAL REPORT are described below. The Company intends to meet its commitments by a combination of raising funds through placement of its securities and through farm out of promoted working interests in its Indonesian properties. There is no guarantee the Company will be successful in raising sufficient capital and the inability to meet its obligations may have material adverse detrimental effects on the Company, its operations and its particular resource properties. See Exhibit-10B: "PSC Terms" for material commitments common to all Indonesian PSC and TAC contracts.

PSC Vendor Promissory Notes

As part of the total consideration paid the original vendors pursuant to the Company's acquisition of its interest in the Bengara-II property the Company issued three promissory notes totaling US\$800,000. As at 7/31/00, the Company had repaid the entire principal and a portion of the interest outstanding on these promissory notes. As at 7/31/02 US\$50,000 of accrued interest remained outstanding.

Bengara-II PSC Commitments

Signature bonuses due Pertamina pursuant to the Bengara-II PSC totaling US\$375,000 have been fully paid. As of January 2003, the Bengara-II PSC is currently in Contract Year-6 ending 12/4/03. During these six PSC contract years ended 12/4/03, the Bengara-II JV is obliged to drill 3 exploration wells and or expend at least US\$ 16,750,000 on petroleum exploratory "Work Commitment" within the Bengara-II contract area pursuant to annual work programs and budgets to be registered with Pertamina and in accordance with the PSC stipulated work commitment expenditure obligations for each of the first ten years of the PSC contact as set forth in the following table. The Company is obliged to provide its Bengara-II Joint Venture 60% share of this amount or US\$7,050,000. The JV reports that has spent an approximately \$2,400,000 towards the Work Commitment as at 10/31/02.

Table of Work Commitment Expenditure Obligations For the First 10-Year Exploration Period of the Bengara-II PSC				
BENGARA-II PSC Contract Year	Unless Extended The PSC Contract Year Ends Upon	PSC Stipulated US\$ Work Commitment Expenditure Obligation	Company's 60% JV Share Of PSC Stipulated Obligations	Co's 60% JV Share of Actual Incurred PSC Expenditure Obligation
Year-1	4-Dec-1998	US\$ 500,000	US\$ 300,000	US\$ 300,000
Year-2	4-Dec-1999	US\$ 500,000	US\$ 300,000	US\$ 300,000
Year-3	4-Dec-2000	US\$ 6,000,000	US\$ 3,600,000	US\$3,600,000
Year-4	4-Dec-2001	US\$ 1,000,000	US\$ 600,000	US\$ 600,000
Year-5	4-Dec-2002	US\$ 5,000,000	US\$ 3,000,000	US\$ 3,000,000
Year-6	4-Dec-2003	US\$ 3,750,000	US\$ 2,250,000	US\$ 2,250,000
Year-7	4-Dec-2004	US\$ 5,250,000	US\$ 3,150,000	
Year-8	4-Dec-2005	US\$ 1,000,000	US\$ 600,000	
Year-9	4-Dec-2006	US\$ 1,000,000	US\$ 600,000	
Year-10	4-Dec-2007	US\$ 1,000,000	US\$ 600,000	
ACCUMULATED TOTALS		US\$25,000,000	US\$15,000,000	US\$7,050,000

Yapen PSC Commitments

As of January 2003, the Yapen PSC is currently in Contract Year-4 ending 9/27/02. Signature bonuses due Pertamina pursuant to the Yapen PSC totaling US\$ 200,000 have been paid in full. During the first four PSC contract years ending 9/27/03, the Yapen JV is obliged to expend at least US\$5,500,000 on petroleum exploratory "Work Commitment" within the Yapen contract area pursuant to annual work programs and budgets to be registered with Pertamina and in accordance with the PSC stipulated work commitment expenditure obligations for each of the first ten years of the PSC contract as set forth in the following table.

Table of Work Commitment Expenditure Obligations For the First 10-Year Exploration Period of the Yapen PSC			
<u>YAPEN PSC Contract Year</u>	<u>Unless Extended The PSC Contract Year Ends Upon</u>	<u>PSC Stipulated US\$ Work Commitment Expenditure Obligation</u>	<u>Incurred Obligation of Yapen JV at Record Date</u>
Year-1	27-Sept-2000	US\$ 500,000	US\$ 500,000
Year-2	27-Sept-2001	US\$ 1,500,000	US\$ 1,500,000
Year-3	27-Sept-2002	US\$ 3,000,000	US\$ 3,000,000
Year-4	27-Sept-2003	US\$ 500,000	US\$ 500,000
Year-5	27-Sept-2004	US\$ 1,500,000	
Year-6	27-Sept-2005	US\$ 4,000,000	
Year-7	27-Sept-2006	US\$ 1,000,000	
Year-8	27-Sept-2007	US\$ 4,000,000	
Year-9	27-Sept-2008	US\$ 1,000,000	
Year-10	27-Sept-2009	US\$ 1,000,000	
ACCUMULATED TOTALS		US\$18,000,000	US\$5,500,000

In accordance with the terms of the farm out agreement dated 11/05/02 with PT Exspan Yapen the Company is relieved of all of its funding obligations to the Yapen JV and the Yapen PSC through all exploration drilling and up to the time Pertamina grants approval for the first commercial development of an oil or gas field found within the Yapen Block. Only beginning from such time the Company would be obliged to pay its net 6% participating interest share of all additional Yapen PSC expenditures.

Consequences of Failure to Fund PSC Commitments

There is no guarantee the Company will be able to raise sufficient funds in a timely fashion to meet its JV share of the funds needed to meet the Work Commitment to Pertamina in either Bengara-II or Yapen PSC. Similarly, there is no guarantee that GeoPetro will be able to raise funds to meet its JV share either. The consequences of a failure or election by either party to not meet JV funding requirements are discussed in the following section. In the event that the either JV is unable to raise sufficient funds to meet its respective accumulated Work Commitment then the JV will have three options.

1. First, the JV could elect to exercise its rights under the PSC and advise Pertamina that it intends to carry forward the under-expended, accumulated Work Commitment into the next Contract Year. In so doing the JV would incur the additional expenditure obligation for the subsequent Contract Year as specified in the PSC. The PSC provides for such carry forward and accumulation of PSC expenditure obligations. Over and under actual expenditures are also accumulated and apply to the cumulative Work Commitment.
2. Second, the JV could elect to advise Pertamina of its difficulties in raising finance to meet the Firm Commitment and request from Pertamina the grant of a one-year extension of the current PSC contract year and corresponding accumulated Work Commitment. Although there is no guarantee that Pertamina would grant such an extension, the request for such is not unusual or uncommon and such requests are commonly granted. In some cases additional annual extensions past the fourth year have been granted. Pertamina may elect to impose certain conditions on such a grant also.
3. Third, the JV could elect to default on its Firm Commitment contractual obligation to Pertamina, so notify Pertamina and thereby terminate the PSC and surrender the contract area back to Pertamina. In such event the JV would lose its sunk costs to date and it is unlikely the JV participants would ever be granted a PSC again. Although there is no precedent on record of its doing so, Pertamina would be contractually entitled to sue and seek payment of the unexpended portion of any Firm Commitment amount from the JV and its participants. There is no guarantee Pertamina would not take this course of action and the JV and the Company might be found liable.

Joint Venture Contract Commitments

GeoPetro Joint Ventures - The Company is party to two identical joint venture contracts with GeoPetro Resources Company of San Francisco, one with respect to the Bengara-II Joint Venture and one with respect to the Yapen Joint Venture ("JV"). Both JV contracts provides for the JV participants to share in all costs and benefits of the respective Bengara-II and Yapen JVs on the basis of 60% share for the Company and a 40% share for GeoPetro subject to "non-consent", "sole-risk" and "default" provisions. Each JV is obliged to meet its respective PSC work expenditure obligations. However, individual participants in each JV, including the Company, have certain flexibility with regard to their individual respective funding obligations to the JV as follows:

1. Each year at the JV annual work program and budget meeting an individual JV participant may elect, but only within fourteen days of annual budget approval deadline, to exercise its JV right of "non-consent" and thereby elect not to participate in a particular operation, activity or budget item project. If the remaining JV parties agree to proceed without the non consenting party then they may do so provided that the non consenting party will not be cash called for his JV share of the non consent activity and the cost of such shall be spread proportionally over the other JV parties. Thereafter if the activity should prove successful and the non-consenting party wishes to rejoin the other JV parties in the non-consent activity then the JV rules of "sole risk" as described in the following paragraph shall apply.
2. The JV agreement provides for actions of less than all the JV participants in accordance with its "sole risk" provisions. In the event of "non consent" by a JV party or in the event that one or more but not all JV parties invoke "sole risk" and agree to bear the full cost of an activity in the JV PSC such as drilling a well, that other parties are not agreeable to, then the operator shall conduct such sole risk activity at the expense of the sole risk parties. In the event the sole risk activity is successful, such as a successful well, then those non-consenting or non sole risk parties may regain their JV participating interest share of such successful well by paying a penalty to the sole risk taking parties equivalent to 500% of the non-consenting party's cost of the sole risk activity if such party had participated in the first place.
3. JV participants who fail to fund their share of agreed JV work programs and budgets within 30 days of a cash call shall be held in "default". If such default is not cured within 60 days the JV agreement provides for the defaulting party to be "squeezed down" and his percentage interest share of the JV is reduced by an amount that bears the same proportion to its percentage interest as the default amount bears to the total JV costs met through that date by the defaulting party.

PT Exspan Yapen Joint Venture - In accordance with the 11/05/02 farm out to PT Exspan Yapen of a 90% interest in the Yapen PSC the Company's 60% owned subsidiary Continental Energy Yapen Ltd. will negotiate and enter into a joint operating agreement (JOA) with PT Exspan Yapen which shall set forth in detail the nature of the joint venture relationship between the two companies. PT Exspan Yapen will act as operator. PT Exspan Yapen is taking a direct assignment of a 90% interest in the Yapen PSC. Such assignment is subject to the approval of Pertamina and the Republic of Indonesia and as of the Record Date of this filing the approval for the assignment has not yet been received although the Company expects it in due course. The JOA will be finalized upon receipt of approval of the assignment. The Company's Continental Energy Yapen Ltd. subsidiary will remain the sole property of the Company and GeoPetro Resources Company and Continental Energy Yapen Ltd. will own an undivided 10% of the Yapen PSC and be party to the JOA with PT Exspan Yapen. The Company expects the PT

Exspan Yapen JOA to be very similar to those similar agreements constituting the GeoPetro joint venture. The Yapen Joint Venture with GeoPetro shall remain in effect to govern the nature of the joint ownership of the 10% Yapen PSC interest by the Company and GeoPetro.

RISK FACTORS

The nature of the Company's business, oil and gas exploration, places the entire foundation of the enterprise in a high-risk category. Oil and gas exploration involves a very high degree of risk. There is no assurance that expenditures to be made, as intended by the Company on exploration or "wildcat" drilling of its properties will result in any discoveries of oil and gas in commercially exploitable quantities. The marketability of any oil and gas actually discovered will be affected by numerous factors beyond the control of the Company including availability of a market for gas, oil and gas market price fluctuations, the proximity and capacity of transport and processing equipment, additional financing and government regulation. Drilling and production operations involve risks including blowouts, oil spills and fires (each of which could result in damage to or destruction of wells, production facilities or other properties, or injury to persons). Completion of a well does not assure a profit on the investment or recovery of drilling, completion and operating costs. In addition, drilling hazards or environmental damage could greatly increase the cost of operations, and various field operating conditions may adversely affect the Company's production from successful wells. Competition among oil companies for quality properties and limited amounts of drilling capital is intense. Specific risk factors are described below and listed in the priority in which Management feels they pose the most serious risk facing the Company.

1. **Financing Risks** - The Company is not generating income or revenue, has generated losses to date and does not presently have sufficient financial resources to undertake by itself all of its planned acquisitions or to fund its contractual commitments. The Company's ability to continue as a going concern depends upon its ability to obtain financing. The Company does not currently have the funds required to completely fulfill its joint venture share of the work commitments due under the Bengara-II, Bangkudulis and Yapen PSC's. There is no assurance that the Company will be able to obtain such financing on acceptable terms, or at all. There is no assurance that the Company will be able to extend or defer its PSC work commitments in the event sufficient funds are not available. It is possible that prolonged inability of the Company to fund its joint venture and PSC work commitments could result in a loss of some or all of its interest in the joint venture or the PSC's. Management is pursuing all available options to raise working capital and funds for its various projects, including possible shares for debt settlements, farm-out of a portion of its interest and commitments on Bengara II and Yapen, and conducting other debt and/or equity offerings. There can be no assurance that the Management will be successful.
2. **Dependency on PSC Contracts** - The Company's principal assets are its three Indonesian properties, the Bengara II and Yapen PSCs and the Bangkudulis TAC. The PSC/TAC contract terms contain requirements regarding quality of service, capital expenditure, legal status of concessionaires, restrictions on transfer and encumbrance of assets and other restrictions. Any failure to comply with these arrangements could result, under certain circumstances, in the revocation of the PSC, and therefore termination of development, production, drilling and other petroleum operations. Furthermore, all plant, property and equipment purchased by Apex Bengara or Apex Yapen for use pursuant to PSC operations becomes the property of Pertamina. The cost of such materials may be recovered only if commercial production is established within the PSC area in accordance with provisions of the PSC.
3. **Political Risks - Indonesia** - The Company's investments in Indonesia are subject to the political risks of foreign investment. These include potential changes in laws affecting foreign ownership, contract and area tenure, government participation, taxation, royalties, duties, rates of exchange and exchange controls. Any new Indonesian government policies adverse to the Company could include a change in crude oil pricing policy, expropriation, nationalization, renegotiation or nullification of existing concessions and contracts, taxation policies, foreign exchange and repatriation restrictions, international monetary fluctuations and currency controls. Direct and indirect effects of the decline in value of the Rupiah have included high levels of domestic inflation, reductions in employment, high interest rates, unavailability of traditional sources of financing, and an overall contraction in production and income levels. These conditions have affected and may continue to affect the operating environment in Indonesia, as well as the cost and availability of financing for natural resources development efforts. Impediments to a return to more normal economic conditions in Indonesia, or shocks to the system that exacerbate current adverse economic conditions, could originate from various sources, including further social unrest, terrorism, Islamic fundamentalism, secessionist provinces, lack of government effectiveness due to political uncertainty, or policy initiatives adverse to foreign investment. Additional information on matters and issues which may have material impact upon the Company's operations on its principal oil and gas properties in Indonesia is included in the Exhibits hereto. Particularly, the reader is referred to ITEM-#19 "Exhibit-10C: Indonesia Update: Part-A: Oil Industry Overview" and "Exhibit-10C: Indonesia Update: Part-B: Social/Economic/Political Situation" together with the external information sources referred to therein for authoritative statistical information and informed third party analysis of the outlook for the Indonesian economy and the petroleum sector.
4. **Oil and Gas Exploration Risks** - Oil and gas exploration involves a high degree of risk. There is no assurance that expenditures to be made by the Company on its oil and gas properties will result in any discoveries of oil and gas in commercial quantities. The Company intends to participate in the drilling of both exploratory and development wells. Exploratory wells have a much greater dry hole risk than do

development wells. The marketability of any oil and gas, which may be acquired or discovered, will be affected by numerous factors beyond the control of the Company. These factors include market fluctuations, the proximity and capacity of oil and gas pipelines and processing equipment, rig availability and government regulation, including regulations relating to prices, taxes, royalties, land tenure, allowable production and environmental protection.

5. **Operating Risks** - Production operations involve risks normally incident to such activities, including blowouts, oil spills and fires (each of which could result in damage to or destruction of wells, production facilities or other properties, or injury to persons). Completion of a well does not assure a profit on the investment or recovery of drilling, completion and operating costs. In addition, drilling hazards or environmental damage could greatly increase the cost of operations, and various field operating conditions may adversely affect the Company's production from successful wells. These conditions include delays in obtaining governmental approvals or consents, shut-in of connected wells resulting from extreme weather conditions, insufficient storage or transportation capacity. While close well supervision and effective maintenance operations can contribute to maximizing production rates over time, production delays and declines from normal field operating conditions cannot be eliminated and can be expected to adversely affect operating revenues from time to time in varying degrees.
6. **Risk of Future Changes in Regulatory Environment** - The Indonesian Government has exercised and continues to exercise significant influence over many aspects of the Indonesian economy, including the oil and gas industry, Indonesian business generally, investments, taxes and the capital markets. The Bengara-II, Yapen and Bangkudulis properties and operations there upon, are governed by Indonesian laws and regulations relating to the development, production, marketing, pricing, transportation and storage of crude oil, taxation and environmental and safety matters. Any new government policies adverse to the Company could include a change in crude oil pricing policy, expropriation, nationalization, renegotiation or nullification of existing concessions and PSC/TAC contracts, taxation policies, foreign exchange and repatriation restrictions, international monetary fluctuations and currency controls. Such changes and developments could have a material adverse effect on private sector entities in general and on the Company in particular.
7. **Management** - The Company depends entirely upon its management to identify, acquire, finance and operate a portfolio of oil and gas exploration properties through which the Company can grow and achieve oil production and a steady income. The Company's management is comprised of a small number of key employees with technical skills and expertise in the business, the loss of any one of whom could harm the Company. The Company does not currently maintain "key-man" insurance for any of its employees.
8. **Director's Conflicts of Interest** - Certain directors of the Company are also directors, officers or shareholders of other companies that are similarly engaged in the business of acquiring, developing and exploiting natural resource properties. Such associations may give rise to conflicts of interest from time to time. The directors of the Company are required by law to act honestly and in good faith with a view to the best interests of the Company and to disclose any interest, which they may have in any project or opportunity of the Company. If a conflict of interest arises at a meeting of the board of directors, any director in a conflict will disclose his interest and abstain from voting on such matter. In determining whether or not the Company will participate in any project or opportunity, the director will primarily consider the degree of risk to which the Company may be exposed and its financial position at that time. Persons considering the purchase of Company securities must appreciate that they will be required to rely on the judgment of these individuals in resolving such conflicts as they may arise.
9. **Currency/Exchange Rate Risk** - All of the Company's material financial obligations and commitments are denominated in US Dollars while its primary source of financing is the issuance of equity securities in Canada for Canadian currency. A substantial amount of the Company's business transactions will be denominated in Indonesian Rupiah due to operations on its principal properties. The Company's financial statements are kept in US Dollars. The Indonesian currency, the Rupiah, has decreased substantially in terms of the US Dollar during the past few years. See historical foreign currency exchange rates set forth in Table-1 above in the section "ITEM-#1: Currency and Exchange Rates". Fluctuations in the Rupiah may have a substantial effect on the Company's financial statements due to related gains or losses due to the rate changes. Until such time as it achieves revenue from petroleum production the Company does not hedge and engage in other strategies to protect itself from adverse fluctuations in the respective exchange rate. Significant variations in exchange rates could have a material adverse effect on the ability of the Company to meet its obligations.

MATERIAL TRENDS AND UNCERTAINTIES

The Company's securities are highly speculative and trade in a specialized market of investors willing to take a large degree of risk for a possibly large reward. Traditionally, junior oil and gas companies and gold companies have thrived in this speculative and high-risk environment. Fiscal 2000 and 2001 saw a shift by investor's away from highly speculative resource issuers to dot.com Internet stocks with similar high speculation characteristics. The effects of this divergence are clear and unmistakable and a large amount of capital formerly available to resource issuers has been diverted. It is uncertain whether this trend will continue and the amount of funds available to the Company

may be affected. A fall out in the dot.com markets could return a lot of risk capital to the resource companies. Continued optimism in the dot.com markets may continue to drain funds formerly available to resource issuers.

World oil prices have risen to relatively high levels in late calendar 2002 largely on fears of supply disruption with possible war in Iraq and appears to have settled in the US\$30 per barrel range. Given the uncertainty, world oil prices are likely to fluctuate and be volatile, possibly over a wide range, with developments as to the resolution of the conflicts with Iraq and international terrorism proceed.

The rebounding of the previously vigorous Asian economies is clearly evident and expected to regain much of its forward momentum this year and next. This rebound offers a triple benefit for the Company as increasing Asian demand buoys world oil prices, international funds and emerging markets funds refocus interest and investment on the region and larger oil companies refocus exploration and development activities in the Asian geographic region.

Resource exploration is a speculative venture involving substantial risk. There is no certainty that the expenditures to be made by the Company in the acquisition of interests in properties under its various agreements will result in discoveries of commercial quantities of oil/gas. Further, the marketability of oil/gas that may be acquired or discovered by the Company will be affected by numerous factors beyond the control of the Company. These factors include market price fluctuations, the proximity and capacity of oil/gas markets, availability of needed equipment, transportation costs and government regulations. The exact effect of these factors cannot be accurately predicted, the combination of these factors may result in the Company not receiving an adequate return on capital.

As of the date of this Annual Report, the Company has insufficient financial resources to undertake all of its planned exploration and development programs. The future development of the properties of the Company shall depend upon the ability of the Company to finance through the joint venturing of projects, debt financing, equity financing or other means. The Company intends to raise required additional funds by selling equity or debt securities, and/or by selling or joint venturing its properties until it develops a cash flow from operations. While the Company has been successful in raising the necessary funds in the past, there can be no assurance it can continue to do so. If such funds cannot be secured, the Company will be forced to curtail its exploration/development efforts to a level for which funding can be secured or relinquish certain of its properties or allow its interest to be diluted pursuant to the terms of the respective joint venture agreements. There is no assurance that the Company will be successful in obtaining such financing. This situation could be exacerbated by additional and continuing acts of international terrorism and a possible war in Iraq; or unforeseen political disturbances in Indonesia.

The Company does not know of any other trends or uncertainties that are reasonably likely to result in, material changes to its liquidity at present or in the foreseeable future. Material increases or decreases in the Company's liquidity are substantially determined by the success or failure of its exploration programs or the acquisition of new or disposition of existing projects.

Additional information on matters and issues which may have material impact upon the Company's operations on its principal oil and gas properties in Indonesia is included in the Exhibits hereto. Particularly, the reader is referred to ITEM -#19 "Exhibit-10C: Indonesia Update: Part-A: Oil Industry Overview" and "Exhibit-10C: Indonesia Update: Part-B: Social/Economic/Political Situation" together with the external information sources referred to therein for authoritative statistical information and informed third party analysis of the outlook for the Indonesian economy and the petroleum sector.

OUTLOOK

The Company is an oil and gas exploration company engaged in the assembly of a portfolio of high potential oil and gas exploration prospects. At present the Company does not have any producing properties and has no revenue. The Company is focusing its efforts in Indonesia where large tracts of acreage can be accumulated, there is a long and positive history of oil exploration success, geological conditions are favorable for hydrocarbon accumulation and the company has acquired rights to two production sharing contract areas covering 3 million acres. Oil exploration involves a high degree of technical and commercial risk and is characterized by a continuous need for fresh capital. The Company will try to minimize and spread this risk by joint-venturing in its oil exploration projects with other companies and participating in properties exhibiting a range of risk profiles to reduce its cost exposure and enhance the possibility of wildcat success while restricting its focus to Indonesia where its limited manpower and technical resources are already deployed. For calendar year 2003, the Company's management has set two important strategic objectives, which it hopes to accomplish during the year. These goals, in order of priority, are:

1. Drill two required TAC commitment wells on Bangkudulis Property and commence commercial production to generate internal revenue. Funds to complete this objective shall be raised by private placement as first preference or by farm-out as second preference, or some combination thereof.
2. Drilling three exploratory commitment wells on the Bengara-II Property. Funds to complete this objective shall be raised by private placement as first preference or by farm-out as second preference, or some combination thereof.

ITEM - # 10.
DIRECTORS AND OFFICERS OF THE REGISTRANT

DIRECTORS

Table No. 6 lists, as at 1/20/03, the names of all the Directors of the Company. The Directors generally serve in their respective capacities from their election on the day of the Annual General Meeting of the Company held in January 2000 and will serve until the next Annual General Meeting or until a successor is duly elected, unless the office is vacated in accordance with the Articles of the Company. Casual vacancies on the Board of Directors are filled by the remaining Directors and the persons filling those vacancies hold office until the next Annual General Meeting at which time they may be re-elected or replaced. The Company distinguishes its directors into two groups, "Executive Directors" and "Non-Executive Directors". Executive Directors are those directors who also serve as Executive Officers employed under contract for such service by the Company. Non-Executive Directors are those directors who do not serve also as Executive Officers of the Company and have not contractual relationship with the Company for personal services outside those of being a director. At the end of Fiscal 2002 and at 1/20/03 the Company has three Non-Executive directors, Mr. Wine, Mr. Barker and Mr. Lichman and two Executive Directors Mr. McAdoo and Mr. Schell.

Table No. 6		
<u>Directors</u>		
<u>Name</u>	<u>Age</u>	<u>Date First Appointed a Director</u>
Gary R. Schell (1) (5)	69	February 1993
Gary Wine (1)	47	May 1997
Richard L. McAdoo (2) (4)	48	January 1999
Lawrence Barker, Jr. (3) (4) (5)	82	October 1999
Stan Lichman (1) (4) (5)	56	July 2001

Notes:

- (1) Resident and citizen of Canada.
- (2) Resident of Indonesia and citizen of the United States.
- (3) Resident and citizen of the United States.
- (4) Members of the Audit Committee of the Board of Directors.
- (5) Members of the Executive Compensation Committee of the Board of Directors.

DIRECTORS COMMITTEES

Audit Committee - The board of directors has created an "Audit Committee" and duly appointed executive director McAdoo and non-executive directors Barker and Lichman to serve on the committee for Fiscal 2002. The Audit Committee is charged with the responsibility of coordinating, reviewing and working with the Company's auditors with respect to the annual fiscal year end audit. Mr. McAdoo serves as the committee's chairman.

Compensation Committee - The board of directors has created a "Compensation Committee" and duly appointed executive director Schell and non-executive directors Barker and Lichman to serve on the committee. The Audit Committee is charged with the responsibility to review and recommend contracts and terms of compensation to be paid to Company management. Mr. Schell serves as the committee's chairman.

RESUME OF DIRECTORS

Richard L. McAdoo holds a Bachelors degree in Geology and Masters degrees in Geophysics from Texas Tech University; and a Masters degree in Business Administration from Boston University. He is registered as a Certified Petroleum Geologist by the American Association of Petroleum Geologists. Actively involved in the international oil exploration and production business for the last 24 years, Mr. McAdoo has held a variety of technical and management positions in exploration and production for Mobil Oil Company, Phillips Petroleum Company, Jackson Exploration, Inc., Triton Energy Corporation, Tracer Petroleum Company, and others in many regions including the North Sea, Middle East, Africa, South America, FSU and Asia. Mr. McAdoo is currently a director, President and CEO of Apex Bengara and Apex Yapen. Prior to their acquisition by the Company Mr. McAdoo held a direct and an indirect share holding interest in both Apex Bengara and Apex Yapen.

Lawrence Barker has had a remarkable career in the international oil business and although unofficially retired is still actively involved in consulting from his San Francisco base. He is a 1942 graduate of Yale with a degree in mathematics. After a short wartime engagement as a meteorology officer in the US Army, he studied geology and petroleum engineering at UCLA/USC before embarking on a career as an

independent oil producer in California, Colorado, Texas, Louisiana and Montana. Today he remains active as a producer of oil and gas and also a director and shareholder of a small private company with activity and some production in California and Texas. With other oilmen in 1963, he founded the Independent Indonesian American Petroleum Company (IIAPCO) and he served as its President and CEO until 1971. IIAPCO signed the first ever Production Sharing Contract in Indonesia in 1966 for an unexplored and rank wildcat area offshore Java. A second contract area adjacent to the first was signed in 1968. Substantial discoveries were made on both of IIAPCO's contracts with cumulative oil production from the two areas to date about 3 billion barrels. IIAPCO merged with Natomas and Mr. Barker became a director of that company in 1971. After several other mergers and acquisitions the Natomas share of the two offshore Java contract areas acquired by fledgling IIAPCO are today still in production and operated by MAXUS, formerly of Dallas but now owned by YPF, the Argentine oil company, and Repsol, the Spanish oil company. Mr. Barker's string of successful Indonesian enterprises continued until 1992 with his participation in the development of fields in the Salawati Basin, Irian Jaya, which have today produced over 500 million barrels.

Gary R. Schell is a professional engineer with over thirty years experience in resource exploration and development. He holds a bachelors degree in Mechanical Engineering from the University of Saskatchewan. He was the founder and President of Jet Energy Corp., Geoquest Resource Corporation, Tylox Resources Inc. and Tracer Petroleum Corporation, all resource companies with a significant focus in the oil and gas sector.

Gary D. Wine is a professional geologist and member of the American Association of Petroleum Geologists as well as the Canadian Society of Petroleum Geologists. He holds a bachelors degree in Geology from University of British Columbia. From 1978 to 1997 Mr. Wine held various positions with the Norcen group of companies, including exploration manager - Argentina, senior geologist responsible for prospect generation and regional studies in Northern Alberta and project geologist for Norcen's permits in Malaysia, Indonesia and Papua New Guinea.

Stan Lichman is a financial services professional and owner operator of an independent financial consultant and tax advisory business with offices in Toronto and Vancouver.

EXECUTIVE OFFICERS

Table No. 7 lists, as at 1/20/03, the names of all of the Executive Officers of the Company. Executive Officers are appointed the Board of Directors and hold office indefinitely at the pleasure of the Board of Directors. The term "Executive Officers" includes and is limited to those Company employees who are members of management who are either President, Vice President, Secretary, Treasurer or Chairman of the Company or of any one or more of the Company's majority or wholly owned subsidiaries.

Table No. 7			
<u>Executive Officers</u>			
<u>Name</u>	<u>Age</u>	<u>Current Title</u>	<u>Date First Appointed a Director</u>
Richard L. McAdoo (1)	48	President	November 1999
Gary R. Schell (2)	69	Chairman & Secretary	February 1993
James C. Haebig (3)	67	Director of Subsidiary	April 1999
Craig Doctor (4)	37	VP Investor Relations	April 2000

Notes:

- (1) He is also President/CEO/Director of the Company's Indonesian subsidiaries: Apex (Bengara-II) Ltd. and Continental Energy Yapen Ltd. and has served as such since the Company's acquisition of these subsidiaries in August 1998. He spends substantially full time on the affairs of the Company and its subsidiaries. He is based in and resident in Jakarta, Indonesia, and is employed under contract to both Apex (Bengara-II) Ltd. and Continental Energy Yapen Ltd.
- (2) He was President of the Company from April 1993 until November 1999 when he resigned as President and was elected Chairman of the board of directors and was appointed Secretary; he resigned as Secretary in December 2000 and was reappointed Secretary in July 2001. He spends substantially full time on the affairs of the Company. He is based in and resident in Vancouver, BC, Canada, and is employed under full time contract to the Company.
- (3) He is not a director or officer of the Company, but he is a director, vice president and also Chief Geophysicist for the Company's majority owned subsidiaries Apex (Bengara-II)Ltd. and Yapen Ltd. and for reporting purposes in this annual report he is considered to be an Executive Officer. He spends substantially full time on the affairs of the Company. He is a resident of Jakarta, Indonesia, and is employed under contract to both Apex (Bengara-II) Ltd. and Continental Energy Yapen Ltd.
- (4) Based in and resident in Vancouver, BC, Canada, and employed under full-time contract to the Company. In December 2001 he resigned his office of Vice President and his full-time employment with the Company. As of the record date of this report he is negotiating with the Company for a part-time employment arrangement in the same capacity.

Orders - No Director or Executive Officer has been the subject of any order, judgment, or decree of any governmental agency or administrator or of any court of competent jurisdiction, revoking or suspending for cause any license, permit or other authority of such person or of any corporation of which he is a Director and or Executive Officer, to engage in the securities business or in the sale of a particular security or

temporarily or permanently restraining or enjoining any such person or any corporation of which he is an officer or director from engaging in or continuing any conduct, practice, or employment in connection with the purchase or sale of securities, or convicting such person of any felony or misdemeanor involving a security or any aspect of the securities business or of theft or of any felony. During the past 10 years, no director, officer or promoter of the Company has been the subject of any penalties or sanctions by a court or securities regulatory authority related to the trading in securities, promotion or management of a publicly traded Company, or theft or fraud, except Richard L. McAdoo, who, as an officer of Triton Indonesia, Inc. ("Triton Indonesia"), consented, without admitting or denying allegations made in a civil complaint filed against Triton Energy Corporation (the US parent of Triton Indonesia), Mr. McAdoo and other officers of that company by the US Securities and Exchange Commission, to the entry of a Final Judgment permanently enjoining him from violating the anti-bribery provision and the internal control and books and records provisions of the Foreign Corrupt Practices Act and the Securities Exchange Act of the United States.

Multiple Directorships - Gary Schell is also a Director of Module Resources Incorp., a CDNX traded public company also involved in oil and gas exploration and development. While the Executive Officers and Directors of the Company are involved in other business ventures and may not spend full time on the affairs of the Company, the Company believes that each devotes as much time to the affairs of the Company as are required to satisfactorily carry out their duty.

Arrangements & Relationships - There are no family relationships between any two or more Directors or Executive Officers. There are no arrangements or understandings between any two or more Directors or Executive Officers, pursuant to which anyone was selected as a Director or Executive Officer.

ITEM - # 11.

COMPENSATION OF OFFICERS AND DIRECTORS

COMPANY DIRECTOR COMPENSATION

The Company does not provide cash compensation its Directors for their service in their capacity as Directors. Directors are entitled to reimbursement for reasonable travel and other out-of-pocket expenses incurred in connection with attendance at meetings of the Board of Directors. The Board of Directors may award special remuneration to any Director undertaking any special services on behalf of the Company other than services ordinarily required of a Director. Other than indicated below no Director received any compensation for his services as a Director of the Company, including committee participation and/or special assignments. During Fiscal 2002, the Company did not pay or accrue any salary, fees or other cash compensation directly or indirectly to any director of the Company solely for their services as directors to the Company. Refer to ITEM -#13.

Non-Executive Director Compensation

Non-Executive Directors are those directors not paid for their service as a director and also not employed or paid by the Company in any capacity such as officer or consultant.

Larry Barker, Non-Executive Director During Fiscal 2002, Mr. Barker served as a Non-Executive Director of the Company and is not entitled to cash compensation for such services. The Company did not pay any cash compensation of any kind to Mr. Barker during Fiscal 2002 does not expect to pay any cash compensation to Mr. Barker for his current services as a Non-Executive Director during Fiscal 2003. See ITEM -#12 for his stock options.

Stan Lichman, Non-Executive Director During Fiscal 2002, Mr. Lichman served as a Non-Executive Director of the Company and is not entitled to cash compensation for such services. The Company did not pay any cash compensation of any kind to Mr. Lichman during Fiscal 2002 does not expect to pay any cash compensation to Mr. Lichman for his current services as Non-Executive Director during Fiscal 2003. See ITEM -#12 for his stock options.

Gary Wine, Director and formerly Geological Consultant During Fiscal 2000, Mr. Wine received cash compensation of C\$17,900 indirectly, through GDW Consulting, a private firm owned by Mr. Wine, for geological evaluation and consulting services performed directly for the Company under written contract to the Company. The services contract was terminated in October 1999. Since the termination, the Company has not paid any cash compensation to Mr. Wine for his services as a Non-Executive director and does not expect to pay any cash compensation for the next fiscal year. See ITEM -#12 for his stock options.

Executive Director Contracts

Executive Directors are those directors who are directly or indirectly employed by the Company and serving as paid officers, advisors or employees of the Company in addition to their unpaid service as a director. The following directors have contracts with the Company or its affiliates and are Executive Directors. All contracts with the Company for directors, officers and senior management are reviewed by the Compensation Committee.

Gary Schell, Director and Chairman The Board of Directors, acting on the recommendation of the Compensation Committee, has approved and authorized signature of a renewed annual employment contract dated 08/01/02 with Milner Downs Equestrian Centre Ltd. (a company controlled by Mr. Schell) for the personal services of the Company's Chairman and Secretary, Mr. Gary R. Schell to provide executive management and corporate administrative services in the Company's Langley headquarters office. The contract provides for a salary of CDN \$ 10,000 (US\$ 6,500) per month.

Richard McAdoo, Director and President The Board of Directors, acting on the recommendation of the Compensation Committee, has approved and authorized signature of a renewed annual employment contract dated 08/01/02 with and for the Company's President, Mr. Richard L. McAdoo to provide executive management from the Company's affiliates' Jakarta office. Under the contract, the Company does not pay any direct salary to Mr. McAdoo but does provide for reimbursement of actual personal family medical insurance premiums. Instead Mr. McAdoo is compensated for his services under direct contract to the Company's joint venture subsidiaries Apex (Bengara-II) Ltd. and Continental Energy Yapen Ltd. pursuant to separate annually renewable personal consulting service agreements in the amounts of US\$ 7,500 (since 08/01/98) and US\$ 2,500 (beginning 01/01/01 and ending 8/31/02) per month respectively. However, Mr. McAdoo's contract and consulting service fees for Continental Energy Yapen Ltd. were terminated effective 8/31/02 in accordance with the farm out of the Yapen Property.

Executive Director Compensation

The compensation of executives of the Company is reviewed by the Compensation Committee and set by a vote of the Board of Directors.

Gary Schell, Director and Chairman During Fiscal 2002/2001/2000, Mr. Schell was paid Canadian dollar cash compensation by the Company indirectly through Milner Downs, respectively, CDN\$75,000, CDN\$60,000, and CDN\$60,000 for his personal services as an Executive Officer of the Company. Commencing 01/01/00 Mr. Schell has been paid or accrued a monthly fee of US\$ 1,000 for his personal services rendered as a director to each Indonesian joint venture subsidiary, Apex (Bengara-II) Ltd. and Continental Energy Yapen Ltd. The Company's pays only its 60% joint venture share of all compensation paid by its joint venture subsidiaries to Mr. Schell.

Richard McAdoo, Director and President At end Fiscal 2000 Mr. McAdoo has not been paid or accrued any cash compensation directly from the Company for his services as an Executive Director and Officer of the Company. Instead Mr. McAdoo is compensated for his services under direct contract to the Company's two joint venture subsidiaries Apex (Bengara-II) Ltd. and Continental Energy Yapen Ltd. pursuant to separate annually renewable personal consulting service agreements in the amounts of US\$ 7,500 (since 8/01/98) and US\$ 2,500 (since 1/01/01) per month respectively. However, Mr. McAdoo's contract and consulting service fees for Continental Energy Yapen Ltd. were terminated effective 8/31/02 in accordance with the farm out of the Yapen Property. See "Note on Compensation of Executive Officers by Indonesian Subsidiaries" below. Mr. McAdoo was directly paid or accrued from Apex (Bengara-II) Ltd., salary compensation of US \$90,000 for the years ended Fiscal 2002/2001/2000 and from Continental Energy Yapen Ltd., salary compensation of US \$30,000, US \$17,500, and nil for the same years respectively. The Company's pays only its 60% joint venture share of all compensation paid by its joint venture subsidiaries to Mr. McAdoo.

EXECUTIVE OFFICER COMPENSATION

The Company employs all of its Executive Officers under direct written contract with the Company and or one or more of its subsidiaries. All Executive Officers are contracted on a full time employment basis and are prohibited from engaging in paid employment from third party's who are also engaged in the oil and gas business. During Fiscal 2002, the Company paid or accrued, directly or indirectly salary or other cash compensation totaling CDN\$36,000 and US\$ 72,000 to two Executive Officers as follows:

Craig Doctor, Vice President Mr. Doctor was employed indirectly by the Company under full time contract to perform investor relations services during Fiscal 2002 through his personal company Doc's Consulting Ltd. During Fiscal 2002/2001/2000 he was paid or accrued CDN\$36,000, CDN\$36,000 and CDN\$36,000 respectively for his services.

James C. Haebig, Director and Officer of Subsidiary At end Fiscal 2002 Mr. Haebig has not been paid or accrued any cash compensation directly from the Company for his services as an Executive Director and Officer of the Company's subsidiaries. Instead Mr. Haebig is compensated for his services under direct contract to the Company's two joint venture subsidiaries Apex (Bengara-II) Ltd. and Continental Energy Yapen Ltd. pursuant to separate annually renewable personal consulting service agreements in the amounts of US\$ 7,500 (since

1/01/98) and US\$ 2,500 (since 1/01/01) per month respectively. However, Mr. Haebig's contract and consulting service fees for Continental Energy Yapen Ltd. were terminated effective 8/31/02 in accordance with the farm out of the Yapen Property. Mr. Haebig was directly paid or accrued from Apex (Bengara-II) Ltd., salary compensation of US \$90,000 for each of the years ended Fiscal 2002/2001/2000 and from Continental Energy Yapen Ltd., salary compensation of US \$30,000, US \$17,500 and nil for the same fiscal years respectively. The Company's pays only its 60% joint venture share of all compensation paid by its joint venture subsidiaries to Mr. Haebig.

Note on Compensation of Executive Officers by Indonesian Subsidiaries

During Fiscal 2002 three of the Company's Executive Officers also serve as directors of both the Company's 60% owned Indonesian Joint Venture Subsidiaries, "Indo JV Subs", namely Apex (Bengara-II) Ltd. and Continental Energy Yapen Ltd. Two Executive Officers, Mr. McAdoo and Mr. Haebig, reside in Jakarta, and are also serving officers of the Indo JV Subs in the capacities of President and Vice President Geophysics respectively. Both also personally provide individual geological, geophysical and management services to each of the Indo JV Subs principally at the Jakarta offices. Such personal services are provided under annually renewable service agreements with each Indo JV Sub. All compensation paid to the Executive Officers by the Indo JV Subs is shared in the proportion of 40% by the Company's Indo JV partners and 60% by the Company.

This treatment of his compensation enables the two joint venture subsidiaries to charge the full amount of his compensation against the respective PSC work commitment and also qualify the entire amount of compensation for reimbursement by Pertamina from oil production as cost recovery as and when commercial production is established. The Company indirectly pays 60% of Mr. Haebig's and Mr. McAdoo's compensation by virtue of its funding its joint venture obligations to both Apex (Bengara-II) Ltd. and Continental Energy Yapen Ltd. The Company's joint venture partner, GeoPetro Resources Company, pays the other 40% of their salaries.

The entire amount of the compensation paid by the Company's Indo JV Subs for personal consulting and management services provided by some of the Company's Executive Officers (and of their own employees and consultants) is expected to 1) qualify and count towards fulfillment of the respective Indo JV Subs' obligatory exploration work expenditure commitment under its Production Sharing Contract ("PSC") with Pertamina and 2) is also expected to qualify as "cost recovery" pursuant to the respective Indonesian production sharing contracts and thereby be reimbursed in full to the Company at some future time. The full amount of such accumulated compensation costs may be recovered by the Indo JV Subs from future petroleum production revenues in accordance with the cost recovery provisions of the respective production sharing contracts in the event that commercial petroleum production is achieved.

Stock Options

The Company does grant incentive stock options to Directors and to Officers and employees; refer to ITEM-#12, "Options to Purchase Securities from Registrant or Subsidiaries". All stock options are set at levels determined from time to time by the board of directors in accordance with applicable securities regulations.

Pension Plans & Stock Ownership Programs

No funds were set aside or accrued by the Company during Fiscal 2002 to provide pension, retirement or similar benefits for Directors or Executive Officers. Except for the stock option program discussed in ITEM-#12, the Company has no material bonus or profit sharing plans pursuant to which cash or non-cash compensation is or may be paid to the Company's Directors or Executive Officers.

Termination Compensation

The Company has no plans or arrangements in respect of remuneration received or that may be received by Executive Officers of the Company in Fiscal 2002 to compensate such officers in the event of termination of employment (as a result of resignation, retirement, change of control) or a change of responsibilities following a change of control, where the value of such compensation exceeds \$100,000 per Executive Officer. No Executive Officer/Director received other compensation in excess of the lesser of US\$25,000 or 10% of such officer's cash compensation, and all Executive Officers/Directors as a group did not receive other compensation which exceeded US\$25,000 times the number of persons in the group or 10% of the compensation.

ITEM - # 12.
OPTIONS TO PURCHASE SECURITIES FROM REGISTRANT OR SUBSIDIARIES

Incentive Stock Options to purchase securities from the Company can be granted only to Directors, Officers, employees and consultants and Employees of the Company in accordance with applicable securities regulations. All optionees other than directors and officers must be employees or consultants or subcontractors engaged under written contract by the Company. The Company has no formal written stock option plan in place as of the date of this Annual Report. At 1/20/03, the Company had 5,630,000 stock options outstanding entitling the holders to purchase 5,630,000 common shares at a price of \$0.15. The names and titles of the Directors and Executive Officers of the Company to whom outstanding stock options have been granted and the number of common shares subject to such options are set forth in Table No. 8, also effective as at 1/20/03, as well as the number of options granted to them as a group. Outstanding options to other employees and consultants as a group is also shown.

Table No. 8 <u>Stock Options Outstanding</u>				
<u>Name of Optionee</u> <u>Directors & Officers</u>	<u>US\$ Option</u> <u>Exercise Price</u>	<u>Date Option Expires</u>	<u>Number of</u> <u>Options</u>	<u>Percent</u> <u>Of Total</u>
Gary Schell	\$0.15	7/30/04	1,000,000	-----
Gary Schell	\$0.15	4/29/04	400,000	25.0%
Richard McAdoo	\$0.15	7/30/04	900,000	16.0%
James Haebig	\$0.15	7/30/04	635,000	11.3%
Gary Wine	\$0.15	7/30/04	200,000	3.5%
Lawrence Barker	\$0.15	7/30/04	100,000	-----
Lawrence Barker	\$0.15	4/29/04	100,000	3.5%
Stan Lichman	\$0.15	7/30/04	100,000	-----
Stan Lichman	\$0.15	4/29/04	100,000	3.5%
Total Directors & Officers as a Group			3,535,000	62.8%
Total Employees & Consultants as a Group			2,095,000	37.2%
Total Stock Options Outstanding at 1/20/03			5,630,000	100.0%

At 1/20/03, the Company had 15,490,606 share purchase warrants outstanding entitling the holders to purchase 15,490,606 common shares at prices between US\$0.15 and CDN\$0.70. All of the share purchase warrant issued by the Company have been issued in conjunction with private placements of Company securities or loans as incentive consideration to the placee or lender. Table No. 9 lists, also as at 1/20/03 share purchase warrants outstanding, the date the share purchase warrants were issued, the exercise price, and the expiration date of the share purchase warrants.

Table No. 9 <u>Share Purchase Warrants Outstanding</u>						
		<u>Warrant Exercise Price US\$</u> <u>As Currently Amended & Exercisable</u>				
<u>Date Warrants</u> <u>Issued</u>	<u>Number of Warrants</u> <u>Originally Issued</u>	<u>Number of Warrants</u> <u>As Amended & Still</u> <u>Outstanding</u>	<u>First Year</u>	<u>Second Year</u>	<u>Third Year or</u> <u>As Revised</u>	<u>Date Warrants</u> <u>Expire</u>
9/10/02	5,408,000	4,656,666	\$ 0.15	\$0.30	\$ 0.60	9/10/05
7/23/02	1,305,000	1,112,000	\$ 0.15	None	None	7/23/03
7/19/02	500,000	500,000	\$ 0.15	\$0.30	\$ 0.60	7/19/05
7/18/02	1,621,840	1,621,840	\$ 0.15	None	None	7/18/03
9/21/00	4,923,000	4,923,000	Revised	Revised	\$0.15	7/30/03
6/22/00	1,829,332	1,096,000	Revised	Revised	\$0.15	7/30/03
3/01/00	350,000	185,000	Revised	Revised	\$0.15	1/31/03
10/12/99	605,600	605,600	Revised	Revised	\$0.15	7/30/03
11/24/99	500,000	500,000	Revised	Revised	CDN\$0.70	11/24/04
8/11/99	290,000	290,000	Revised	Revised	\$0.15	7/30/03
Total		15,490,606	Outstanding at 1/20/03			

ITEM - # 13.
INTEREST OF MANAGEMENT IN CERTAIN TRANSACTIONS

The following are the only transactions since 8/1/00, or proposed transactions, which have materially affected or will materially affect the Company in which any director, officer, or beneficial holder of more than 10% of the outstanding common shares, or any of their respective relatives, spouses, associates or affiliates has had or will have any direct or material indirect interest. Management believes the transactions referenced below were on terms at least as favorable to the Company as it could have obtained from unaffiliated parties.

Richard L. McAdoo, President/Director - During fiscal 2002 but effective from 8/01/00 the Company closed its purchase of the Bangkudulis property; see ITEM-#1, "Acquisition/Disposition of Principal Properties". At the effective date of the sale, Dimensions West Energy, Inc., the seller of the Bangkudulis property, and the Company shared a common director, Richard McAdoo, who up until his resignation on 1/30/01 was also President and a director of the seller. Mr. McAdoo declared his conflict of interest and abstained from voting as a director of both the Company and the seller with regard to the Bangkudulis Acquisition. Up until his resignation from the seller's board in January 2000 preceding the sale effective date, the Company also shared a second director, Gary Schell with the seller. Neither Mr. McAdoo nor Mr. Schell have or expect to have any personal beneficial interest in the contemplated transaction other than that of being shareholders.

Lawrence J. Barker, Jr., Non-executive Director - During Fiscal 2000, the Company farmed out a 40% interest in both its Apex (Bengara-II) Ltd. property and its Continental Energy Yapi Ltd. property to a privately held American company, GeoPetro Resources Company of San Francisco. See "ITEM-#1, "Acquisition/Disposition of Principal Properties". At the time of the farm out Mr. Barker was a director of both the Company and GeoPetro. Mr. Barker did not participate in the negotiations and did not personally receive any personal material benefit or compensation from either the Company or GeoPetro as a result of the farm out.

P A R T - II
ITEM - # 14.
DESCRIPTION OF SECURITIES TO BE REGISTERED

Debt Securities to be Registered.

This Form-20F filing is made as an annual report and matters ordinarily required to be disclosed pursuant to this sub-heading of this ITEM-#14 are not applicable. A description of the Company's currently registered securities is made in ITEM-#4, ITEM-#5 and ITEM-#12.

American Depositary Receipts.

This Form-20F filing is made as an annual report and matters ordinarily required to be disclosed pursuant to this sub-heading of this ITEM-#14 are not applicable. A description of the Company's registered securities is made in ITEM S#4, #5 and #12.

Other Securities to be Registered.

This Form-20F filing is made as an annual report and matters ordinarily required to be disclosed pursuant to this sub-heading of this ITEM-#14 are not applicable. A description of the Company's registered securities is made in ITEM S-#4, #5 and #12.

P A R T - III
ITEM - # 15.
DEFAULTS UPON SENIOR SECURITIES

This Form-20F filing is made as an annual report and matters ordinarily required to be disclosed pursuant to this ITEM-#15 are not applicable.

ITEM - # 16.
CHANGES IN SECURITIES AND CHANGES IN SECURITY FOR REGISTERED SECURITIES

This Form-20F filing is made as an annual report and matters ordinarily required to be disclosed pursuant to this ITEM-#16 are not applicable. A description of the Company's currently authorized, issued, outstanding and registered securities is made in ITEMS-#4, #5 and #12.

P A R T - I V
ITEM - # 17.
FINANCIAL STATEMENTS

For its 7/31/02 year-end reporting, and all subsequent reporting, the Company adopted the U.S. currency as its reporting currency and has prepared its financial statements on that basis. Accordingly, all prior years' figures presented have been recast in the U.S. currency through a "translation of convenience" whereby all amounts appearing are restated from Canadian to U.S. currency using the exchange rate prevailing at 7/31/01. The accounts of the Company are now prepared in U.S. funds and the Company's Canadian operations are translated into U.S. dollars under the temporal method. The Company's financial statements are prepared in accordance with Canadian Generally Accepted Accounting Principles (GAAP), the application of which, in the case of the Company, conforms in all material respects for the periods presented with US GAAP, except as disclosed in footnotes to the financial statements.

AUDITORS

The Company's auditors are Staley Okada & Partners (formerly Staley, Okada, Chandler & Scott), ("SOAP") Chartered Accountants, of Surrey, British Columbia, Canada. SOAP or its predecessor firm has been the Company's auditors since the Company's incorporation in 1984. SOAP has been a member of the Institute of Chartered Accountants of British Columbia since 1962. The firm is also a member of the Canadian Institute of Chartered Accountants. In addition, two SOAP partners are qualified American CPA's in good standing in the states of Illinois and Washington.

AUDITS, REPORTING, FINANCIAL STATEMENTS AND THEIR APPLICABILITY

The Company prepares annual audited consolidated financial statements as at its year end date of 31 July. These consolidated financial statements include the accounts of the company and its majority owned subsidiaries. During the course of the Company's fiscal year management prepares unaudited quarterly consolidated financial statements as at 31 October, 31 January and April 30 which are filed with British Columbia Securities Commission within 60 days of the quarter's end.

ELECTRONIC FILING AT THE SEDAR WEBSITE

The Company makes continuous financial and corporate information disclosure as well as statutory filings with Canadian securities regulators on the internet at the "SEDAR" website www.sedar.com. SEDAR is the "System for Electronic Document Archiving and Retrieval" employed by Canadian securities regulatory commissions to enable publicly traded companies to electronically file and archive documents and filings in compliance with applicable laws and securities trading regulations. The Company began filing electronically on SEDAR in 1997.

STATEMENT ON AGE OF FINANCIAL STATEMENTS

The audited financial statements included pursuant to this ITEM-#17 are effective as at the Company's Fiscal 2002 year end at 7/31/02. The cut-off Date of the auditors report for the same Fiscal 2002 financial statements is 12/06/02. The Record Date of this Form-20F annual report is 1/20/03. The actual electronic filing of this Form-20F report and its Exhibits, including the Fiscal 2002 financial statements, on EDGAR is scheduled for 1/24/03. Elapsed time between the fiscal year end effective date of the financial statements and the effective date of the auditors report is 128 days. Elapsed time between the fiscal year end effective date of the financial statements and the Record Date of this report is 168 days.

FINANCIAL STATEMENTS ATTACHMENT & INDEX

The auditors report and audited financial statements as required under ITEM-#17 are included attached hereto, are incorporated herein and are found immediately following the text and Exhibits of this Form-20F Annual Report starting on page 81 and continuing as shown in the table of contents following.

Consolidated Financial Statements Table of Contents Attached Hereto as Pages Numbered 42 to 84		<u>Page</u>
1.	Financial Statement Title Page	56
2.	Auditor's Report, dated 12/06/02	56
3.	Management's Responsibility For The Financial Statements	57
4.	Consolidated Balance Sheet at 7/31/02 and 7/31/01	58
5.	Consolidated Statement of Shareholders' Equity (Deficiency) for the Years ended July 31, 2000, 2001 and 2002	59
6.	Consolidated Statement of Loss for the Years ended July 31, 2000, 2001 and 2002; and cumulative from 08/01/93 (date of development stage)	60
7.	Consolidated Statement of Cash Flows for the Years ended July 31, 2000, 2001 and 2002 and cumulative from August 1, 1993 (date of development stage)	61
8.	Consolidated Schedule of Resource Property Costs for the Years ended July 31, 2000, 2001 and 2002 and cumulative from 08/01/93 (date of development stage)	62
9.	Consolidated Schedule of Administrative Expenses for the Years ended July 31, 2000, 2001 and 2002 and cumulative from 08/01/93 (date of development stage)	63
10.	Notes to the Consolidated Financial Statements	64
11.	Auditors' Report on Schedules Title Page	80
12.	Auditor's Report on Schedules, 12/06/02	80
13.	Schedules to the Consolidated Financial Statements	81
	Schedule I: Indebtedness of and (to) Related Parties – Current	82
	Schedule II(a): Property, Plant and Equipment	83
	Schedule II(b): Accumulated Amortization	84
	Note to Report on Schedules	84

ITEM - # 18.
FINANCIAL STATEMENTS

The Registrant has elected to provide its financial statements pursuant to ITEM -#17.

ITEM - # 19.

EXHIBITS

INDEX TO EXHIBITS

The following Exhibits are incorporated herein this Form-20F and are attached hereto following the signatures below at the page number indicated on the following index:

<u>ATTACHED EXHIBITS TITLE & CONTENTS</u>	<u>Page</u>
Exhibit-1: <u>Certificate, Memorandum & Articles of Incorporation</u>	48
Exhibit-2: <u>Shareholder Rights Instruments</u>	48
Exhibit-3: <u>Voting Trusts</u>	48
Exhibit-4: <u>Material Contracts</u>	48
Exhibit-5: <u>Foreign Patents</u>	48
Exhibit-6: <u>Statement How Earnings Per Share Calculated</u>	48
Exhibit-7: <u>Statement How Reported Ratios Are Calculated</u>	48
Exhibit-8: <u>Diagram of Parent & Subsidiaries</u>	49
Exhibit-9: <u>Statement on Age of Financial Statements</u>	49
Exhibit-10: <u>Other Exhibits</u>	49-55
Exhibit-10A: <u>Glossary: Part-A: Oil Industry Technical Terms Definitions</u>	49
Exhibit-10A: <u>Glossary: Part-B: Abbreviations and Units of Measurement</u>	51
Exhibit-10A: <u>Glossary: Part-C: Conversion Factors</u>	52
Exhibit-10B: <u>PSC-Terms: Part-A: Production Sharing Contract Terminology</u>	53
Exhibit-10B: <u>PSC-Terms: Part-B: Production Sharing Terms & Conditions</u>	55
Exhibit-10C: <u>Indonesia Update: Part-A: Oil Industry Overview</u>	58
Exhibit-10C: <u>Indonesia Update: Part-B: Social/Economic/Political</u>	59
Exhibit-10C: <u>Indonesia Update: Part-C: Indonesian Crude Oil Pricing</u>	61
Exhibit-11: <u>Financial Statements Attachments Pursuant to ITEM-#17:</u>	61-90
Exhibit-11A: <u>Auditors Report - End Fiscal Year 2002, 7/31/02, Financial Statements</u>	62
Exhibit-11B: <u>Financial Statements - Audited at End Fiscal Year 2002 7/31/02</u>	64
Exhibit-11C: <u>Notes to Financial Statements - Audited at End Fiscal Year 2002 7/31/02</u>	70
Exhibit-11D: <u>Supplemental Schedules to Financial Statements - Audited at End Fiscal Year 2002 7/31/02</u>	86

COPIES & DISTRIBUTION

This Form-20F annual report and its Exhibits, including the audited financial statements at Fiscal 2002 year end 7/31/02, was submitted for filing electronically on the US-SEC's EDGAR system on 1/24/03.

SIGNATURES

The registrant hereby certifies that it meets all of the requirements for filing on Form 20-F and that it has duly caused and authorized the undersigned to sign this Form-20F Fiscal 2002 year end 7/31/02 annual report on its behalf.

Continental Energy Corporation
(Registrant)

"Gary R. Schell"

By: Gary R. Schell
Director & Chairman of the Board
Record Date: January 20, 2003

E X H I B I T S 1 THROUGH 11
To The US-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

E X H I B I T - 1 : CERTIFICATE, MEMORANDUM & ARTICLES OF INCORPORATION.

Materials required to be filed under this Exhibit-1 are incorporated into this Annual Report Form-20F filing by this reference. Exhibit-1 consists of those copies of the Company's certificate of incorporation, certificates of name changes, memorandum of incorporation, and articles of association already filed with the US-SEC under cover of previous filings made with the US-SEC by the Company including but not limited to the most recently filed Form-20F Annual Report for the Company's last fiscal year (as amended), Form-20F Annual Report filings for previous years; and any information filed as part of a Form-6K periodic filings of material changes, news, contracts or events affecting the Company.

E X H I B I T - 2 : SHAREHOLDERS RIGHTS INSTRUMENTS

The disclosure required to be filed under this Exhibit-2 is incorporated into this Annual Report Form-F filing by the following reference. Exhibit-2 disclosure consists of the same materials referred to and incorporated by reference under Exhibit-1 above. All instruments defining rights of holders of Company securities instruments are therein described or presented in their entirety. A statement commenting upon shareholder rights instruments issued by the Company relative to the date of this annual report is also set forth in the body of this Form-20F within ITEM -#4 under the section entitled "Shareholders Rights Instruments".

E X H I B I T - 3 : VOTING TRUSTS

The disclosure required to be filed under this Exhibit-3 is incorporated into this Annual Report Form-F filing by the following reference. Exhibit-3 disclosure consists of the same materials referred to and incorporated by reference under Exhibit-1 above and/or under Exhibit-4 below. Any share voting trusts, pooling agreements or similar agreements regarding collaborative voting of the Company's shares are therein described or presented in their entirety. A statement commenting upon voting trusts known to the Company relative to the date of this annual report is also set forth in the body of this Form-20F within ITEM -#4 under the section entitled "Voting Trusts".

E X H I B I T - 4 : MATERIAL CONTRACTS

Materials required to be filed under this Exhibit-3 is incorporated into this Annual Report Form-F filing by this reference. Exhibit-3 consists of those copies of the Company's material contracts already filed with the US-SEC under cover of previous filings made with the US-SEC by the Company (as amended); including, but not limited to, the most recently filed Form-20F Annual Report for the Company's last fiscal year, Form-20F Annual Report filings for previous years; and any information filed as part of a Form-6K periodic filings of material changes, news, contracts or events affecting the Company.

E X H I B I T - 5 : FOREIGN PATENTS

The Company does not have any foreign patents. The Company has never registered any foreign patents in the past and does not contemplate registering any foreign patents in the future.

E X H I B I T - 6 : STATEMENT HOW EARNINGS PER SHARE CALCULATED

The disclosure required to be filed under this Exhibit-6 is incorporated into this Annual Report Form-F filing by the following reference. A statement commenting upon how earnings per share are calculated is set forth in the body of this Form-20F within ITEM -#8 under the section entitled "How Earnings Per Share Are Calculated".

E X H I B I T - 7 : STATEMENT HOW REPORTED RATIOS ARE CALCULATED

The disclosure required to be filed under this Exhibit-7 is incorporated into this Annual Report Form-F filing by the following reference. A statement commenting upon how any ratio of earning to fixed charges, any ratio of earnings to combined fixed charges and preferred stock dividends or any other ratios in the registration statement or report is set forth in the body of this Form-20F within ITEM -#8 under the section entitled "How Reported Ratios Are Calculated".

E X H I B I T - 8 : SUBSIDIARIES

The disclosure required to be filed under this Exhibit-8 is incorporated into this Annual Report Form-F filing by the following reference. The identity and relationship of the registrant as parent company and its subsidiaries is set forth in the body of this Form-20F within ITEM -#1 under the section entitled "Subsidiaries".

E X H I B I T - 9 : STATEMENT ON AGE OF FINANCIAL STATEMENTS

The disclosure required to be filed under this Exhibit-9 is incorporated into this Annual Report Form-F filing by the following reference. A statement commenting upon the age of the attached audited financial statements of the Company relative to the date of this annual report is set forth in the body of this Form-20F within ITEM -#17 under the section entitled "Statement on Age of Financial Statements".

E X H I B I T - 10A : GLOSSARY

To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART-"A" : TECHNICAL TERMS DEFINITIONS

The meaning of certain technical words, terms and phrases commonly used in the oil and gas industry and also used in the above referenced filing are described in the glossary below. Described terms are capitalized below and throughout the reference filing only when capitalized in standard usage.

<u>Condensate</u>	Liquid hydrocarbons at atmospheric pressure and temperature which are condensed out of and recovered from natural gas production. Condensate typically has a high gravity and if recovered in sufficient quantities it can be sold at a premium to crude oil. If recovered in low quantities insufficient to justify the cost of separate handling it is commonly 'spiked' into and mixed with crude oil production.
<u>DHI</u>	A DHI is a Direct Hydrocarbon Indication. A seismic DHI is a specific type of "seismic anomaly" widely accepted by the petroleum exploration industry to be a seismically derived direct indication of the actual presence of hydrocarbons, especially gas. Examples of seismic DHI include bright spots, flat spots and gas chimneys. Another example of a DHI is the presence of naturally occurring oil and gas seepage observed currently or historically at a specific land or seabed location or encountered in old wells or diggings.
<u>Drillable Prospect</u>	"Prospects" are commonly sub-classified as "Drillable Prospect" ("DP") when in the opinion of the geologist and geophysicist making the determination, no further seismic or geological supporting investigation is necessary and the prospect is ready for drilling. Prospects not classified as drillable are legitimate potential drilling targets which require some additional element of confidence building investigation and substantiation prior to being elevated to "drillable prospect" status.
<u>Dry Hole</u>	Any wellbore drilled but found to be incapable of producing crude oil or natural gas in a quantity sufficient to justify completion as an oil or gas production well.
<u>Exploration Well</u>	A well that is designed to test for and to confirm the presence of hydrocarbons.
<u>Hydrocarbons</u>	Complex chemical compounds in which hydrogen and carbon are predominant; in the oil and gas industry, usually synonymous with petroleum and meant to include both crude oil and natural gas.
<u>Indonesian Crude Price or "ICP"</u>	An index price of Indonesian crude oil used in the Indonesian oil and gas industry.
<u>Lead</u>	A "lead" is promising geological or seismic indication of a potential petroleum resource that may, subject to additional investigation, develop into a drilling target or prospect. A lead may be an interesting geological feature, an event observed on seismic, an anomaly recognized on remote sensing data, from surface geological field work and/or from geochemical or geophysical surveys which, subject to further investigation, definition and quantification may be upgraded to a prospect or even a "play".
<u>LNG</u>	Liquefied Natural Gas, generally limited to liquefied methane and produced from natural gas by application of external industrial processing after first removing natural gas liquids.
<u>LPG</u>	Liquefied Petroleum Gas, generally restricted to propane and butane extracted from natural gas by application of an external industrial process.

<u>Natural Gas Liquids or "NGL"</u>	Liquids removed from natural gas either by atmospheric condensation or industrial process, including condensate and LPG.
<u>Plan Of Development or "POD"</u>	A comprehensive technical and financial justification and plan for commercial exploitation of a petroleum discovery prepared by PSC Contractors, filed with and subject to review and approval by Pertamina.
<u>Play</u>	A "play" is a distinctive and recognized geological feature, unique to a given geographic area, having demonstrable regional extent that exhibits geologic characteristics favorable for the entrapment of a petroleum resource possibly in multiple accumulations. It would be expected that further geological investigation into the feature or area or concept defined and recognized as a play will yield multiple drilling targets ("prospects") and other promising indications ("leads") pointing to the potential presence of a petroleum resource. Plays may or may not be restricted to one or more rock or time stratigraphic interval and may overlap one another in map view. A play is normally conceptualized, named and carefully described soon after it is recognized. The extent of any play is limited only by its definition.
<u>Possible Reserves</u>	Petroleum reserves that are estimated to be recoverable in the same manner as probable reserves, but with a lesser degree of certainty than probable reserves.
<u>Potential Resource</u>	An estimation of the amount of petroleum resource that may be present in a mapped geological feature with apparently favorable geological conditions for the entrapment of petroleum but which is not confirmed by any well or factual information and is inherently a highly speculative estimation. See "resource" and "reserves".
<u>Probable Reserves</u>	Petroleum reserves that are estimated to be recoverable from undeveloped tracts or zones where the potential has been reasonably confirmed by geological or geophysical control and engineering data, but where available data has been insufficient to establish proof that these reserves would be economically recoverable under current technology and existing market conditions (thereby justifying inclusion within proven reserves).
<u>Prospect</u>	A "prospect" is a seismically or geologically mapped feature, with evident closure of its boundary limits, exhibiting geological characteristics favorable as a drilling target in the search for a commercially exploitable petroleum resource accumulation. Normally several prospects are found within the larger area of the geological feature and concept of a defined play. Some prospects may even test two overlapping plays at a single location. Prospects may or may not be restricted to one or more stratigraphic intervals and multiple prospects may overlap one another in map view.
<u>Proven Reserves</u>	Petroleum reserves that are reasonably expected to be recoverable on the basis of adequate tests, geological or geophysical control and engineering data and other information from existing wells given current technology and existing market conditions.
<u>Reserves</u>	The portion of a resource that is realistically considered to be economically recoverable using existing technology, taking into account costs, sales prices and other prevailing economic conditions; often used with the modifying terms proven, probable or possible as an indication of the degree of confidence of the estimate. All statements of reserves are only estimates and there is no guarantee any such estimates will be realized. See the 85-page 1998 revision of the Society of Petroleum Evaluation Engineers (SPEE) Monograph I, Guidelines for Application of the Definitions for Oil and Gas Reserves, which was approved by the World Petroleum Congress in October 1996 and by the Society of Petroleum Engineers in March 1997 to illustrate the "Petroleum Reserves Definitions" codified in 1988 by the Reserves Definitions Committee, Society of Petroleum Evaluation Engineers for detailed description of reserves classifications.
<u>Reservoir Conditions</u>	Conditions of temperature and pressure existing within a given oil or gas reservoir in the subsurface. Commonly used in estimates of oil and gas resource, in-place within the reservoir without adjustments for recovery.
<u>Resource</u>	An accumulation of petroleum, regardless of amount or extent, that may be commercially exploitable. Only that portion of a resource which is commercially exploitable may be called reserves in accordance with the definitions of reserves classifications. The resource is the total amount of petroleum contained in the accumulation without downward adjustment for the amount of petroleum which is not technically feasible to extract and the amount which is not economically justifiable to extract. Resource is commonly referred to in the oil industry as OOIP, Original Oil in Place or OGIP, Original Gas in Place.
<u>Seismic Anomaly</u>	"Seismic anomalies" are a specific type of lead recognized only on seismic profiles. Seismic anomalies may also be a part of and/or characteristic of a particular prospect or play, but are not, of themselves, either prospects or plays. Examples of seismic anomalies include bright spots, dim spots, flat spots, polarity reversals, attribute anomalies, character changes, fade-outs and gas chimneys.
<u>Sumatran Light Crude or "SLC"</u>	Sumatran Light Crude, a category of crude oil with defined characteristics produced from the Minas field in Sumatra, Indonesia, and used as a benchmark in ICP oil price statistics and calculations and crude oil delivery contracts.
<u>Wellbore</u>	The hole produced by drilling. Otherwise referred to as a 'well' if successfully producing or being utilized or as a 'dry hole' if plugged and abandoned.
<u>Wellhead</u>	The accumulation of caps and valves at the surface end of the wellbore.

E X H I B I T – 10A : GLOSSARY
To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART–“B” : ABBREVIATIONS OF UNITS OF MEASURE

In the above referenced filing the following commonly used oil and gas industry abbreviations for certain units of measure have the meanings ascribed to them below.

<u>API Gravity</u>	A measure as defined by the American Petroleum Institute of the specific gravity or density of Crude Oil or Condensate.
<u>B</u>	As a part of an abbreviation, a reference meaning billion as in BCF (billion cubic feet).
<u>Barrel</u>	A volumetric measure of oil equal to 42 United States gallons at a temperature of 60 degrees Fahrenheit
<u>bbl or bbls</u>	Barrel or barrels
<u>bbls/d or BPD</u>	Barrels per day
<u>BO</u>	Barrels of oil
<u>BOE</u>	Barrels of Oil Equivalent. An energy content equivalence measure used to compare oil and gas volumes. One BOE is defined in the Production Sharing Contract as 6,000 standard cubic feet of natural gas, given that the natural gas is predominantly methane with a calorific value of 1,000 BTU per cubic foot of gas. For Natural Gases with significantly different calorific values other conversion factors may apply.
<u>BOPD</u>	Barrels of oil per day, a flow rate
<u>BTU</u>	British Thermal Unit. A measure of energy content used in the evaluation of natural gas, defined as the amount of heat required to raise the temperature of one pound of water by one degree Fahrenheit
<u>BWPD</u>	Barrels of water per day; a flow rate
<u>CF or cf</u>	cubic feet
<u>cf/d or CFD or CFPD</u>	cubic feet per day
<u>EOR</u>	Enhanced Oil Recovery
<u>GOR</u>	Gas to Oil Ratio, a measure of the amount of natural gas produced relative to the amount of crude oil produced, commonly measured in standard cubic feet of gas per barrel of oil
<u>KBE</u>	Kelly bushing elevation. Located on the floor of the drilling rig, the kelly bushing is the usual reference for all depth measurements during drilling operations
<u>M</u>	As a part of an abbreviation, a reference meaning thousand, as in MBbls (thousand barrels) or MBO (thousand barrels oil)
<u>MM</u>	As a part of an abbreviation, a reference to million, as in MMCF (million cubic feet) or MMBO (million barrels oil)
<u>Mscf/d</u>	thousands of standard cubic feet of gas per day; gas production rate
<u>psia</u>	pounds per square inch absolute
<u>Rupiah or Rp</u>	Rupiah. Currency of the Republic of Indonesia
<u>SCF</u>	Standard cubic feet; referring to a volume of gas at standard conditions
<u>SCF/d or SCFPD</u>	Standard cubic feet per day; a flow rate
<u>Standard Conditions</u>	Conditions of temperature and pressure, defined by the American Petroleum Institute as 14.65 pounds per square inch at 60 degrees Fahrenheit, commonly used as a standard for volumetric measures of oil and gas at time and point of sale
<u>STB/d</u>	Stock Tank Barrels of oil per day; oil production rate
<u>STB</u>	Stock Tank Barrel. A barrel of liquid at stock tank conditions
<u>STBO</u>	Stock tank barrels of oil
<u>Stock Tank Conditions</u>	Naturally existing atmospheric conditions of temperature and pressure. Recoverable oil reserves are commonly quoted in 'stock tank barrels' (STBO)
<u>T</u>	As a part of an abbreviation, a reference to trillion as in TCF (trillion cubic feet)

E X H I B I T – 10A : GLOSSARY
To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART-“C” : CONVERSION FACTORS

Conversion factors between common, internationally used, oil and gas industry units of measure and traditional American “oilfield” units are set forth in the following table and conform to “American Petroleum Institute” or “API” definitions. These conversions may be utilized for units expressed throughout this filing. Traditionally oil and gas are measured and described in terms of mass, volume and heat content. Mass and volume are not equivalent at all terms of pressure, temperature and composition and the API’s commonly accepted conversion factors either quote the actual composition, temperature and pressure at the time of measurement or adjust them to traditional “standard” conditions of 32 API degree gravity for oil, and standard temperature of 60 degrees Fahrenheit and standard pressure of 14.695 pounds per square inch.

<u>To Convert From</u>	<u>To</u>	<u>Multiply By</u>
Barrels of Oil	Cubic Meters (m ³) (oil)	0.159
Cubic Meters (m ³) (oil)	Barrels (oil)	6.290
Cubic Meters (gas)	Cubic Feet (gas)	35.305
Tonnes (1,000 kilograms)	Barrels of 32° API oil	7.280
Barrels of Oil	Tonnes	0.137
Feet	Meters	0.305
Meters	Feet	3.281
Miles	Kilometers ("km")	1.609
Kilometers (km)	Miles	0.621
Acres	Hectares	0.405
Hectares	Acres	2.471
Acres	Square kilometers	0.005
Square kilometers	Acres	247.1

--- End Exhibit-10A---

E X H I B I T – 10B : PSC TERMS
To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART–“A” : PRODUCTION SHARING CONTRACT TERMINOLOGY & DEFINITIONS

Certain words, terms, phrases and concepts used to describe Continental's Indonesian oil and gas concessions, and the Production Sharing Contracts from whence they are derived, are unique or contractually defined and for the benefit of the reader are defined and described below.

<u>AWP&B</u>	Annual Work Program and Budget, a planning document prepared by a Production Sharing Contractor, submitted annually in advance of each calendar year to Pertamina for its review and approval
<u>Bangkudulis</u>	The name of a petroleum concession in Indonesia in which the Company has a participating interest. Often used to define Block or PSC, as in Bangkudulis Block or Bangkudulis TAC.
<u>Bengara-II</u>	The name of a petroleum concession in Indonesia in which the Company has a participating interest. Often used to define Block or PSC, as in Bengara-II Block or Bengara-II PSC.
<u>Bengara-II Block</u>	The 3,650 square kilometer area defined as the Contract Area within the Bengara-II PSC, located mostly onshore and partially offshore in the Indonesian province of East Kalimantan and within which the Contractor has exclusive rights to explore for and produce petroleum.
<u>Budget</u>	Cost estimates of all items included in the Work Program. The Budget is submitted annually in advance of each calendar year to Pertamina, for Pertamina's approval together with the Work Program on which it is based. Together with the Work Program, it is referred to as the AWP&B.
<u>Contract Year</u>	A period of twelve (12) consecutive months according to the Gregorian Calendar counted from the effective date of the PSC or from an anniversary of such effective date.
<u>Contract Area</u>	The geographic area in which exploration, development and other petroleum production activities are permitted to be conducted under a particular PSC.
<u>Contractor</u>	Unless the context otherwise indicates, the party contracting with Pertamina under a PSC; a Production Sharing Contractor.
<u>Cost Oil or Cost Gas</u>	Any oil or gas which is sold to recover the capital and non-capital operating costs incurred by the Production Sharing Contractor in the performance of oil and gas production activities within the Contract Area. Cost Oil and Cost Gas are not taxable and all of the sales proceeds of Cost Oil and Cost Gas accrue to the benefit of the Production Sharing Contractor and not to Pertamina.
<u>Cost Recovery</u>	The signature feature of the PSC contract is the concept and PSC provision for "Cost Recovery". Cost Recovery is a PSC provision which provides that any Contractor making a commercial discovery of petroleum within his PSC contract area is allowed to fully recover ALL of expenditures, both accumulated past costs and all current costs as they are realized, within the contract area out of first revenue from oil and gas production prior to sharing any remaining revenue with Pertamina and the State.
<u>Crude Oil</u>	Crude Oil means crude mineral oil, asphalt, ozokerite and all kinds of hydrocarbons and bitumens, both in solid and in liquid form, in their natural state or obtained from Natural Gas by condensation or extraction.
<u>Domestic Market Obligation or "DMO"</u>	An obligation set forth in the PSC in which the Contractor is required to sell a portion (normally 25%) of its share of Profit Oil to Pertamina at a reduced price (normally 15% of market price) under certain circumstances (normally commencing after five years of production).
<u>Firm Commitment</u>	Typically, a PSC Contractor may not relinquish the PSC during the first three Contract Years. The accumulated Work Commitment for the initial three Contract Years becomes the minimum amount the Contractor is obliged to expend under the Bengara-II PSC and is referred to as the "Firm Commitment". The Firm Commitment for the Bengara-II PSC is US\$7,000,000. The Firm Commitment for the Yapen PSC is US\$5,000,000.
<u>First Tranche Petroleum or "FTP"</u>	A PSC term and obligation set which sets aside a fixed percentage of the first portion of Crude Oil or Natural Gas produced from a Contract Area, 15% in the case of Yapen PSC and 20% in the case of Bengara-II PSC, and subjects it to sharing between Pertamina and the Contractor at the PSC – specified shareable Profit Oil or Profit Gas rate, prior to making any deductions from production for cost recovery.
<u>Force Majeure</u>	A PSC term referring to delays or default in performance under the PSC caused by circumstances beyond the control and without the fault or negligence of Pertamina and/or Contractor that may affect economically or otherwise continuing operations under the PSC, including but not restricted to Acts of God or the public enemy, perils of navigation, fire, hostilities, war (declared or undeclared), blockade, labor disturbances, strikes, riots, insurrections, civil commotion, quarantine restrictions, epidemics, storm, earthquakes, or accidents.
<u>Indonesian Income Tax Law</u>	The prevailing Republic of Indonesia Tax Code including all the related and applicable regulations.

<u>Investment Credit</u>	A provision of some PSCs (and a term of the Bengara-II PSC but not included in the Yapen PSC) pursuant to which Pertamina allows the Contractor to recover an extra percentage of the costs of certain expenditures for depreciable assets, as an incentive to stimulate production-enhancing expenditures. Separate investment credit percentages may apply for oil and for gas production assets.
<u>MIGAS</u>	An Indonesian oil and gas regulatory authority, the Directorate General of Oil and Gas, a divisional office of the Ministry of Mines and Energy. Production Sharing Contractors to Pertamina are subject to regulations issued by MIGAS, particularly with regard to safety and environmental issues.
<u>MME</u>	The Ministry of Mining and Energy of the Republic of Indonesia. Production Sharing Contractors to Pertamina are subject to regulations issued by MME and decrees issued by the Minister, particularly with regard to governmental policy and PSC contractual issues.
<u>Natural Gas</u>	All associated and/or non-associated gaseous hydrocarbons produced from wells, including wet mineral gas, dry mineral gas, casing head gas and residue gas remaining after the extraction of liquid hydrocarbons from wet gas.
<u>Operating Costs</u>	Expenditures made and obligations incurred in carrying out Petroleum Operations in the Contract Area pursuant to the PSC and as determined in accordance with the Accounting Procedure prescribed by the PSC.
<u>Pertamina</u>	Perusahaan Pertambangan Minyak dan Gas Bumi Negara, the Indonesian national oil company; a signatory to, with administrative authority over, all PSC's.
<u>Petroleum Operations</u>	All exploration, development, extraction, producing, transportation, marketing, abandonment and site restoration operations authorized or contemplated under the PSC
<u>Petroleum</u>	Includes both crude oil and natural gas
<u>Production Bonus</u>	An amount payable in cash by the Contractor to Pertamina pursuant to the PSC upon reaching certain cumulative production levels from within the PSC contract area. Production bonuses are not cost recoverable from production pursuant to the PSC
<u>Production Sharing Contract or "PSC"</u>	The standard type of contract through which the Republic of Indonesia grants oil and gas exploration and production concessions to private industry.
<u>PSC, Standard Terms</u>	The standard type of contract through which the Republic of Indonesia grants oil and gas exploration and production concessions to private industry. The Company's Bengara-II PSC is a standard terms PSC.
<u>PSC, Frontier Terms</u>	The modified form of the standard type of contract through which the Republic of Indonesia grants oil and gas exploration and production concessions to private industry with commercial terms substantially more favorable to the production sharing contractor for certain PSC contract areas in which the RI government wishes to provide added incentive to stimulate exploration. The Company's Yapen PSC is a Frontier Terms PSC.
<u>Production Sharing Contractor</u>	The name given any signatory company or group of companies acting in singular capacity as the Production Sharing Contractor to Pertamina pursuant to a Production Sharing Contract.
<u>Profit Oil or Profit Gas</u>	In accordance with the provisions of the PSC, crude oil or natural gas produced from a PSC contract area which is shared between Pertamina and the Contractor according to a production sharing split set forth in the PSC after first subtracting any applicable Cost Oil or Cost Gas.
<u>Signature Bonus</u>	A provision of any PSC obliging the Contractor to pay to Pertamina in cash or other consideration such as goods or services, at or near the time of signature of the PSC. Signature bonuses are not cost recoverable from production pursuant to the PSC.
<u>Technical Assistance Contract or "TAC"</u>	A modified form of the standard type of contract through which the Republic of Indonesia grants oil and gas exploration and production concessions involving reduced exploration risk properties to private industry. The Company's Bangkudulis Field Development project is held under a TAC.
<u>Uplift</u>	A term used to describe a provision unique to the Company's Bangkudulis TAC under which the TAC Contractor is permitted to recover 130% of its actual drilling costs as an incentive to drill in a prescribed portion of the TAC contract area.
<u>Work Commitment</u>	The obligation of a Contractor to expend at least a minimum monetary amount for work within the Contract Area for each of the first 10 Contract Years of the 30 year PSC term. The Work Commitment is set forth in the PSC and, although the monetary amount may also be accompanied with a description of contemplated work activity for the stipulated amount, the term "Work Commitment" is used to refer to the monetary amount as opposed to the activity. Both under-expenditures and over-expenditures of any single Contract Year's Work Commitment may be carried forward into successive Contract Years provided that on entering each successive Contract Year the Contractor incurs the Work Commitment for that Contract Year. Once a Contractor has fulfilled the Firm Commitment, provided that it has satisfied the Work Commitment for a Contract Year, the Contractor may relinquish the PSC Contract Area at the end of any Contract Year and thereby be released from all subsequent Contract Year Work Commitments.
<u>Work Program</u>	A defined term in the PSC meaning a statement itemizing the Petroleum Operations to be carried out in the Contract Area by the Contractor during any calendar year. The Work Program is conceived and prepared by the Contractor and is submitted annually in advance of each calendar year to Pertamina, for Pertamina's approval together with a Budget. The Work Program and Budget are together referred to as the "Annual Work Program & Budget" or AWP&B. The Work Program which is a discretionary calendar year plan conceived by the PSC Contractor is distinguished from the Work Commitment which is a fixed monetary expenditure stipulated on a Contract Year basis.

<u>Yapen</u>	The name of a petroleum concession in Indonesia in which the Company has a participating interest. Often used to define Block or PSC, as in Yapen Block or Yapen PSC.
<u>Yapen Block</u>	The 9,500 square kilometer area defined as the Contract Area within the Yapen PSC, located entirely offshore in the Indonesian province of Irian Jaya and within which the Contractor has exclusive rights to explore for and produce petroleum.

E X H I B I T – 10B : PSC TERMS
To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART–“B” : PRODUCTION SHARING TERMS AND CONDITIONS

The following discussion of the production sharing contract concession arrangement pursuant to which the Company holds its principal oil and gas properties is provided for background and purposes of additional descriptive reference from the discussion in the main body of the Form-20F annual reports filed by the Company.

PRODUCTION SHARING CONTRACTS

To achieve national development objectives in the oil/gas sector, the government of the Republic of Indonesia actively sought foreign oil companies for their risk capital and technical expertise. As an instrument through which to grant concessions to foreign oil companies, the Production Sharing Contract (“PSC”) was introduced in 1966 replacing an older “Contract of Work” system. Indonesia pioneered the original PSC. Similar PSC-type systems are now employed by many countries worldwide. Since the first introduction of the PSC, Indonesia has evolved a variety of slightly modified forms of the basic PSC theme with the cooperation of foreign and national Contractors. These include Technical Assistance Contracts (“TAC’s”), Joint Operation Body agreements (“JOB Agreements”), Joint Operating Agreements (“JOA’s”), First Tranche Petroleum (“FTP”) agreements and Enhanced Oil Recovery Agreements (“EOR’s”). PSC’s of these separate varieties are often referred to by the name of the Contract Area and the type of PSC rather than by the more generic “PSC” term.

The production sharing-based system used in Indonesia and other countries may be contrasted with royalty-based systems used in the United States and many other developed countries. Under production sharing systems, exploration, development and production costs are recouped from the value of petroleum produced. The balance is “shared” between the private sector operator and the government (which, in the case of Indonesia, includes Pertamina). Under royalty systems, the private sector operator must pay a fixed amount for each barrel (or other measure) of petroleum produced regardless of overall costs and whether they have been recouped. In production sharing systems, the government receives less revenue when costs are high or prices are low, thus protecting operators from some development and market risks. In royalty systems, operators are given more independence because the government has no interest in costs incurred, and operators retain more benefits of cost savings and high market prices.

Pertamina enjoys a government-granted monopoly on the exploitation of all hydrocarbon resources in Indonesia. PSC’s are the mechanism by which it shares its rights with private sector participants. Pertamina effectively has the power to negotiate, issue and sign PSC’s in all their forms, although the signing of PSC’s requires presidential and ministerial approval. Payments of cost recovery and production sharing to PSC Contractors are generally administered and effected by Pertamina.

Production Sharing Contract Term & Administration - The Production Sharing Contract or PSC is the form by which the government of Indonesia grants oil and gas concession rights. All PSC contracts are negotiated, awarded and supervised by Pertamina, the Indonesian state oil company. Pertamina refers to holders of its PSC’s as production sharing “Contractors”. In return for a commitment to conduct an exploration work program of specified, annual minimum expenditure monetary amounts, a “Work Program”, a Contractor signing a PSC will receive exclusive rights to explore for and exploit and produce any petroleum found within his PSC contract area.

Cost Recovery - The signature feature of the PSC contract is the concept and PSC provision for “Cost Recovery”. Cost Recovery is a PSC provision which provides that any Contractor making a commercial discovery of petroleum within his PSC contract area is allowed to fully recover ALL of his prior expenditures within the contract area out of first revenue from oil and gas production prior to sharing any remaining revenue with Pertamina and the State. Other important PSC terms are described below.

Expiry - If a Contractor has not established commercial petroleum production from a PSC contract area within the ten year Exploration Period, the PSC expires and terminates at the end of the tenth year or any extensions thereof. If commercial petroleum production is established within the contract area, the PSC will remain in full force and effect for the full 30-year term. Any remaining or future Work Commitment expenditure obligations are cancelled at the time of a commercial discovery.

PRODUCTION SHARING CONTRACTS - VARIATIONS

The Bengara-II PSC is a “Standard” PSC of the form employed by Pertamina for all oil and gas concessions in Indonesia. The Yapen PSC is a slightly modified form of the “Standard” PSC, referred to as a “Frontier Terms PSC”, and is intended to provide incentives to oil companies to

stimulate exploration in the underdeveloped areas of eastern Indonesia and deep water offshore. With the exception of the differing production sharing percentage splits and the absence of investment credits in "Frontier Terms PSC's" there is otherwise no difference in the two types of PSC contracts. However, the differing production sharing terms are substantial. Under Frontier Terms, Apex Yapen is entitled to receive, after cost recovery, a production share of 62.5% of all oil produced and 71.4% of all gas produced from the Yapen Block as compared to "Standard Terms" under which Apex Bengara shall receive 26.7% of oil produced and 62.5% of all gas produced from the Bengara-II Block.

TAC Variation - The Technical Assistance Contract or "TAC" is a slightly modified form of the standard PSC employed by Pertamina when granting concessions to contract areas that contain some element of proven reserves and expressly DO NOT involve exploration. Commonly TAC contracts are granted to cover rehabilitation of old oil fields, implementation of enhanced oil recovery programs in existing fields and lastly to develop small or marginal oil or gas discoveries which have been made but not yet exploited. The contractor's work commitment is focused on development, appraisal and exploitation of the existing reserves rather than exploration. TAC's normally cover much smaller geographic contract areas than PSC's limited to the agreed extents of the subject field or find and TAC's carry a term of 20-years as opposed to the PSC's 30-year term. Otherwise the TAC's are almost identical to standard PSC's including the production sharing terms. The Company's Bangkudulis Block property is held under a TAC. Actual production sharing terms for Bangkudulis TAC are the same as for the Bengara-II PSC, 26.7% of oil produced and 62.5% of all gas produced. See section below entitled "Material Events Occurring Since Last Fiscal Year End - GAT Bangkudulis Acquisition Signed".

Annual Work Program & Budget - Each Contractor is responsible to Pertamina for the preparation and submission of an AWP&B for each separate Contract Area. Each AWP&B must be approved by Pertamina annually in advance of work program execution. Typically the amount spent by a Contractor in any given year will correspond to the Contractor's work commitment for the PSC contract year. The Contractor is responsible for the execution of each year's work program. The Contractor files monthly, quarterly and annual financial statements and technical work progress reports with Pertamina.

Plan of Development for Discoveries - For each significant discovery within a Contract Area, the Contractor may file a Plan of Development ("POD") with Pertamina. The POD is a comprehensive technical and financial justification and plan for the commercial exploitation of the discovery, and is subject to Pertamina approval. Once Pertamina approves the POD, the Contractor may commence commercial petroleum production and sell that production to Pertamina according to the PSC terms.

Audits - As a general rule, Pertamina does not audit a Contractor's activities until it grants POD approval. The Contractor can expect a Pertamina audit annually thereafter. The primary purpose of Pertamina audits is to ensure that any funds claimed by the Contractor to be "cost recoverable," or subject to cost recovery from Contract Area production, were actually expended by the Contractor in the performance of petroleum operations within the Contract Area and in accordance with the PSC and prevailing Pertamina regulations. The Pertamina auditors certify which expenditures by the Contractor are cost recoverable and which, if any, are not. Once certified, the Contractor may recover costs in accordance with the PSC terms without further comment by Pertamina. In theory, Pertamina may reject cost recovery of expenditures by the Contractor which were not spent (1) pursuant to a Pertamina approved AWP&B, (2) in accordance with Pertamina procurement regulations, or (3) in accordance with PSC terms and conditions. In practice, if the Contractor can document and show good cause for any expenditures in support of petroleum operations in the Contract Area, it is likely that such expenditures will be approved by the auditors and therefore recoverable, even though apparently not meeting all of these three conditions.

Cost Recovery - The key defining feature of a PSC is that all "operating costs" (as defined) incurred by the Contractor within the Contract Area pursuant to a Pertamina-approved Work Program are fully recoverable from petroleum production in the Contract Area. This concept is referred to as "cost recovery." "Operating costs" for the current year, as defined in PSC accounting procedure, consist of all current-year non-capital costs, plus current-year depreciation on capital costs, plus any operating costs not yet recovered from prior years. Capital costs include expenditures for assets which have useful lives in excess of one year, and are depreciated commencing in the year the asset is placed in service at rates which vary depending on the type of asset. Generally, assets are depreciated using a declining balance method, and are fully depreciated within seven years. All costs that do not meet this description, although possibly amortizable under GAAP, are expensed and not depreciated. Subject to limitations imposed by the PSC's First Tranche Petroleum ("FTP") provision, if any, the Contractor is entitled to take and sell for his own account, tax-free, as much oil produced by the Contractor from the Contract Area as is required to recover its current and past period operating costs. Production sold to recover allowable costs is referred to as "Cost Oil." Cost recovery in a particular Contract Area is "ring-fenced," meaning that unrecovered costs from one PSC cannot be recovered from revenue of another PSC. Generally, "ring fencing" arises from the requirement that each PSC must have a separate corporate entity as Contractor.

Production Sharing - Following deduction of Cost Oil, the balance of total production is known as "Profit Oil." Profit Oil is shared between Pertamina and the Contractor on the basis of a fixed percentage set out in the PSC. Two basic Profit Oil splits are commonly used by Pertamina in PSC's, one known as "standard terms" and one known as "frontier terms" which is more favorable to the Contractor and is restricted in use to PSC's covering remote or frontier areas to provide the Contractor additional economic incentive to explore there. In "standard terms" PSC's including Bengara-II PSC, the Profit Oil split, or production share, for crude oil is fixed at 73.2143% for Pertamina and 26.7857% for the Contractor. For natural gas the standard PSC split is 37.5% for Pertamina and 62.5% for the Contractor. In "frontier terms" PSC's such as Yapen PSC the Profit Oil split is 37.5% for Pertamina and 62.5% for the Contractor and gas is shared 71.4286% for the Contractor and 28.5714% for Pertamina.

First Tranche Petroleum - PSC's typically provide that each year, a quantity of production referred to as First Tranche Petroleum ("FTP"), is first shared according to the production sharing split before recovery of costs and deduction of Cost Oil is permitted. FTP is set at 20% in the Bengara-II

PSC and 15% in the Yapen PSC. This limits the amount of cost recovery to the remaining balance of production (usually 80%) in any given year. Recovery of unrecovered operating costs carried over from prior years may also be limited to a percentage of the value of total production within the overall limitation on cost recovery.

Investment Credits. - As an incentive for the Contractor to make capital investments and recover such expenditures through depreciation over time rather than directly in the case of a non-capital investment, an investment credit provision has been introduced into PSC's. The investment credit provides that the Contractor may recover a certain percentage of its capital expenditures from total Contract Area production after FTP but before cost recovery. The investment credit percentage is 15.78% in the Bengara-II PSC for costs associated with crude oil development projects and 102.14% for certain capital investment costs in gas development projects. There is no investment credit provision in frontier terms PSC's such as Yapen PSC.

Domestic Market Obligation. - After five years from the date of first production from any given field within a Contract Area, the Contractor is required to fulfill a Domestic Market Obligation ("DMO"). The DMO provision obliges the Contractor to sell 25% of the Contractor's share of Profit Oil to Pertamina to supply the Indonesian domestic market at a pre-agreed discount price equivalent to 25% of the prevailing market price in the case of the Yapen PSC and 15% of market price in the case of the Bengara-II PSC. The DMO does not apply to natural gas production.

Marketing of Petroleum Produced. - Generally, PSC production of oil and gas may be taken in kind and freely marketed and exported by the Contractor. If Pertamina's share of production is marketed by the Contractor to third parties, normally the crude oil sold is valued at the net realized price (f.o.b. Indonesia) received by the Contractor. If a more favorable price is available to Pertamina, then the Contractor must meet such price if notified by Pertamina, or Pertamina can choose to market its share of such petroleum itself. In any year in which the quantity of petroleum to which Pertamina is entitled is less than 50% of total production, Pertamina has the option to market up to 50% of the total petroleum produced from the Contract Area for the recovery of operating costs, provided that the net realized price for such petroleum is not less favorable than that available to the Contractor.

--- End Exhibit-10B PSC TERMS ---

E X H I B I T – 10C : INDONESIA UPDATE
To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART-“A” : SOCIAL, ECONOMIC AND POLITICAL SITUATION

SOURCES OF INDONESIAN OIL SECTOR, POLITICAL AND ECONOMIC ANALYSIS DATA

The following sections describe the current overview status of the Indonesian petroleum industry. The source of the following information is contained in periodically updated form at three important internet websites. The Company urges readers interested in a comprehensive overview of the current status of the Indonesian energy industry and world oil markets to refer to and utilize these authoritative, independent source of current and periodically updated information on the Indonesian petroleum sector.

1. A **“Country Analysis Brief”** for Indonesia is updated annually, usually in December, and published by the USA Department of Energy's (“DOE”) Energy Information Administration (“EIA”) unit. The brief is freely available in its entirety by download from the DOE-EIA at its internet website located at URL <http://www.eia.doe.gov/emeu/cabs/indonesia.html>.
2. The **“Annual Petroleum Report – Indonesia”** is researched, compiled and published annually, usually in October, by the staff of the US Embassy in Jakarta. This authoritative annual publication is a fine reference for specific and detailed statistical information about the current Indonesian petroleum sector. It also contains knowledgeable information about the current political situation in Indonesia and possible effects upon the Indonesian petroleum industry. The US Embassy also produces other reports valuable for reference on the Indonesian economic and political situation. The entire text of the latest “Annual Petroleum Report – Indonesia” may be downloaded in “PDF” file format from its internet website located at URL <http://www.usembassyjakarta.org>.
3. An **“Annual Statistics Bulletin”** for members is published by the Organization of Petroleum Exporting Countries” and is available for download in “PDF” file format from its internet website <http://www.opec.org/Publications/AB/AB.asp>.

ADDITIONAL INDEPENDENT SOURCES OF INFORMATION

Additional independent sources of information regarding the Indonesia social, political and financial situation as well as economic analysis and political issues can be found at the following recommended websites.

1. General economic and factual information on Indonesia in an overview form is published in the CIA's World Fact Book at their website URL <http://www.odci.gov/cia/publications/factbook/geos/id.html>
2. The US State Department publishes “Country Background Notes” summarizing useful economic, political and commercial background information on most countries, which is usually updated annually. For Indonesia are found at website http://www.state.gov/www/background_notes/indonesia_0010_bgn.html.
3. A comprehensive 1992 US Library of Congress study on Indonesia provides informative detailed historical information on Indonesia but due to its publication date is somewhat out of date on current events. It can be found at Internet URL: <http://lcweb2.loc.gov/frd/cs/idotoc.html>.
4. The link to the website for the Jakarta Post, a leading English language daily newspaper published on-line is particularly recommended for daily news reporting on current Indonesian events. The Jakarta Post can be reached directly at URL <http://www.thejakartapost.com>.
5. See URL <http://www.usembassyjakarta.org/econ/economicstrends.html> for a report on “Economic Trends” updated at least annually, usually in July, by US Embassy staff in Jakarta with economic facts, analysis and opinions on future Indonesian economic developments.
6. See URL <http://www.usembassyjakarta.org/econ/investment.html> for the latest report on the Indonesian “Investment Climate” with useful information updated at least annually, usually in May, by US Embassy staff in Jakarta for those investors considering investments in Indonesia.

E X H I B I T – 10C : INDONESIA UPDATE
To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART–“B” : OVERVIEW OF INDONESIAN OIL AND GAS INDUSTRY

The following sections describes the current overview status of the Indonesian petroleum industry and is derived entirely from the three internet sources described and cited in the preceding section.

Introduction - Indonesia currently holds proven oil reserves of 5 billion barrels. This represents a 14% decline in proven reserves since 1994. Much of Indonesia's proven oil reserve base is located onshore. Central Sumatra is the country's largest oil producing province and the location of the large Duri and Minas oil fields. Other significant oil field development and production is located in accessible areas such as offshore northwestern Java, East Kalimantan, and the Natuna Sea. Indonesian crude oil varies widely in quality, with most streams having gravities in the 22° to 37° API range. Indonesia's two main export crudes are Sumatra Light Crude (“SLC”), often referred to as “Minas” crude after the name of the field of origin, having a 35° API gravity, and the heavier, 22° API gravity “Duri” field crude.

Petroleum Potential - Indonesia's aggregate explored and unexplored oil and gas resources were estimated as of January 1997 at 73.30 billion barrels of oil and 307.95 trillion cubic feet of natural gas. The country has 60 tertiary sedimentary basins and sub-basins with potential for trapping and accumulation of significant hydrocarbon resources, covering an area of more than 2 million square kilometers. Of the 60 basins, 38 have been thoroughly explored, of which 24 yielded significant reserves and 14 are currently producing.

Exploration - The majority of Indonesia's producing oil fields are located in the central and western sections of the country. Therefore, the focus of new exploration has been on frontier regions, particularly in eastern Indonesia. Sizeable, but as of yet unproven, reserves may lie in the numerous, geologically complex, pre-tertiary basins located in eastern Indonesia. These regions are much more remote and the terrain more difficult to explore than areas of western and central Indonesia.

Production Enhancement - Companies producing from existing fields are investing in programs to increase recovery rates and to prolong the life of the fields. Caltex, a Chevron/Texaco joint venture, which has the largest operation of any multinational oil company in Indonesia, is undertaking a steam injection project at the Duri field on Sumatra.

Oil Production - In 2000, Indonesia ranked seventeenth among world oil producers, with approximately 1.9 percent of the world's production. At 1.41 million barrels of oil per day (MMB/D) in 2000, Indonesia's production of crude oil and condensate continued a gradual decline from 1.55 MMB/D in 1999 and 1.56 MMB/D in 1998. Caltex is the largest single independent Indonesian oil producer with some 758,000 BOPD or 48% of the nation's total. Other large producers are Maxus with 9%, Conoco, Arco and Unocal with 6% each, Mobil with 5%, Vico and Total with 4%, and Pertamina with 3%.

World Rank - Indonesia is the second largest oil producer, after China, in the Asia Pacific Region with over 27% of the region's oil production. Indonesia ranks 17th, behind Canada, in the list of the world's largest oil producers, accounting for about 2.4% of the world's annual production. Indonesia is the third oldest oil producer in the world after the United States and Russia, with production having commenced more than a century ago.

OPEC Position - Of the 11 members of the Organization of Petroleum Exporting Countries (“OPEC”), Indonesia ranks ninth among OPEC producers with production about 5% of the OPEC total. Statistics available for last quarter of 2001 indicate that Indonesia's OPEC quota for crude oil production is 1,125,000 barrels per day. Its actual crude oil production for the same period was 1,170,000 barrels per day (not counting lease NGL's) and its total forecast production capacity for the same period is 1,250,000 barrels per day. The Organization of Petroleum Exporting Countries (OPEC) was founded in Baghdad, Iraq, in September 1960, to unify and coordinate members' petroleum policies. OPEC members' national oil ministers meet regularly to discuss prices and, since 1982, to set crude oil production quotas. Original OPEC members include Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela. Between 1960 and 1975, the organization expanded to include Qatar (1961), Indonesia (1962), Libya (1962), the United Arab Emirates (1967), Algeria (1969), and Nigeria (1971). Ecuador and Gabon were members of OPEC, but Ecuador withdrew in December 1992, and Gabon followed suit in January 1995. US-DOE Energy Information Agency estimates the current eleven OPEC members account for roughly 40% of world oil production and about 77% of the world's proven oil reserves.

Crude Oil Reserves - Indonesia currently has proven oil reserves of slightly under 5 billion barrels, approximately 14% less than in 1994. Indonesia's proven and probable reserves were reported at year-end 2000 at 9.8 billion barrels of oil.

Petroleum Exports - The dollar value of Indonesia's oil and gas exports rose to \$14.4 billion in 2000, compared with \$9.8 billion in 1999. In the first six months of 2001, oil and gas exports reached \$7.0 billion. Oil and gas imports also rose in 2000 to \$6.0 billion, compared with \$3.7 billion a year earlier. In the first six months of 2001, crude oil and oil product imports totaled \$3.1 billion. The leading destinations for export of Indonesian crude oil and condensate in 1996 were Japan (37%), China (18%), South Korea (12%), Singapore (8%) and the United States (7%). Major purchasers of LNG are Japan (63%), South Korea (26%) and Taiwan (11%).

Petroleum Imports - Indonesia imports crude oil from other countries as additional feedstock for its refineries. Domestic crude oil tends to be heavy and waxy and not suitable as the sole feedstock for production of middle distillates such as gasoline, kerosene and diesel, which are in high demand in the domestic market. Indonesian crude oil is often blended with lighter crudes, mostly from the Middle East. In 2000, Indonesian crude oil imports rose to 90 million barrels from 71.8 million barrels in 1996, largely from Saudi Arabia, Nigeria and Malaysia, with imports valued at \$2.3 billion for the year. Fuel product imports rose significantly to 90 million barrels valued at \$3.0 billion for the year due to increased domestic demand and limited refinery capacity. Domestic fuel consumption increased to 54.8 million kiloliters in 2000, about 4.5 million kiloliters higher than pre-crisis levels.

Refineries - Indonesia has eight refineries, all owned by Pertamina, with a combined design capacity of 1.02 Million BOPD. The largest refineries are in Cilacap, Central Java; Balikpapan, East Kalimantan; and Balongan, West Java, with capacities of 285,000 BOPD, 241,000 BOPD and 125,000 BOPD respectively. In 2000 Indonesian refineries processed crude oil equal to approximately 96% of total design capacity.

Natural Gas - Indonesia's Natural Gas Reserves are estimated at 170.3 trillion standard cubic feet (TSCF) - 94.7 TSCF proven and 75.6 TSCF potential. Gross annual Natural Gas production is 2.9 trillion cubic feet (TCF). Export revenue from LNG and LPG totaled \$7.2 billion in calendar year 2000. Indonesia has natural gas reserves of over 170.3 trillion standard cubic feet (TSCF) Under production sharing contracts (PSCs) with Pertamina, Indonesia's largest Natural Gas producers are Exxon/Mobil, Total, Vico, Arco and Unocal.

LNG & LPG - The largest percentage of Indonesia's natural gas production is processed into LNG and LPG, followed by use by the electricity and petrochemical sectors. LNG production at Arun and Badak (Bontang) for 2000, at 27.3 million metric tons (MMT), fell slightly from 1999 production of 29.8 MMT, reversing the upward trend of the preceding years. Japan, South Korea and Taiwan were the key markets for LNG. LPG production fell further to 2.1 MMT from 2.3 MMT in 1999, with exports also falling from a 1993 peak to 1.31 MMT to the four top customers -- Japan, Hong Kong, Taiwan and Australia. The government continued actively to promote BP's giant Tangguh gas field, under development in Papua, with a particular focus on potential markets in China and India and awarded a contract for development of Bontang's Train I expansion.

New Investment - Expenditures by petroleum companies declined 10 percent in 2000 to US \$3.6 million from \$4.0 million in 1999. The upward trend in the number of new exploration and appraisal wells peaked in 1998 at 145, then fell to 89 in 1999 and 82 in 2000, significantly below the 100 wells drilled in both 1996 and 1997. Unocal continued its ambitious drilling program in the deepwater, offshore East Kalimantan blocks. The number of new production sharing contracts (PSC's) signed improved to 5 in 2000, from 4 in 1999 and the record 29 signed in 1997. Ten contracts were totally relinquished. Three new PSC's were signed in May 2001.

Fiscal Policy - In line with its International Monetary Fund (IMF) commitments, the government is still pressing forward with plans to hike domestic fuel prices and secure passage of a new draft oil and gas law in the face of earlier failures on both fronts. Fuel prices were raised across the board first in October 2000, then again in June 2001. The Indonesian Parliament (DPR) agreed to a further 30-percent increase planned for January, which, if implemented in a timely fashion, would reduce the GOI's fuel subsidy burden from Rp 53.7 trillion in 2001 (21 percent of central government spending) to Rp 32.3 trillion (13 percent). As a step forward toward reforming Indonesia's petroleum sector, the DPR passed the oil and gas bill into law in a plenary session on October 23, replacing Oil and Gas Law No. 44/1960 and Law for Pertamina No. 8/1971. The new law changes the face of Indonesia's petroleum sector by radically redefining the role of state-owned Pertamina and deregulating the downstream market. The Government was also implementing Law 22/1999 on Regional Autonomy and Law 25/1999 on Fiscal Decentralization, which entered into force in January 2001. While current petroleum contracts would be grandfathered, uncertainty over details of the implementing regulations and specific policies continued to create uncertainty and diminished the level of new investments in the petroleum sector."

Legal Framework Of The Indonesian Oil And Gas Sector - The basic philosophy of the Indonesian government regarding natural resources is set out in Article 33 of the 1945 Constitution, which places natural resources in the jurisdiction of the State, for the benefit of the people. The government controls the exploitation of all natural wealth including petroleum resources within the country. The development and exploitation of petroleum resources within Indonesia is under the supervisory control of Pertamina. Pertamina was incorporated in 1968 for the purpose of taking responsibility for all petroleum activities in the country including exploration, production, refining, transportation and marketing. Its exclusive rights are set out in Law No. 8 of 1971. Pertamina operates under the broad authority of the Directorate General for Oil and Gas ("MIGAS") of the Ministry of Mining and Energy of the Indonesian Government ("MME"). The Indonesian Government is currently in the process of amending Law No. 8 of 1971. It is very likely that under the revised law, responsibility for granting and supervising oil and gas exploration and production concessions through Production Sharing Contracts ("PSC's"), and the right to share in production revenues under PSC's with oil company operators, will be transferred from Pertamina to the MME. Following a transition period, Pertamina's involvement in oil/gas exploration and production would be as a PSC Contractor to MME in its own right. Generally, foreign investment in Indonesia is regulated by Law No. 1 of 1967 and subsequent amending legislation, which is administered by the Capital Investment Coordinating Board ("BKPM"). One of the most significant exceptions to this rule is oil and gas exploration and exploitation, for which foreign companies are able to negotiate and operate directly with Pertamina without BKPM approval or oversight.

Production Sharing Contracts - See Exhibit-10B: PSC-Terms for a comprehensive overview of the production sharing contract oil and gas concession system as used in Indonesia.

E X H I B I T – 10C : INDONESIA UPDATE
To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART-“C” : Indonesian Crude Oil Prices

Crude oil prices in Indonesia are posted monthly by Pertamina and represent the prices that Pertamina will pay Contractors for production for that month. The benchmark price posted by Pertamina is referred to as the Indonesian Crude Price (“ICP”), and represents the price of a certain crude oil produced from the Minas Field in South Sumatra, known in international markets as Sumatran Light Crude (“SLC”). All other crude oils produced in Indonesia are priced by Pertamina by adding to or subtracting from the ICP, based on the quality and value of each crude oil compared to SLC. Pertamina posts prices for over 40 varieties of Indonesian crude oil each month. The posted ICP may differ from the prevailing SLC market price on any given day. Historically the ICP tends to fall midway between the two world spot prices for West Texas Intermediate crude and North Sea Brent crude. The Company has the full right to export all production it may realize from the Bangkudulis, Bengara-II or Yapen PSC Contract Areas and sell it on the world market at prevailing world market prices or the most favorable long term contract it can negotiate. Alternatively the Company could sell its production to Pertamina and be assured of getting the posted ICP price suitably adjusted for quality. Bangkudulis crude is expected to sell at a price equal to SLC without any adjustment. Pertamina’s crude oil marketing arm “Perta” posts a variety of prices for several Indonesian crudes including SLC or ICP located at its internet website at URL: <http://www.perta.com/icp.html>. The following table shows the actual posted price for SLC/ICP for each month during the calendar year 2001.

Indonesian Crude Oil Pricing				
Posted For Production During Month Shown				
Posted For Month Of	US\$/Barrel ICP or SLC Crude Price			
	<u>Year</u> <u>2002</u>	<u>Year</u> <u>2001</u>	<u>Year</u> <u>2000</u>	<u>Year</u> <u>1999</u>
January	18.63	24.12	24.57	10.99
February	18.84	25.49	26.24	10.56
March	22.62	25.39	27.19	12.03
April	25.35	27.35	24.12	15.20
May	25.37	28.18	27.96	16.29
June	24.28	27.83	30.68	16.27
July	24.97	25.04	30.16	18.11
August	25.64	24.77	29.95	19.55
September	27.13	24.68	32.95	21.80
October	28.83	19.35	32.47	22.38
November	27.76	18.06	31.06	23.74
December	31.84	17.60	25.03	24.22

--- End Exhibit-10C INDONESIA UPDATE ---

E X H I B I T – 11 : ATTACHMENT TO ITEM-#17 FORM-20F
FINANCIAL STATEMENTS AT END FISCAL 2000 7/31/02
To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART-“A” : AUDITOR'S REPORTS

CONTINENTAL ENERGY CORPORATION
(A DEVELOPMENT STAGE COMPANY)
CONSOLIDATED FINANCIAL STATEMENTS
31 JULY 2002
U.S. Funds

STALEY, OKADA & PARTNERS

Chartered Accountants

INDEPENDENT AUDITORS' REPORT

To the Directors of Continental Energy Corporation:

We have audited the consolidated balance sheet of Continental Energy Corporation (a development stage company) as at 31 July 2002 and 2001 and the consolidated statements loss and cash flows for the years ended 31 July 2002, 2001 and 2000 and cumulative from 1 August 1993 through 31 July 2002 and the consolidated statement of shareholders' equity (deficiency) from 31 July 1999 through 31 July 2002. These consolidated financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on these consolidated financial statements based on our audit.

We conducted our audit in accordance with United States generally accepted auditing standards for the years ended 31 July 2002 and 31 July 2001. All prior years were audited in accordance with Canadian generally accepted auditing standards. Those standards require that we plan and perform an audit to obtain reasonable assurance whether the financial statements are free of material misstatement. An audit includes examining, on a test basis, evidence supporting the amounts and disclosures in the financial statements. An audit also includes assessing the accounting principles used and significant estimates made by management, as well as evaluating the overall financial statement presentation.

In our opinion, these consolidated financial statements present fairly, in all material respects, the financial position of the Company as at 31 July 2002 and 2001 and the results of its operations and the changes in its cash flows for the years ended 31 July 2002, 2001 and 2000 and cumulative from 1 August 1993 through 31 July 2002 in accordance with Canadian generally accepted accounting principles. As required by the Company Act of British Columbia, we report that, in our opinion, these principles have been applied on a basis consistent with that of the preceding year.

"Staley, Okada & Partners"

Surrey, B.C., Canada
6 December 2002

STALEY, OKADA & PARTNERS
CHARTERED ACCOUNTANTS

COMMENTS BY AUDITORS FOR U.S. READERS ON CANADA-U.S. REPORTING CONFLICT

To the Directors of Continental Energy Corporation:

In the United States, reporting standards for auditors require the expression of a qualified opinion when financial statements are affected by significant uncertainties such as those referred to in Note 1 to these financial statements. The above opinion on our report to the shareholders dated 6 December 2002 for the years ended 31 July 2002, 2001 and 2000 is not qualified with respect to, and provides no reference to these uncertainties since such an opinion would not be in accordance with Canadian reporting standards for auditors when the uncertainties are adequately disclosed in the financial statements.

Surrey, B.C., Canada
6 December 2002

"Staley, Okada & Partners"

STALEY, OKADA & PARTNERS
CHARTERED ACCOUNTANTS

MANAGEMENT'S RESPONSIBILITY FOR THE FINANCIAL STATEMENTS

The preparation and presentation of the Company's financial statements is the responsibility of management. The financial statements have been prepared in accordance with Canadian generally accepted accounting principles and necessarily include estimates which are based on management's best judgments. Information contained elsewhere in the Annual Report is consistent, where applicable, with that contained in the financial statements.

Management is also responsible for installing and maintaining a system of internal controls to provide reasonable assurance that assets are safeguarded and that reliable financial information is produced for preparation of financial statements.

Staley, Okada & Partners, a firm of chartered accountants, were appointed by the shareholders as external auditors to conduct an independent examination and express their opinion on the financial statements. The Auditors' Report outlines the auditors' opinion and the scope of their examination.

The Board of Directors is responsible for ensuring that management fulfills its responsibilities for financial reporting and internal control. The Board exercises these responsibilities with the assistance of the Audit Committee of the Board.

E X H I B I T – 11 : ATTACHMENT TO ITEM-#17 FORM-20F
FINANCIAL STATEMENTS AT END FISCAL 2000 7/31/02
To The Us-Sec Form-20f For Fiscal 2002 Ended 7/31/02 Filed In January 2003 By
CONTINENTAL ENERGY CORPORATION

PART-“B” : FINANCIAL STATEMENTS

Statement 1

Continental Energy Corporation

(A Development Stage Company)

Consolidated Balance Sheet

As at 31 July

U.S. Funds

ASSETS	2002	2001
		<i>(Note 14)</i>
Current		
Cash	\$ 225,414	\$ 27,399
Accounts and GST receivable	42,199	24,894
	<u>267,613</u>	<u>52,293</u>
Prepaid Resource Property Costs <i>(Note 4)</i>	<u>29,000</u>	-
Prepaid Share Issuance Costs <i>(Note 4)</i>	<u>43,500</u>	-
Capital Assets		
Investment in and advances to GAT Bangkudulis Company Ltd. <i>(Note 5b)</i>	-	1,589,472
Resource property costs - <i>Schedule 1 (Notes 1 and 5)</i>	945,717	1,397,772
Furniture, equipment and computer <i>(Note 6)</i>	13,252	20,623
	<u>958,969</u>	<u>3,007,867</u>
	<u>\$ 1,299,082</u>	<u>\$ 3,060,160</u>
LIABILITIES		
Current		
Accounts payable and accrued liabilities <i>(Note 9b)</i>	\$ 6,713,877	\$ 1,256,883
Cash received in advance of share issuance <i>(Note 8e)</i>	150,000	-
Promissory notes payable <i>(Note 7)</i>	50,000	50,000
	<u>6,913,877</u>	<u>1,306,883</u>
Continued Operations <i>(Note 1)</i>		
Contingencies and Commitments <i>(Note 12)</i>		
SHAREHOLDERS' EQUITY (DEFICIENCY)		
Share Capital - <i>Statement 2 (Note 8)</i>	18,972,265	18,370,981
Contributed Surplus - <i>Statement 2</i>	116,550	116,550
Deficit - <i>Statement 2</i>	<u>(24,703,610)</u>	<u>(16,734,254)</u>
	<u>(5,614,795)</u>	<u>1,753,277</u>
	<u>\$ 1,299,082</u>	<u>\$ 3,060,160</u>

ON BEHALF OF THE BOARD:

_____"Gary R. Schell"_____, Director

_____"Richard L. McAdoo"_____, Director

- See Accompanying Notes -

Site Visit Instructions

Check-in will begin at 10:00 AM (EST) on 01 April 2026 in the lobby of NUWCDIVNPT Building 80, Newport, RI.

The following rules apply for the site visit:

- This will be an Unclassified site visit.
- The site visit will be scripted, no questions will be allowed during the tour.
- Follow-up questions can be submitted via email to amy.k.prisco.civ@us.navy.mil and Michael.N.Ouellette.civ@us.navy.mil
- The use of any type of camera during the site visit is strictly prohibited.
- All interested attendees must pre-register by email to amy.k.prisco.civ@us.navy.mil and Michael.N.Ouellette.civ@us.navy.mil
- In Addition to pre-registration, all participants MUST have an approved Visit Request.
- Instructions for submitting a visit request can be found at the following link:
<http://www.navsea.navy.mil/Home/Warfare-Centers/NUWC-Newport/Contact-Us/Mandatory-Visit-Request/>
- Access to NUWC Division Newport WILL NOT be granted under any circumstances without a valid visit authorization request on file. Please list Michael Ouellette, Code 0223, as the Government POC when submitting the visit request.
- Visitor Control - NUWC Newport Building 80 Lobby outside of NUWC Gate 23 Is open Monday- Friday from 7 a.m. to 3:30 p.m. (except holidays)

NUWC NEWPORT DIVISION VISITOR REQUEST INFORMATION

- A site visit request is **MANDATORY** for all visitors to NUWC Division Newport; government and civilian, regardless of the duration or classification of the visit. Access to NUWC Division Newport will not be granted under any circumstances without a valid visit authorization request (VAR) on file.
- If you can submit your VAR via JPAS, you do not need to send a VAR on company letterhead.
- Due to the number of visit requests received, please allow 7 business days for processing time, whether received by JPAS or on Company Letterhead via FAX or EMAIL.
- Visitors who arrive prior to the 7 working day processing window may not be guaranteed access.
- Please note, all individuals not in possession of a common access card (CAC) or Teslin Card (Retired Military/Dependent ID), will require a DBIDS card for access.
- No unescorted access will be granted without a valid DBIDS card or ID mentioned above.

determines that the Company has cash available (or may draw on unfunded loan commitments to the extent permitted by the Financing Documents) after payment of the Special Distribution, Priority Distributions, required operating expenses and liabilities of the Company, Reimbursable Member Expenses and Reimbursable Project Management Expenses, the Company shall distribute such available cash, or draw on any unfunded loan commitments to the extent permitted by the Financing Documents, and distribute all or any portion of that excess cash and drawdown funds to the Members Pro Rata, subject to the provisions of the Financing Documents.

2.3. **Appendix A (Glossary of Defined Terms)**. Appendix A to the LLC Agreement is hereby amended by (a) amending the definition of “Electric Interconnection Agreements” to be and read as follows:

“**Electric Interconnection Agreements**” means (a) the Amended and Restated Interconnection and Operating Agreement, dated January 4, 2001, between the Company and Entergy Gulf States and (b) the Interconnection and Operating Agreement, made effective July 1, 2001, between Cleco Power and the Company.

and (b) adding, in proper alphabetical order, new definitions of “Cleco Power” and “Entergy Gulf States”, which shall be and read as follows:

“**Cleco Power**” means Cleco Power LLC.

“**Entergy Gulf States**” means Entergy Gulf States, Inc., a Texas corporation.

3. **Effectiveness**. The LLC Agreement, as amended hereby, shall continue to be, and shall remain, in full force and effect.

4. **Headings**. The headings of the Sections of this Amendment are included for convenience only and shall not be deemed to constitute a part of this Amendment.

5. **Counterparts**. This Amendment may be executed in any number of separate counterparts, each of which when so executed and delivered shall be deemed an original, but all such counterparts shall together constitute one and the same agreement. All signatures need not be on the same counterpart.

6. **Governing Law**. This Amendment shall be governed by, and construed in accordance with, the laws of the State of Delaware, without reference to the conflict of laws rules thereof that would direct the application of the laws of another jurisdiction.

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EXECUTIVE SUMMARY

1. Barbados has articulated a strategic plan to strengthen the reliability, efficiency, and resilience of its water and sanitation services. To support this effort, the World Bank is preparing a Program-for-Results (PforR) operation namely the **Barbados Water Security and Sector Performance Program (P508674) (the Program)** to be implemented by the **Barbados Water Authority (BWA)**. The Program aims to strengthen the operational capacity of BWA to improve service delivery, water resources management, and pollution control through institutional strengthening, digital modernization, and targeted operational improvements and this **Environmental and Social Systems Assessment (ESSA)** evaluates Barbados' legal and institutional capacity to manage the associated environmental and social risks.

2. As these challenges are not unique to Barbados, select additional countries in the Caribbean Region, and the World Bank have agreed to develop a regional approach to leverage synergies for overcoming these common barriers. In this context, a Caribbean Regional Water Security Multi-Phased Approach (MPA) Program is proposed across several Caribbean States connected through an overarching regional platform and water security framework. The Barbados Program is the first activity of that MPA to move forward. While this ESSA is stand-alone for the Barbados Program, the implementation of the program, including ESSA recommendations will benefit from Barbados's participation in the MPA.

Program Description

3. The Program supports the implementation of BWA's Strategic Plan 2025–2029, structured around three results areas:

- a. **Results Area 1 – Building BWA capacity for efficient service delivery:** modernization of BWA's operations, reduction of non-revenue water (NRW), improved customer service, and enhanced financial management.
- b. **Results Area 2 – Pollution management through country-wide safely managed sanitation services:** expanding safely managed sanitation through regulatory updates, treatment improvements, and low-cost sewer pilots.
- c. **Results Area 3 – Foundations for long term water security:** strengthening institutional coordination, groundwater monitoring, water resource management, and disaster response.

4. The total national program cost is approximately US\$80 million, with US\$50 million to be financed by the World Bank through the PforR.

Environmental and Social Benefits

5. The Program is expected to result in positive outcomes for Barbados both for its the citizens and overall environment. Key environmental and social benefits include:

- a. *Improved Services:* more reliable water and sanitation services for the population as a whole;
- b. *Enhanced Institution:* strengthened institutional and staff capacity;
- c. *Stronger Environmental Management:* reduced groundwater abstraction that will improve water resources management and better wastewater management to reduce pollution and improve public and environmental health
- d. *Improved Adaptation:* better water conservation and lower energy use resulting in enhanced system resilience and efficiency.

Key Risks and Mitigation

6. All works are expected to be small-scale and limited to existing rights-of-way; therefore, activities requiring resettlement, affecting sensitive habitats, or using hazardous substances are excluded from financing under the Program. Identified risks are moderate and manageable through national systems and BWA procedures. Main risks include:



- a. *Community impacts*: temporary access disruptions, traffic interference, and nuisance during civil works;
- b. *Construction-related impacts*: dust, noise, minor soil disturbance, and waste generation;
- c. *Occupational risks*: worker safety during excavation and exposure to traffic and heat; and
- d. *Social risks*: short-term business disruption, exclusion from digital services, insufficient and ineffective communication and stakeholder engagement, potential for sexual exposure and abuse/sexual harassment (SEA/SH) incidents, and potential dissatisfaction among BWA staff due to organizational reforms.

National Systems Assessment

- 7. Barbados has a comprehensive legal framework for environmental protection, occupational health and safety, land use, public health, and labor rights. Key national agencies include the Planning and Development Department (PDD), Environmental Protection Department (EPD), and Coastal Zone Management Unit (CZMU). Additionally, BWA's Water Resources and Environmental Management Unit (WREMU) oversees environmental management as well as providing support to occupational safety enforcement and customer service.
- 8. The assessment finds strong alignment between national systems and World Bank PforR Core Principles, particularly regarding environmental protection, worker safety, and grievance redress.
- 9. However, the assessment indicates that while there is alignment, BWA's institutional capacity and operational procedures may be insufficient to ensure consistent application of national environmental and social regulations because staffing, standardized Environmental Codes of Practice (ECOPs), standard operating procedures (SOPs), and monitoring systems seem limited.
- 10. Practical implementation requires reinforcement on:
 - a. Documented SOPs for environmental and social management and contractor oversight;
 - b. Construction-phase customer and public engagement including robust and monitorable grievance and incident reporting mechanisms; and
 - c. Standardized procedures for inclusive service access and stakeholder communication (e.g., documented offline/assisted options for users who can't use digital tools, plus clear, timely notifications and feedback channels during works).

Conclusions and Recommendations

- 11. The ESSA concludes that Barbados maintains a strong environmental and social regulatory framework, supported, among others, by the Planning and Development Act (2019, as amended 2020), the Environmental Impact Assessment (EIA) Regulations (2021), and sectoral instruments covering occupational safety, pollution control, and coastal management. These instruments collectively provide a sound legal foundation for managing environmental and social risks, define clear permitting procedures, and empower technical agencies—such as the Environmental Protection Department, Coastal Zone Management Unit, and Planning and Development Board—to impose mitigation and monitoring conditions as part of development approvals.
- 12. However, the assessment finds that institutional capacity and operational procedures within the BWA require strengthening to ensure consistent application of these safeguards at the program level. Legal provision implementation could be constrained by limited environmental and social staffing, the absence of standardized ECOPs, SOPs, and ad hoc monitoring and reporting systems.



13. Preliminary findings suggest that many of the required SOPs, templates, and field-level protocols are missing or not yet standardized. Confirmation of these elements will depend on additional inputs and documentation from BWA. Once this information is received, the ESSA will be updated to refine the analysis of institutional readiness.

14. Key recommendations to BWA include:

- a. Strengthening BWA's environmental and social management procedures and staff capacity by clarifying institutional roles, allocating staff and resources for environmental and social oversight, and establishing internal procedures for supervision, monitoring, and reporting of environmental and social performance;
- b. Formalizing screening and exclusion processes for ineligible activities;
- c. Strengthening the current grievance mechanism to include construction-phased options for interaction with the public;
- d. Integrating SEA/SH prevention and response protocols throughout operations;
- e. Enhancing contractor OSH and environmental supervision; and
- f. Ensuring accessibility of digital services and communication, particularly for vulnerable customers.

15. To assist in operationalizing these broader recommendations, the following four environmental and social items are included in the Program Action Plan (PAP):

- a. Boosting BWA's environmental and social capacity by December 31, 2026;
- b. Developing standard environmental and social protocols and SOPs prior to first tender or within six months of effectiveness, whichever comes first;
- c. Launching a stakeholder engagement plan for wastewater reuse by December 31, 2026; and
- d. Standardizing community engagement procedures to include grievance monitoring and redress for construction activities by December 31, 2026.



1. INTRODUCTION

16. Barbados has articulated a strategic plan to strengthen the reliability, efficiency, and resilience of its water and sanitation services. To support this effort, the World Bank is preparing a Program-for-Results (PforR) operation, the **Barbados Water Security and Sector Performance Program (P508674)** (the Program), to be implemented by the **Barbados Water Authority (BWA)**.

17. The Program will advance key priorities, including non-revenue water reduction, metering improvements, small-scale sewer pilots, and institutional strengthening such as digital customer service modernization.

18. As part of the due diligence required for PforR preparation, the World Bank conducts a suite of assessments covering technical, economic, fiduciary, climate/disaster resilience, and environmental and social aspects to confirm program feasibility, strengthen risk management, and align systems with World Bank Group Core Principles. Within this framework, this Environmental and Social Systems Assessment (ESSA) examines the national legal and institutional arrangements that govern the management of environmental and social effects, identifies strengths and gaps in existing systems, and recommends actions to enhance their capacity to address risks proportionate to the Program. The ESSA serves both as a diagnostic and as a practical tool to inform the Program design, including the Program Action Plan (PAP) and Disbursement Linked Indicators (DLIs).

19. As these challenges are not unique to Barbados, select additional countries in the Caribbean Region, and the World Bank have agreed to develop a regional approach to leverage synergies for overcoming these common barriers. Addressing the Region's complex water challenges requires highly specialized, continuous technical engagement, which most countries are not currently equipped to provide. Instead, they often rely on short-term external consultants or regional programs with limited scope, which leads to high costs, fragmented implementation, and limited continuity in reforms. This skills gap constrains countries' ability to develop strategic investments with an adequate level of readiness; to implement, operate and maintain critical infrastructure; and respond to climate-related pressures in a timely and sustainable manner. While several regional institutions provide support in water-related areas, regional gains of scale have been limited. Addressing this systemic constraint calls for a coordinated, wider-Caribbean approach to technical assistance, capacity building, and peer learning—shaped in the form of a regional water center of excellence. In this context, a Caribbean Regional Water Security Multi-Phased Approach (MPA) Program is proposed across several Caribbean States connected through an overarching regional platform and water security framework. The Barbados Program is the first activity of that MPA to move forward¹. While this ESSA is stand-alone for the Barbados Program, the implementation of the program, including ESSA recommendations will benefit from Barbados's participation in the MPA.

A. Government Program

20. Barbados is among the most water-scarce countries in the world, with renewable freshwater resources estimated at only 304 m³ per person per year, well below the international threshold for absolute scarcity. Nearly 90 percent of the national water supply depends on groundwater abstraction from fragile limestone aquifers, which are increasingly exposed to overextraction, saline intrusion, and contamination.

21. To address this structural vulnerability, the Government has developed the Barbados Water Authority Strategic Plan (SP) 2025–2029, which defines a comprehensive framework for building a resilient, efficient, and climate-smart water and sanitation sector.

¹ The first mover countries under this MPA are Barbados, St. Lucia, and Jamaica, with potential interest to be confirmed by Belize and Grenada; working together for overarching regional support with a regional Caribbean agency. Please see the World Bank Project PAD for the Program (P508674) for more details on the MPA. [\[INCLUDE LINK TO FINAL PAD IN PUBLISHED ESSA\]](#)



22. The Strategic Plan's overarching objective is to ensure equitable access to safe water and sanitation while preserving the island's scarce water resources. Its implementation is led by the BWA under the oversight of the Office of the Prime Minister.

23. The Strategic Plan combines investment in infrastructure with institutional modernization and policy reform. Its main lines of action include:

- a. Reducing non-revenue water (NRW), which is currently around 55 percent, through improved asset management, leak detection, and digital monitoring systems;
- b. Rehabilitating water supply infrastructure to enhance reliability and service quality;
- c. Expanding sewerage coverage, which currently reaches only 5 percent of the population, using low-cost and decentralized sanitation technologies;
- d. Improving septage collection, treatment, and reuse to protect groundwater resources; and
- e. Upgrading of BWA's two wastewater treatment plants to tertiary level and reuse capacity.

24. However, BWA faces persistent financial and investment constraints that limit its ability to deliver these reforms. High operating expenses, which are mainly driven by personnel and energy costs for pumping and desalination, have eroded the utility's financial space. Between 2018 and 2024, BWA's annual revenues averaged US\$86 million, enough to cover direct operating costs but insufficient to fund capital works. Average annual investment of US\$15 million between 2021 and 2023 is vastly below the US\$1 billion in sector investment needs estimated for the coming decades. At this rate, it would take more than 70 years to achieve current expansion objectives.

25. The Barbados Water Security and Sector Performance Program, with a total cost of about US\$80 million, of which US\$50 million is financed by International Bank for Reconstruction and Development (IBRD), is intended to help mitigate sector financing constraints and advance implementation of BWA's Strategic Plan (2025–2029) by channeling resources to infrastructure rehabilitation, digital modernization, and institutional strengthening.

B. ESSA Objective

26. The ESSA aims to evaluate the adequacy of Barbados' existing legal, regulatory, and institutional frameworks to manage the environmental and social risks and impacts associated with the Program. The ESSA identifies areas where national systems and BWA's practices are well aligned with the World Bank's PforR Core Principles, highlights potential gaps or capacity constraints, and recommends measures to strengthen risk management, enhance stakeholder engagement, and ensure that Program activities promote environmental sustainability, social inclusion, and protection of public and worker health and safety.

C. ESSA Methodology

27. This ESSA has been prepared in line with the World Bank's Program-for-Results Financing Environmental and Social Systems Assessment Guidance (2020). Successive drafts of ESSA were reviewed by BWA and stakeholders with consultations being held on December 11, 2025, and January 20, 2026. Comments and recommendations from those consultations are herein reflected. See Annex 1 for a summary of those consultations.

28. The methodology involved three main steps: (i) screening of potential environmental and social risks and impacts of Program-supported activities; (ii) assessment of the adequacy of Barbados' national systems and BWA's institutional arrangements to manage these risks in line with the Bank's Core Principles; and (iii) identification of opportunities for strengthening systems through targeted recommendations.

29. The assessment focused on identifying the environmental and social effects most likely to arise from Program activities. The identified effects were then organized into four categories: potential social benefits, potential social risks and impacts, potential environmental benefits, and potential environmental risks and impacts.



30. The assessment further examined the national policy, legal, and regulatory framework governing water and sanitation projects in Barbados, with attention to how these instruments apply to the types of works and impacts expected under the Program. Institutional roles and responsibilities of BWA and relevant oversight agencies were analyzed, alongside the established procedures for environmental screening, permitting, and compliance monitoring.

31. The review also considered BWA's track record in managing environmental and social aspects of both capital and maintenance works. Multiple sources of information were reviewed to inform the assessment, including:

- a. Policies, technical guidelines, and procedures issued by BWA and the Planning and Development Department (PDD) on environmental and social management;
- b. Environmental and social management plans, monitoring reports, and implementation records, including those prepared for internationally financed projects;
- c. BWA's procedures for client communication and grievance handling; and
- d. Additional documentation relevant to national environmental and social systems and BWA operations.

32. To complement the document review, a questionnaire was administered to BWA, and follow-up interviews were conducted with key personnel. These interviews, held remotely, provided further clarification on institutional practices, responsibilities, and implementation challenges.

33. The findings from this process were analyzed against the World Bank's Core Principles for PforR operations. The Core principles establish the minimum requirements a program must meet to ensure that environmental and social impacts are properly managed. They also aim to strengthen country systems and promote sustainable development outcomes.

- a. **Core Principle #1:** Program environmental and social management systems are designed to (i) promote environmental and social sustainability in the Program design; (ii) avoid, minimize, or mitigate adverse impacts; and (iii) promote informed decision-making relating to a Program's environmental and social effects.
- e. **Core Principle #2:** Program environmental and social management systems are designed to avoid, minimize, or mitigate adverse impacts on natural habitats and physical cultural resources resulting from the Program. Program activities that involve the significant conversion or degradation of critical natural habitats or critical physical cultural heritage are not eligible for PforR financing.
- f. **Core Principle #3:** Program environmental and social management systems are designed to protect public and worker safety against the potential risks associated with (i) the construction and/or operation of facilities or other operational practices under the Program; (ii) exposure to toxic chemicals, hazardous wastes, and otherwise dangerous materials under the Program; and (ii) reconstruction or rehabilitation of infrastructure located in areas prone to natural hazards.
- g. **Core Principle #4:** Program environmental and social systems manage land acquisition and loss of access to natural resources in a way that avoids or minimizes displacement and assists affected people in improving, or at the minimum restoring, their livelihoods and living standards.
- h. **Core Principle #5:** Program environmental and social systems give due consideration to the cultural appropriateness of, and equitable access to, Program benefits, giving special attention to the rights and interests of Indigenous Peoples/Sub-Saharan African Historical Underserved Traditional Local Communities, and to the needs or concerns of vulnerable groups.
- i. **Core Principle #6:** Program environmental and social systems avoid exacerbating social conflict, especially in fragile states, post-conflict areas, or areas subject to territorial disputes.



34. The ESSA determines whether all core principles, or only a subset, are applicable to the Program. It then analyzes the system by comparing it against these principles to assess alignment and identify any gaps that may require additional measures.

35. Gaps between the current national and institutional systems and the Bank's requirements were identified. Where noted, recommendations were formulated to strengthen both system-level and project-level practices. These recommendations will be consolidated into the Program Action Plan (PAP), which specifies actions, responsibilities, and timeframes for improving the management of environmental and social aspects during Program implementation. Several of these actions will be supported through the Modernization Contract and can be reinforced by relevant DLIs.

2. PROGRAM DESCRIPTION

A. Program-for-Results

36. The Barbados Water Security and Sector Performance Program (P508674) will be implemented under the PforR financing instrument, with a total program cost of approximately US\$80 million, of which US\$50 million will be financed by IBRD.

37. The Program supports the implementation of the BWA Strategic Plan 2025–2029, focusing on strengthening the operational, institutional, and financial capacity of the BWA to enhance service delivery, improve water resources management, and expand safely managed sanitation.

38. The PforR will be structured around three results areas, each supported by a set of DLIs and Disbursement-Linked Results (DLRs) that aim to incentivize improved performance, institutional reforms, and financial sustainability.

39. **Results Area 1 – Building BWA capacity for efficient service delivery:** This area focuses on the modernization of BWA's operations and systems to improve service delivery, customer engagement, and financial and operational performance.

Table 1 – Disbursement Linked Indicators for Result Area 1

Disbursement Linked Indicator	Disbursement-Linked Results (DLRs)
DLI 1: Improved BWA's services continuity capacity and reduced water losses	DLR 1.1 – BWA's disaster recovery and business continuity plan approved by BWA's Board DLR 1.2 – 24 priority District-Metered Areas (DMAs) operationalized DLR 1.3 – Eighty thousand (80,000) customer water meters installed and automated to a central BWA system country-wide DLR 1.4 – 24 DMAs reached 30% (or less) physical water losses
DLI 2: Improved BWA's customer responsiveness and financial performance	DLR 2.1 – Digital customer portal is integrated with commercial management systems through: contracting of a service provider for the design, establishment, and integration of said portal; and the installation and operationalization of said portal DLR 2.2 – 25% increase of customer complaints investigated within twenty (20) days of receipt DLR 2.3 – 20% increase in BWA's operating revenues



Disbursement Linked Indicator	Disbursement-Linked Results (DLRs)
DLI 3: BWA's organizational reform, systems digitalization, and staff enhancement program are implemented	DLR 3.1 – Performance accountability mechanism for water supply and sanitation services established through: the signing of a performance and service level agreement between the Borrower and the PIE; and the publication of the annual performance report of the PIE's website. DLR 3.2 – Integrated utility-wide MIS is able to capture data and generate reports across three (3) priority modules (technical, financial, and WRM) DLR 3.3 – BWA's organizational and staffing reform implemented through: BWA's Board approval of the new BWA organizational structure; 30% of key staff positions filled; and 100 employees accredited

40. **Results Area 2 – Pollution management through country-wide safely managed sanitation services:** This area promotes the transition to safely managed sanitation across Barbados by strengthening wastewater and septage management systems and enabling water reuse. It will also support regulatory and operational improvements for septage collection, treatment, and reuse, as well as the development of low-cost sewerage pilots and stakeholder engagement programs to promote acceptance of wastewater reuse.

Table 2 – Disbursement Linked Indicator for Result Area 3

Disbursement Linked Indicator	Disbursement-Linked Results (DLRs)
DLI 4: Reduced unsafe disposal of untreated wastewater and septage	DLR 4.1 – Water reuse regulation and an amendment to the septage management regulation are published in the official gazette DLR 4.2 – Two thousand (2,000) functional household sewerage connections established through lower-cost solutions DLR 4.3 – One hundred thirty-two thousand (132,000) KL of septage collected and treated

41. **Results Area 3 – Foundations for long-term water security:** This area strengthens Barbados's capacity to manage its limited water resources sustainably and adapt to climate variability. Activities will operationalize the Water and Wells Management Committee (WWMC) and enhance the Water Resources and Environmental Management Unit (WREMU), establish high-level coordination mechanisms across sectors, and improve data and monitoring systems.



Table 3 – Disbursement Linked Indicator for Result Area 3

Disbursement Linked Indicator	Disbursement-Linked Results (DLRs)
DLI 5: Implemented institutional, legal and technical mechanisms for water security management and reporting	DLR 5.1 – Institutional and regulatory water security mechanisms are implemented through the Cabinet approval on the establishment of a high-level coordination mechanism on water security; and BWA’s Board approval of the WWMC regulation². DLR 5.2 – Country-wide groundwater monitoring and reporting system is implemented through an updated inventory of wells in Barbados; and public disclosure of annual water resource status based on WRIMS³.

B. Institutional and Implementation Arrangements

42. BWA is a statutory corporation established under the Barbados Water Authority Act (1980, amended 2023), mandated to manage the national water supply and sewerage services that will serve as implementing agency for the Program. Its organizational structure includes technical and administrative departments such as Engineering and Planning, Operations, Quality Assurance, Human Resources, and Corporate Services.

43. BWA plans to establish a small Program Management Unit (PMU) to manage the overall administration of the Program, coordination across Departments, and coordination of the Modernization Contract implementation. BWA already has established a ‘Transformation Committee’ which comprises members of its board, directors and selected technical staff. BWA could utilize this same committee to ensure close coordination and synergized work between the Modernization Contract and BWA management and staff. Works are expected to be undertaken both directly by BWA staff and through private contractors, with BWA engineers and site supervisors anticipated to oversee compliance with technical specifications.

44. Regarding environmental and social responsibilities within BWA, the Water Resources and Environmental Management Unit (WREMU) holds the central mandate for water resources management, including the protection and sustainable use of groundwater, while occupational health and safety is managed by the Occupation Safety and Health (OSH) Committee and the OSH function within the Human Resources Department. Customer complaints and service-related grievances are handled through the Customer Service Division, with statutory oversight of tariffs and service quality provided by the Fair-Trading Commission (FTC).

45. External regulatory oversight rests with national authorities. The PDD manages development screening and permitting, with final permissions issued by the Planning and Development Board (PDB). For projects with potentially significant impacts, technical review is carried out by the multi-agency EIA Committee, which includes the Environmental Protection Department (EPD), the Coastal Zone Management Unit (CZMU), and the Ministry of Health and Wellbeing. However, the types of works anticipated under the Program, like NRW reduction measures, metering, and small sewer pilots, are small in scale and located within existing rights-of-way; therefore, they are unlikely to require a full EIA.

² The Barbados Water Authority (Amendment) Act, 2023 (amending Cap. 274A) provides for the temporary alteration of conditions attached to abstraction licenses during officially declared drought events, upon recommendation of the Water and Wells Management Committee.

³ The WRM module under DLR 3.2 will be implemented through a Water Resources Information Management System (WRIMS), envisioned as an integrated system for managing and visualizing water resources information, linked to associated decision-support tools and other relevant systems managed by BWA for WRM. WRIMS will also enables data-sharing and integration with relevant national agencies involved in water resources management.



46. BWA has prior experience implementing projects financed by multilateral development banks (MDBs) and bilateral agencies, which required compliance with international environmental and social frameworks. These include the Inter-American Development Bank (IDB)–financed Water Sector Resilience Nexus project and mains rehabilitation programs supported by the Caribbean Development Bank (CDB). Through these operations, it is expected that BWA has gained exposure to MDB requirements in areas such as stakeholder consultation, grievance redress mechanisms, and the preparation and implementation of environmental management plans.

3. POTENTIAL ENVIRONMENTAL AND SOCIAL EFFECTS

47. The Barbados Water Security and Sector Performance Program is expected to generate overall positive outcomes in terms of improved service reliability, water conservation, and institutional strengthening. However, negative environmental and social impacts and risks must be identified, assessed, and managed appropriately. The following sections provide a summary of the identified impacts, risks, and benefits.

A. Exclusions

48. According to Paragraph 10 of the World Bank Policy on Program-for-Results Financing and Paragraph 14 of the World Bank Guidance: Program-for-Results Financing – Environmental and Social Systems Assessment, activities that are likely to have significant adverse impacts that are sensitive, diverse, or unprecedented on the environment and/or affected people are ineligible for PforR financing.

49. Such activities are excluded to ensure that the Program operates within a manageable environmental and social risk profile consistent with Core Principles 1–6 of the PforR Policy.

50. Based on this guidance and the scope of the Barbados Water Security and Sector Performance Program, the following activities are excluded from Program financing:

- a. Works requiring land acquisition, physical displacement, or involuntary resettlement;
- b. Works located within or adjacent to protected areas, marine reserves, coastal buffer zones, or other ecologically sensitive sites designated under national law or international conventions;
- c. Activities causing significant conversion or degradation of critical natural habitats, such as mangroves, wetlands, or turtle nesting beaches;
- d. Works situated in areas prone to severe natural hazards (coastal erosion, landslides, or flooding) unless robust engineering and climate resilience measures are fully integrated in design;
- e. Works involving asbestos-containing materials, lead-based paints, or other hazardous or prohibited substances or technologies;
- f. Activities that could adversely affect physical cultural heritage, including disturbance or destruction of antiquities, relics, or burial sites;
- g. Activities with transboundary environmental implications (e.g., marine or groundwater pollution across national boundaries); and
- h. Any intervention that, based on screening, would be classified as High risk under the World Bank Environmental and Social Framework (ESF) or equivalent national standards.

51. The Program must include a screening mechanism, to be institutionalized within the BWA WREMU, that confirms all proposed investments comply with these exclusions prior to financing. Any activity falling under these categories will be screened out during preparation and replaced by eligible interventions of comparable value and scope.



B. Potential Environmental and Social Benefits

52. Overall, the Program is designed to deliver broad environmental, social, economic, and institutional benefits, improving the quality of life for consumers, supporting sustainable development, and fostering greater equity and resilience within the community.

53. The Program delivers key environmental benefits by reducing water losses through leak detection and improved metering, which limits groundwater extraction and enhances water security. Efficiency measures lower energy consumption and the carbon footprint of water supply. Additionally, the sewer pilot reduces groundwater contamination risks by minimising reliance on on-site sanitation. Collectively, these initiatives support more sustainable water management and increase resilience to climate variability. Specific environmental benefits include:

- a. *Water Resources Conservation:* The Program focusses on reducing NRW through leak detection, repair, pressure management, and loss control, along with improved metering. These measures collectively decrease physical water losses in the distribution system, which in turn limits the need for groundwater extraction. Sustained NRW control not only ensures a more reliable water supply for customers but also contributes significantly to long-term national water security.
- b. *Improved Efficiency of Water Distribution Systems:* Measures such as more accurate metering and improved leak control leads to reduced physical water losses and decreases the need for excessive pumping. These actions not only optimize energy usage but also result in sustained long-term efficiency gains and help lower the carbon footprint of the water supply system.
- c. *Reduced Risk of Groundwater Contamination:* The sewer pilot aims to reduce the risk of groundwater contamination by decreasing dependence on on-site sanitation systems and informal waste disposal. This initiative initially provides localized benefits by lowering the likelihood of untreated effluent seeping into soils and groundwater, with the potential for broader impact if successfully scaled up.
- d. *Strengthened Water Security and Integrated Water Resources Management:* The Program strengthens water security and integrated water resources management by implementing targeted initiatives such as NRW reduction through leak detection and repair, pressure management, enhanced metering, a sewer pilot, and operational modernization. These measures collectively improve the reliability and sustainability of water supply, support better measurement and monitoring, bolster institutional capacity for evidence-based decisions, and increase resilience to droughts and climate variability, while also reducing the risk of groundwater contamination.

54. The potential social benefits include improved reliability and continuity of water services, better public health and hygiene, and temporary job creation during construction. The Program also aims to enhance staff skills and promote a safer, more inclusive work environment, particularly for women. Additional positive impacts include increased resilience to climate and disaster events, greater transparency and customer confidence in billing, and indirect economic gains in tourism due to improved water security. The Program delivers several significant social benefits including:

- a. *Improved Reliability and Continuity of Water Services:* Upgrades to commercial services, such as the Customer Information System and Commercial Management System, support more consistent water supply and transparent billing practices. This reduces disputes and reinforces consumer trust in water charges.
- b. *Enhanced Public Health and Hygiene:* By ensuring a reliable supply of potable water and better wastewater management through sewer pilots and pollution control, the Program helps to mitigate health risks and fosters safer, cleaner communities.
- c. *Temporary Job Creation:* Construction and implementation phases provide short-term employment opportunities, contributing to local economic development and supporting livelihoods during project execution.



- d. *Better Jobs through Capacity Building*: The Program offers technical and leadership training, adopts fair policies, and provides facilities like lactation rooms and separate bathrooms. It aims to boost staff skills, support career growth, and create a safe, female-friendly, and respectful workplace.
 - e. *Climate and Disaster Resilience*: Infrastructure modernization and improved water management practices increase the community's ability to withstand and recover from adverse events linked to climate change and natural disasters.
 - f. *Improved Customer Service and Transparency*: The digitalization of commercial services leads to more open and accurate billing systems, reducing the potential for disputes and enhancing customer confidence in the fairness of water charges.
 - g. *Gender Empowerment and Inclusion*: Through capacity building and the adoption of institutional policies, the Program actively promotes the involvement of women in technical and managerial roles within the Barbados Water Authority (BWA). This results in enhanced professional development, career progression for female staff, improved gender balance, and a more diverse and effective organisational culture.
 - h. *Tourism and Local Economic Benefits from Improved Water Security*: By reducing Non-Revenue Water (NRW), improving service reliability, and controlling pollution, the Program indirectly boosts tourism competitiveness. A reliable supply of safe water and improved wastewater management strengthen the tourism sector and generate long-term local economic benefits, including job creation and increased revenue.
55. Annex 2 contains detailed descriptions of the Program's potential environmental and social benefits.

C. Potential Environmental and Social Risks, Impacts, and Mitigation Measures

56. Potential environmental risks and impacts associated with the Program are minor and the impacts are generally short-term, localized, and manageable with appropriate mitigation measures. Specific risks and impacts include:
- a. *Vegetation Loss*: Vegetation and minor soil disturbance may occur during trenching and pipe-laying for the sewer pilot, potentially impacting local aesthetics and causing limited erosion. Mitigation measures include surveying and marking vegetation, minimising removal, restoring disturbed areas, and ensuring compensatory replanting. Soil disturbance from excavation is localised and low-risk, addressed by careful planning, prompt reinstatement, erosion control, and re-vegetation of affected areas.
 - b. *Minor Soil Disturbance and Erosion*: Minor soil disturbances as well as localised dust, noise, and vibration, may occur during small-scale excavation, leak repair, metering, and sewer pilot works. These impacts are generally limited due to the shallow depth and urban settings, and can be effectively managed by planning works during dry periods, promptly reinstating surfaces, implementing simple erosion controls, and using measures to minimise dust and noise. Regular inspections and use of well-maintained equipment further help to mitigate any adverse effects.
 - c. *Dust, Noise, and Vibration*: Activities such as leak repair, metering, and sewer pilot works can result in dust emissions, elevated noise, and ground vibrations, with recommended mitigation measures including dust suppression, noise control, and advance notifications to nearby residents. Minor water pollution risks may arise from improper spoil or wastewater disposal, but these impacts are typically short-term and localised; proper collection, disposal, and use of silt fences are advised. Additionally, small-scale generation of solid, construction, and hazardous waste should be managed responsibly to prevent environmental harm.
 - d. *Minor Water Pollution Risks*: Leak repairs, metering, and sewer pilot works can lead to temporary dust, noise, vibration, minor water pollution, and small-scale waste generation. These impacts are short-term and localised, with recommended mitigation including dust suppression, noise control, proper wastewater disposal, and waste segregation. Training for contractors and advance notifications for residents are advised to minimise environmental harm.



- e. *Small-Scale Waste Management*: Ensuring proper waste management practices and risk mitigation measures for sewer pilot works and well monitoring installations. It emphasises the segregation and safe handling of construction and hazardous waste, proper disposal, recycling where possible, and contractor training. For monitoring equipment installation in wells, it details precautions to maintain well integrity and water quality, such as using dedicated observation wells, upgrading sanitary seals, disinfecting wells, and securing cables to prevent operational issues.
 - f. *Production Well Integrity and Water Quality Risks from Monitoring Installation*: Proper installation and maintenance of well monitoring equipment are crucial to prevent sanitary risks, operational disruptions, and water quality issues. Key practices include using dedicated observation wells, ensuring sanitary seals and waterproof connectors, disinfecting wells, and keeping thorough records. Additionally, works should be scheduled to minimize service interruptions. Temporary access disturbances, such as blocked sidewalks or driveways, may occur during activities like leak repair and metering, but these impacts are generally minor and short-term, especially in crowded urban environments.
57. The social risks associated with the Program are also limited in nature, generally short-term, and easily mitigated with proper planning, communication, and capacity building. Specific social risks and impacts include:
- a. *Temporary Access Disturbances*: Minor and short-term access disturbances may occur during leak repairs, metering, and sewer pilot works, particularly in dense urban areas, due to trenching or excavation near property entrances. Measures such as advance notification, safe walkways, clear signage, and scheduling outside peak hours are recommended to reduce disruption to residents, businesses, and community facilities. Additionally, business livelihood impacts can be mitigated by briefing shop owners, providing temporary vending locations, and prioritising restoration of access points.
 - b. *Minor and Short-term Business Livelihood Disruption*: Program related Construction works, leak repair, metering, and the sewer pilot are expected to cause temporary business disruption, particularly for small shops and roadside vendors, as well as minor traffic and mobility impacts such as lane closures and congestion. Mitigation measures include engaging directly with affected businesses, providing clear signage and alternative vending spots, prioritising the restoration of access, and implementing approved traffic management plans with safe pedestrian routes.
 - c. *Minor and Short-term Traffic and Mobility Impacts*: Construction activities, leak repairs, metering, and the sewer pilot may cause temporary business, traffic, and mobility disruptions, which will be mitigated through direct engagement, signage, access restoration, and approved traffic management plans.
 - d. *Cultural Heritage Risks*: While the risk of disturbing cultural heritage is low since all works will take place in previously developed areas, strict chance-find procedures will be enforced if subsurface materials are encountered.
 - e. *Occupational Health and Safety Risks*: Work activities for metering and sewer pilot projects expose workers to occupational risks, such as traffic hazards, machinery accidents, and heat stress, necessitating strict OSH standards, PPE, and climate-adapted safety measures.
 - f. *Community Health and Safety Risks*: Hazards and mitigation measures associated with sewer pilot projects, such as risks from unsecured trenches, construction machinery, and community interaction, including the potential for SEA/SH. Mitigation includes inclusive community engagement, strategies for worksite safety, traffic management, dust and noise control, and waste handling.
 - g. *Digital Divide, i.e. the Exclusion of Vulnerable Users*: Efforts to modernise commercial services and water management include the adoption of digital platforms and operational committees, which raise concerns about accessibility for vulnerable users. Hybrid service delivery, inclusive design, targeted outreach, and fallback mechanisms are recommended to ensure equitable access, while transparency, early engagement, and support are key to addressing private users' concerns regarding data and monitoring procedures. Monitoring and feedback channels will help identify and resolve issues during implementation.



- h. *Concerns from Private Users and Operators:* Private users may have concerns about new data and monitoring requirements as water management committees become operational, but clear communication, transparency, and support can mitigate potential conflicts. Additionally, the removal of informal water connections as part of NRW reduction measures may cause temporary disruption, with strategies in place to notify affected users and provide pathways for regularisation and service continuity.
- i. *Disruption and Elimination of Informal or Unmarked Connections:* Measures are in place to address the impacts of removing illegal or informal water connections during NRW reduction and metering works, including pre-works surveys, communication, regularisation options, temporary service continuity, and grievance mechanisms.
- j. *Service disruptions due to underground Utility Works:* There are risks of accidental damage to underground infrastructure during sewer and metering works, which can be mitigated through careful planning, consultation, advanced detection methods, and clear emergency protocols.
- k. *Potential Opposition or Disruption Among BWA Staff:* Organizational reforms within BWA may face resistance, which is being managed through transparent communication, staff engagement, and capacity-building strategies.

58. Annex 3 contains detailed descriptions of the Program's potential environmental and social risks, impacts, and mitigation measures.

4. NATIONAL SYSTEMS ASSESSMENT

59. In line with the World Bank's PforR Policy, the assessment of program systems considers the institutional, organizational, and procedural arrangements in place for managing environmental and social effects. This includes the laws, regulations, policies, and implementing guidelines applicable to the Program, as well as the mandates of the institutions responsible for applying and enforcing them. The assessment also takes into account inter-agency coordination where responsibilities for oversight and compliance are shared across entities.

60. The analysis was undertaken to evaluate whether the existing national and BWA systems are adequate to manage the range of potential risks and impacts identified under the ESSA.

A. National Legal Framework

61. Barbados maintains a comprehensive body of legislation that governs environmental, social, labor, and health aspects relevant to water and sanitation projects. This framework combines overarching development control instruments—such as the Planning and Development Act (2019, amended 2020) and Environmental Impact Assessment Regulations (2021)—with sectoral laws addressing environmental protection, coastal and natural resource management, cultural heritage, public health, labor rights, and workplace safety. Together, these instruments establish the legal authority for environmental and social screening, permitting, compliance, and enforcement. Annex 4 summarizes the main elements of the national legal framework and their applicability, highlighting the extent to which they provide safeguards for Program activities.

62. Barbados's key environmental laws and regulations relevant to the Program and summarises their applicability. The Planning and Development Act 2019 and the Environmental Impact Assessment (EIA) Regulations 2021 set out procedures for development control, including requirements for environmental review, particularly for projects with significant potential impacts. For the Program's activities—such as non-revenue water (NRW) reduction, metering improvements, and small sewer pilots—full EIAs are typically not needed due to the limited scale and use of existing rights-of-way, but further assessment may be required if works are close to sensitive areas like heritage or conservation sites.

63. The Environment Protection Act 1997 provides the framework for environmental management, empowering authorities to regulate pollution, emissions, and waste. Program activities must comply with these rules, especially regarding the management of construction waste and hazardous materials (such as oils and asbestos), to prevent environmental harm.



64. With regards to care needed for Program related works, the Electric Light & Power Act, Cap. 278, and the National Petroleum Corporation Act, Cap. 280, mandate that the Program must coordinate with the respective utility licensees, obtain their consent, and comply with safety and technical conditions before executing works near electricity or gas lines. The Road Traffic Act, Cap. 295, requires securing permits for any works affecting public roads, implementing approved traffic management plans, and ensuring prompt restoration of roadways and sidewalks. Additionally, the Preservation of Antiquities and Relics Act, 2011 governs the identification, protection, and management of antiquities, cultural heritage, outline the responsibilities of relevant authorities and the legal obligations for safeguarding these assets. Collectively, these laws ensure that the Program operates safely, legally, and with minimal disruption to public services and safety.

65. The legal and regulatory framework for water and sanitation management in Barbados is established through several acts including the Barbados Water Authority Act, 1980 empowers the BWA to oversee water resources, implement key activities such as non-revenue water (NRW) reduction and metering, and manage infrastructure, ensuring the Program operates with clear institutional authority. The Utilities Regulation Act, 2002, and its 2010 Amendment, place the BWA under the Fair Trading Commission's regulatory oversight, ensuring transparency, consumer protection, and proper tariff setting, which supports the Program's financial and operational sustainability. Finally, the Barbados Water Authority (Water Services) Regulations, 1982, define the obligations of both the BWA and consumers regarding water distribution, supply, metering, and billing, providing practical guidelines for the Program's day-to-day implementation and consumer engagement. The Water Reuse Act, 2023 (Act 2023-10) Creates a permit system to produce, supply, or sell reclaimed (non-potable) water. Establishes inspection, enforcement, suspension/revocation, offences, and administrative penalties. Establishes a Water Reuse Committee to advise on permits and compliance. Defines allowable uses of reclaimed water (e.g., irrigation, equipment cooling, groundwater recharge). The Act aims to promote sustainable water management by regulating the safe and beneficial use of reclaimed wastewater.

66. The workplace health and safety legislation relevant to the Program primarily consists of the Safety and Health at Work Act, 2005 (as amended in 2022) and its subsidiary regulations. This framework obliges employers, including BWA and its contractors, to ensure safe working conditions through risk assessments, worker training, provision of personal protective equipment (PPE), and adequate sanitary facilities. The 2022 Amendment broadened employer and worker responsibilities and strengthened inspectorate powers, aligning local standards with international conventions. Specific regulations address risks such as noise exposure, hazardous materials, heavy equipment operation, and heat stress, requiring targeted preventive measures, monitoring, and compliance with official directives. Additionally, the Employment Rights Act, 2012-19 guarantees fundamental labour rights for all staff and contractors, including fair terms of employment and protections against unfair dismissal, supporting the overall objective of safeguarding health, safety, and welfare in the workplace.

67. The Land Acquisition legislation relevant to the Program sets out the legal framework for the compulsory purchase of land required for public works, such as those undertaken by the Barbados Water Authority (BWA) and its contractors. It outlines the procedures for identifying affected properties, notifying landowners, and determining fair compensation for acquired land. The legislation ensures that acquisitions are conducted transparently and in accordance with due process, safeguarding the rights of property owners while enabling essential infrastructure projects to proceed efficiently. This framework is crucial for minimising disputes and facilitating timely delivery of Program works.

68. In addition to the legal framework there are two key national initiatives in Barbados related to the Program, the Integrated Solid Waste Management Program (ISWMP), and the Comprehensive Disaster Management (CDM) Policy. ISWMP, supported by international partners, aims to modernise waste management practices, covering landfill operations, recycling, hazardous waste, and institutional capacity, though it is not legally binding. The CDM Policy provides strategic guidance for integrating disaster risk reduction and climate resilience in the water sector, encouraging the incorporation of emergency preparedness and resilient design. Both initiatives offer important frameworks for sustainable development, despite not carrying legal obligations.



B. International Legal Framework

69. Barbados is party to several international conventions and treaties that are relevant to the Program. The majority of these aim to safeguard environmental and occupational health and safety standards across borders. A comprehensive list of and analysis of how they relate to the Program is found in Annex 5. They serve as authoritative guidance for marine protection, hazardous waste management, and occupational health and safety, each providing specific obligations and recommended practices for member states. Collectively, these agreements establish minimum standards and procedures designed to ensure responsible stewardship of natural resources and the safety of workers.

70. For instance, the Cartagena Convention offers a comprehensive structure for the protection and sustainable use of marine and coastal environments in the Wider Caribbean Region. Its protocols address issues such as land-based sources of pollution, conservation of specially protected areas and wildlife, and oil spill response. Similarly, the Basel Convention regulates the international movement and disposal of hazardous wastes, requiring strict controls and prior informed consent to prevent environmental harm. These conventions are complemented by instruments like the ILO C155, which sets global benchmarks for workplace health and safety, further reinforcing the commitment to sustainable development and human well-being.

71. In practice, these international agreements not only set overarching legal requirements but also promote cooperation and information sharing among nations. They encourage the adoption of best practices in areas such as waste handling, pollution prevention, and occupational risk management. For programs operating within Barbados, adherence to these frameworks ensures alignment with global standards, reduces the risk of environmental degradation, and supports the creation of safer workplaces. This legal context ultimately underpins the responsible execution of development activities, contributing to the long-term health of both people and the environment.

C. National System Capacity to Respond to Identified Environmental and Social Risks

72. As described above and detailed in Annexes 4 and 5, Barbados' legal framework provides comprehensive authority and clear institutional responsibilities for addressing environmental and social risks and impacts relevant to the Program. However, in preparing the ESSA some implementation gaps were identified that will need to be addressed during program implementation. The major recommendations are included in the PAP. Annex 6 contains the detailed analysis done for making the overall recommendations and preparing the environmental and social actions in the PAP.

73. The Barbados Water Authority Act and Water Services Regulations lay out BWA's responsibilities for asset management, lawful connections, service continuity, and the procedures for planned service interruptions, ensuring a baseline of operational stability and public notification. This helps mitigate environmental and social risks related to service disruptions, such as lack of access to essential water services.

74. Sewerage Regulations require formal submissions and approvals for sewer works, enforcing system compliance and promoting inter-agency coordination. This is crucial for minimizing public health risks and environmental hazards, particularly in densely populated areas. The regulations also support the prevention of unsanitary conditions in the event of infrastructure strikes or failures, as further reinforced by the Health Services Act and Nuisances/Building Regulations, which empower authorities to act when public health is at risk due to utility-related incidents.

75. Permit requirements under the Road Traffic Act and Highways Act, along with the necessity for Traffic Management Plans, ensure that road works are conducted safely and with minimal disruption to the public. These measures address social risks such as restricted mobility, increased accident risks, and the economic impact of prolonged road closures. Similarly, crossing or working near electric and gas lines requires adherence to the respective licensee's safety standards and explicit consent, protecting both workers and the general public from potential hazards.

76. The Planning and Development Regulations provide for the imposition of protective conditions, such as method statements and work sequencing, which can extend to cross-utility coordination and approvals. This fosters a culture of safety and accountability, reducing the risk of accidental strikes and service interruptions that could negatively affect



communities and businesses. However, the effectiveness of these mechanisms depends on robust enforcement and active collaboration among utility providers.

77. Despite the strengths of the national system, there are notable gaps, particularly related to the absence of reviewed Standard Operating Procedures (SOPs) for BWA. Formalizing and approving SOPs should include construction management; maintenance operations; site and dig safety; waste management including segregation, labelling, temporary staging, collection, and reuse-recycling; schedules; record-keeping; are either unclear or undocumented for these types of waste. Such actions would improve transparency and community outreach; consistency in managing third-party consents; wayleaves; pre-dig coordination.

78. Additionally gaps in practical implementation were noted within the BWA concerning small-scale, dispersed construction wastes. Specific operational procedures

79. Contractor orientation is another area of concern, as there is no evidence of training or requirements for housekeeping, litter control, or secure storage measures to prevent windblown debris and scattering at construction sites. This oversight increases the risk of environmental harm, particularly from small quantities of waste generated during rolling works, which may not be subject to the same scrutiny as larger projects.

80. Finally, the existing grievance and communication channels are primarily designed for billing and service issues, with no mechanism in place for advance public notices or for receiving and resolving complaints specifically related to construction-phase activities. Within BWA, efforts are needed around the formalization and standardization of utility coordination and strike prevention protocols.

81. To address these gaps, the PAP recommends updating BWA's SOPs to align fully with national regulations and to include a formalised process for recording utility consents and wayleaves, conducting coordinated pre-dig activities, and enforcing protection standards at utility crossings. The SOPs should also establish clear protocols for utility-strike response, including mark-out standards and safe-dig specifications. Enhanced competency-based training for supervisors and plant operators, along with the introduction of field inspection checklists and key performance indicators, will further strengthen compliance and reduce social risks associated with utility works.

82. In summary, the Barbados national system provides a solid foundation for managing the social risks linked to utility works, with multiple layers of regulation and inter-agency coordination. However, improvements in SOP transparency, process standardisation, and workforce training are necessary to close existing gaps and ensure that social and environmental risks are managed proactively and effectively.

83. The Program, including the PAP, aims to enhance both the national and BWA systems' capacity to manage environmental and social risks effectively.

5. CONCLUSIONS AND RECOMMENDATIONS

A. Alignment of National Environmental and Social Systems

84. The World Bank's PforR policy outlines six guiding principles for managing the environmental and social aspects of the programs it finances. The assessment of the environmental and social systems applicable to the Program indicates that Barbados's legal and regulatory framework is generally aligned with the core principles and planning elements of the PforR Directive. This framework adequately addresses environmental, social, and occupational health and safety considerations. However, certain gaps have been identified, along with opportunities for institutional strengthening.

85. Key recommendations to BWA based on the World Bank's six PforR environmental and social Core Principles are detailed below.



86. Core Principle #1: Promoting Environmental and Social Sustainability - Program environmental and social management systems are designed to promote sustainability within the Program's design, avoid or mitigate adverse impacts, and support informed decision-making regarding environmental and social effects. Recommendations include:

- a. *Early Screening of Potential Impacts.* Barbados' Planning and Development Act and EIA Regulations require systematic project screening to determine if an IEE or full EIA is needed, with criteria including location and potential impacts. Small-scale utility works, such as those in the Program, are usually exempt or require only IEEs unless near sensitive sites. The Barbados Water Authority's WREMU provides technical guidance, and SOPs exist for managing construction impacts; however, there is no consistent tool to screen all environmental and social risks. It is recommended to adopt a checklist for all activities and exclude projects with significant adverse impacts not aligned with World Bank PforR policies.
- b. *Consideration of Strategic, Technical, and Site Alternatives.* The Planning and Development Act (2019) and EIA Regulations (2021) require EIAs to include analysis of project alternatives, covering technical approaches, site locations, and a "no development" scenario. Most Program activities are unlikely to trigger a full EIA and may only require an IEE, which does not consistently require detailed consideration of alternatives. Recommendations involve reviewing or developing BWA internal guidelines for considering alternatives for all Program activities and institutionalizing documentation of the "no action" scenario. Strengthening the link between sectoral planning tools and BWA decision-making is also advised.
- c. *Assessment of Induced, Cumulative, and Transboundary Impacts.* EIA Regulations (2021) require EIAs to evaluate significant effects, including interactions between impacts, providing a basis for considering cumulative impacts. However, comprehensive cumulative impact assessment and strategic environmental assessment are not explicitly mandated. For small-scale works, induced, cumulative, or transboundary impacts are unlikely, though localized overlaps could occur. Recommendations include integrating questions on overlaps in timing and location into BWA's screening tool, requiring contractors to prepare traffic and waste management plans, and ensuring marine safeguards for sewer pilots near coastal areas.
- d. *Identification of Mitigation Measures.* Barbados' legal framework mandates identification and implementation of mitigation measures through EIA Regulations and sectoral laws such as the Environment Protection Act, Coastal Zone Management Act, Safety and Health at Work Act, Health Services Act, and Road Traffic Act. These frameworks require pollution control, erosion management, occupational health and safety, and traffic safety measures. Mitigation is generally required when projects undergo an IEE or EIA, with systematic mitigation for small-scale interventions not automatically activated unless imposed by authorities. Recommendations include developing standardized Environmental Codes of Practice (ECoPs) and SOPs, integrating them into contracts, enhancing monitoring capacity, and providing training for staff and contractors.
- e. *Articulation of Institutional Responsibilities and Resources.* The Planning and Development Department (PDD), Planning and Development Board (PDB), and the EIA Committee form the core regulatory mechanism for integrating environmental and social considerations into project approvals. BWA has an Environmental Unit, but lacks a documented, Program-wide definition of roles and resources to implement plans. There is no evidence of standard ECoPs or integrated social SOPs. Recommendations focus on developing clear Program-level procedures, providing capacity building, and designating environmental and social focal points within BWA.
- f. *Responsiveness and Accountability through stakeholder consultation and grievance Redress mechanisms.* Legal provisions require public notice, opportunities for written submissions, and public hearings for projects requiring IEE or EIA. At the Program level, BWA engages with customers but lacks mechanisms specifically addressing environmental and social issues arising from works. Grievance handling exists through consumer complaint processes and statutory pathways but is not tailored to construction-phase risks. Recommendations include standardizing consultation and disclosure for all works, establishing a Program-level grievance redress mechanism (GRM),



embedding consultation and GRM requirements in contracts, setting service standards and escalation protocols, and providing resources and training for implementation.

87. Core Principle #2: Protection of Natural Habitats and Cultural Resources - Program environmental and social management systems aim to avoid, minimize, or mitigate adverse impacts on natural habitats and physical cultural resources. Activities involving significant conversion or degradation of critical natural habitats or cultural heritage are not eligible for PforR financing. Recommendations include:

- g. *Identification and Mitigation of Impacts on Biodiversity and Cultural Resources.* Barbados' legal framework includes instruments for screening and mitigating potential impacts on natural habitats and cultural resources, with specific controls and conditions imposed by regulatory agencies. Most Program activities are unlikely to trigger significant impacts, but localized risks may occur. Screening is reactive, with no standardized procedure within BWA. Recommendations include developing screening checklists and ECoPs, strengthening collaboration with technical authorities, using biodiversity mapping tools, training staff and contractors, establishing procedures for chance finds, and excluding high-risk activities from the Program.
- h. *Protection, Conservation, and Rehabilitation of Natural Habitats.* Regulatory instruments such as the Coastal Zone Management Act, Physical Development Plan, Trees (Preservation) Act, Soil Conservation Act, and National Conservation Commission Act provide statutory mechanisms for habitat protection. Program activities are expected to occur mainly within urban rights-of-way, but risks may arise near sensitive areas. Proactive conservation measures are not systematically embedded, and monitoring is limited. Recommendations include introducing standardized screening tools, excluding high-risk activities, enhancing monitoring, and providing capacity building.
- i. *Avoidance of Significant Conversion or Degradation of Critical Natural Habitats.* Program works are expected in urban areas, with direct interaction with critical habitats unlikely. However, interventions extending into sensitive areas are subject to biodiversity studies and specialized surveys. No mechanism exists within BWA for verifying potential overlaps before works are designed. Recommendations focus on excluding high-risk activities from financing.
- j. *Protection of Physical Cultural Property.* National legislation prohibits unauthorized excavation or removal of antiquities and integrates heritage into development control. The likelihood of direct impacts on cultural resources is low, but excavation could encounter relics. Current procedures rely on permitting triggers, with no systematic screening tool for managing such risks. Recommendations include developing screening tools, integrating chance-find procedures into SOPs and contracts, providing training, and incorporating standard heritage protection clauses.

88. Core Principle #3: Safety of Public and Workers - Program environmental and social management systems are designed to protect public and worker safety during construction, operation, and maintenance, and from exposure to hazardous materials or natural hazards. Recommendations include:

- a. *Community, Individual, and Worker Health, Safety, and Security.* Legislation covers occupational and community safety, with mandatory risk assessments, training, and protective equipment. No published standard OSH/EHS procedures under BWA were identified, nor formalized contractor guidance. Recommendations include developing and enforcing standard Environmental, Health, and Safety Codes of Practice, incorporating mandatory clauses into contracts, training staff and contractors, strengthening supervision, and establishing incident response protocols.
- b. *Child and Forced Labor.* Legislation prohibits child and forced labor, with minimum working age set at 16 years. World Bank standards require no employment of minors under 18 in hazardous activities. No explicit contractual clauses were identified. Recommendations include including standard labor clauses, verifying the age of workers, collaborating with the Labor Department for inspections, and providing orientation and training.
- c. *Management of Hazardous Materials.* Regulation is provided by the Environment Protection Act and other sectoral laws, but systems are designed for large-scale waste streams and do not address small-scale hazardous waste typical of Program activities. No Program-specific procedures or training protocols were identified. Recommendations



include developing standardized protocols, strengthening internal guidance, building contractor capacity, formalizing incident response, and clarifying safe disposal pathways.

- d. *Integrated Pest Management*. Pesticide control is regulated for agricultural use, but there are no measures promoting integrated pest management for infrastructure works. No guidance or training exists within BWA. Recommendations include prioritizing non-chemical pest management measures and developing contractor guidance notes.
- e. *Worker Training for Hazardous Chemicals*. General obligations for risk assessment and training exist, but specific protocols for hazardous materials are lacking. Recommendations include developing tailored training modules, integrating international good practice references, and requiring contractors to document worker orientation.
- f. *Risk Management in Areas Prone to Natural Hazards*. The Emergency Management Act provides the national framework for preparedness, with sectoral laws addressing specific risks. Program activities are not expected to be highly vulnerable, but construction phase risks exist. No specific procedures for integrating hazard preparedness were identified. Recommendations include introducing hazard-screening steps, incorporating resilience measures, and developing contractor guidance notes.

89. Core Principle #4: Managing Land Acquisition and Access to Natural Resources - Program systems aim to avoid or minimize displacement and assist affected people in restoring or improving livelihoods and living standards. Recommendation include:

- a. *Land Acquisition and Impacts*. Activities requiring involuntary land acquisition or physical displacement are ineligible for financing. Program investments are expected within existing public rights-of-way, with only temporary access restrictions anticipated. No systematic procedures exist for screening or mitigating short-term disruptions. Recommendations include clarifying exclusion of activities requiring land acquisition, screening for access restrictions, minimizing disruptions through contractual requirements, and ensuring accessible grievance mechanisms.

90. Core Principle #5: Attention to Vulnerable Groups and Equitable Access - Program systems must consider the cultural appropriateness of benefits, equitable access, and the needs or concerns of vulnerable groups. No Indigenous Peoples are present in Program areas. Exclusion risks for other vulnerable groups are assessed under Core Principle 1. No further action is required.

91. Core Principle #6: Avoiding Social Conflict - Program systems must avoid exacerbating social conflict, particularly in fragile or post-conflict areas. Recommendations include:

- a. *Management of Conflict Risks*. The Program could potentially exacerbate labor conflicts due to institutional changes. Barbados' labor legislation recognizes the right to unionize, and BWA employees are represented by the Barbados Workers' Union. Formal communication and engagement strategies incorporating union participation have not been developed. Recommendations include establishing structured participation mechanisms and procedures for engaging workers' representatives as part of the broader communication and engagement strategy during the modernization process.

92. See Annex 7 for the full analysis of the core principles by assessing the extent of the alignment, identifying gaps and providing related recommendations.

B. Elements to Integrate into the Program Action Plan

93. To assist in operationalizing these broader recommendations, the following four environmental and social items are included in the Program Action PAP:

- a. Boosting BWA's environmental and social capacity;
- b. Developing standard environmental and social protocols and standardizing SOPs;
- c. Launching a stakeholder engagement plan the Program; and



- d. Standardizing community engagement procedures to include grievance monitoring and redress for construction activities.

94. The four environmental and social actions contained in the PAP, as detailed in Table 4, below, are based on the recommendations made in the ESSA. In carrying out these actions, BWA is expected to adhere to those recommendations as outlined above and included in Annexes 3, 6, and 7.

Table 4 – Environmental and Social Actions for the Project Action Plan

	PAP Item	Source	Responsibility	Timing	Completion Measurement
1	Boost BWA's capacity by hiring or appointing E&S risk management and gender experts. And training relevant BWA staff in environmental and social risk mitigation, community health and safety, Occupational Health and Safety, and SEA/SH protocols.	ESSA	BWA	December 31, 2026	Specialists are engaged or appointed for the Program with the required skills per agreed Terms of Reference. Initial BWA training has been conducted, with attendance documented and materials archived. Additional training will be held throughout the program.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)

Environmental and Social Systems Report

	PAP Item	Source	Responsibility	Timing	Completion Measurement
2	Develop standardized E&S Codes of Practice, Standard Operating Procedures for small-scale water and sanitation works, and a screening. with the Bank to identify E&S risks and requirements for inclusion in bidding documents.	ESSA	BWA	Prior to the first tender launch or within 6 months after Program effectiveness, whichever comes first.	Environmental Codes of Practice, Standards Operating Procedures and screening checklist developed and adopted by BWA. BWA's environmental and social requirements are incorporated into all civil works bidding and contracting documents.
3	Launch a stakeholder engagement plan for the overall Program, including digitalization, septage and wastewater reuse uptake, and groundwater management.	ESSA	BWA	December 31, 2026	Provision of completed Stakeholder Engagement Plan are developed and satisfactory to the Bank.
4	Standardize consultation and disclosure procedures for all water and sanitation small works that includes a Program-level incident reporting and grievance mechanism for stakeholders to report issues related to implementation.	ESSA	BWA	December 31, 2026	Procedures for consultations and disclosure of water and sanitation small works including incident reporting and grievance mechanism adopted and implemented by BWA.



ANNEX 1 – SUMMARY OF STAKEHOLDER CONSULTATIONS

95. The Barbados Water Authority (BWA) organized stakeholder consultations on the draft Environmental and Social Systems Assessment (ESSA) for the Barbados Water Security and Sector Performance Program (P508674) (the Program) on December 9, 2025. A second stakeholder consultation both on the Program and the draft ESSA was held on January 20, 2026. Internal and external BWA Stakeholders were included in both consultations including all BWA Departments, Unions representing the water sector, the Prime Minister's Office, Ministry of Finance, Ministry of Health, and Ministry of Environment. A summary of those discussions follows.

96. Comments and clarifications received during these consultations enabled the ESSA Team to better understand how the environmental and social compliance works on the ground, identify missing elements, and clarify the recommendations in order to develop a robust and realistic Project Action Plan (PAP).

97. **Project Design and Disbursement Linked Indicators (DLIs):** The World Bank team presented the overall technical design of the water sector Program, focusing on the structure, objectives, and the five main DLIs, while BWA participants raised questions about targets, funding, and implementation details.

- a. **Program Structure and Objectives:** the Program's high-level objectives, including water security, pollution management, and improved performance of BWA, with a \$50 million loan based on BWA's strategic plan were outlined. The Program is part of a regional initiative, with Barbados as the front runner, and includes plans for a regional center of water excellence.
- b. **DLIs Overview and Flexibility:** It was explained that the Program is structured around five DLIs, each with sub-indicators and scalable targets, allowing for partial disbursement as milestones are achieved. The DLIs cover areas such as reducing water losses, commercial improvement, institutional strengthening, wastewater management, and water security.
- c. **Funding Allocation and Value for Money:** The allocation of funds to each DLI is based on the relevance and value for money rather than direct costs, with flexibility for BWA to reallocate funds as needed, provided overall targets are met. The Bank presented a breakdown of the US\$50 million allocation, clarifying that these figures are not fixed and can be adjusted during implementation.
- d. **Clarification on Metering and NRW Targets:** BWA participants questioned the feasibility of installing 80,000 smart meters and achieving a 25% non-revenue water (NRW) target, citing funding constraints and the need for realistic targets. After discussion, the Bank agreed to consider revising the NRW target to 30% and to clarify whether the target should include both physical and commercial losses.
- e. **Implementation Arrangements:** The Program's implementation structure was outlined, including oversight by the Prime Minister's Office, the BWA board, and a transformation committee, with BWA responsible for hiring an independent verification agent and establishing a Project Execution Unit (PEU). BWA requested updates to reflect their internal nomenclature and committee structure.

98. **Environmental and Social Systems Assessment (ESSA) and Program Action Plan:** The Bank team led a review of the ESSA, highlighting the assessment of Barbados's environmental and social systems, identified gaps, and the proposed Program action plan, with BWA participants providing feedback and requesting clarifications on timelines and responsibilities.

- a. **ESSA Objectives and Exclusions:** It was explained that the ESSA evaluates the adequacy of Barbados's legal, regulatory, and institutional frameworks for managing environmental and social risks under the Program, with exclusions for activities requiring land acquisition, resettlement, or work in protected areas.



- b. **Identified Risks and Benefits:** The team outlined expected social and environmental benefits, such as improved service reliability and water conservation, as well as risks including temporary access disruptions, occupational health and safety, digital exclusion, and potential concerns from private well users and staff.
 - c. **Key Gaps and Recommendations:** Gaps identified include limited environmental and social staffing, fragmented monitoring systems, absence of standardized codes and practices, inconsistent consultation procedures, and lack of a documented grievance mechanism. Recommendations include hiring or assigning staff for environmental and social risk management, developing standardized procedures, and improving stakeholder engagement.
 - d. **Program Action Plan (PAP) and Next Steps:** The PAP includes four main actions: strengthening environmental and social capacity, developing standard operating procedures, launching a stakeholder engagement plan, and standardizing consultation and disclosure procedures. BWA requested draft terms of reference for new roles and clarification on the integration of these actions into their processes.
 - e. **Timelines and Document Review:** The World Bank team emphasized the need for BWA to review and provide feedback on the ESSA and Program action plan by the end of the week, with flexibility on timelines for implementation but a firm deadline for agreeing on the actions themselves. BWA highlighted the importance of consulting with the union and internal stakeholders before finalizing commitments.
99. **Stakeholder Engagement, Union Involvement, and Document Sharing:** BWA participants, including union representatives, raised concerns about receiving Program documents for the first time, requested additional time for review, and emphasized the need for further consultation and clarification on key indicators, gender targets, and implementation details.
- a. **Document Access and Review Process:** BWA and union representatives noted that some documents, such as the Program document and NRW reduction plan, were being seen for the first time, and requested time to review and discuss them internally before providing feedback or agreement.
 - b. **Union Consultation and Clarification Requests:** Union representatives asked for clarification on gender balance targets, the scope of accreditation indicators, and the timeline for achieving these targets, with the Bank confirming that the gender accreditation target is 40% women by the end of the five-year Program and is scalable.
 - c. **Sharing of Supporting Documents:** The World Bank team committed to sharing slides, the ESSA, and other relevant documents with BWA and union representatives, and to providing examples and draft terms of reference for new roles as requested.
100. **Coordination with External Agencies and Next Steps:** The Bank and BWA agreed on coordination with external agencies such as the Ministry of Health, FTC, and PIU, and outlined next steps for finalizing Program documents, setting appraisal dates, and preparing for upcoming meetings.
- a. **Coordination with Ministry of Health and FTC:** BWA confirmed participation in upcoming meetings with the Ministry of Health regarding septage management regulations and noted FTC's involvement in the Program, with the Bank welcoming their input.
 - b. **Finalization of Program Documents:** The Bank emphasized the need to finalize the DLIs and Program action plan by Friday to meet internal World Bank deadlines, with flexibility for further adjustments during appraisal and negotiation.



ANNEX 2 – POTENTIAL ENVIRONMENTAL AND SOCIAL BENEFITS

Potential Environmental Benefit	DLIs and Activities Likely to Enable It	Nature of Benefit (scope, duration, likelihood)
Water Resources Conservation	• DLIs 1, 3, 4, and 5. • NRW reduction (leak detection and repair; pressure management; loss control measures). • Metering.	NRW reduction measures reduce physical losses in the distribution network, thereby decreasing the need for groundwater abstraction. The benefits to both consistent water supply to customers and long-term national water security are cumulative, if NRW control measures are sustained over time.
Improved Efficiency of Water Distribution Systems	• DLIs 1, 3, and 5. • NRW reduction. • Metering.	More accurate metering and effective leak control reduce physical losses and the need for excessive pumping, leading to optimized energy use. These improvements generate long-term efficiency gains and contribute to reducing the overall carbon footprint of the water supply system.
Reduced Risk of Groundwater Contamination	• DLI 4. • Sewer pilot.	By reducing reliance on on-site sanitation systems and informal disposal practices, the pilot lowers the probability of untreated effluent infiltrating soils and groundwater. The benefits are initially localized, but may be scaled up if the approach is replicated.
Strengthened Water Security and Integrated Water Resources Management (WRM)	• DLIs 1, 2, 3, 4, and 5. • NRW reduction (leak detection and repair; pressure management; loss control measures). • Metering. • Sewer pilot. • Operational modernization and digitalization of commercial services.	Improved reliability and sustainability of water supply through better measurement, monitoring, and regulation of water resources. Enhanced data and institutional capacity enable evidence-based decision-making, reduced abstraction pressures, and improved resilience to droughts and climate variability.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)
Environmental and Social Systems Report – Annex 2

Potential Social Benefit	DLIs and Activities Likely to Enable It	Nature of Benefit
Improved Service Reliability and Continuity	• DLIs 1, 2, 3, 4, and 5. • NRW reduction (leak detection and repair; pressure management; loss control measures). • Metering.	Improved reliability and continuity of water services, including a reduction in leaks, fewer unplanned service interruptions, and more consistent water pressure across the distribution network.
Enhanced Public Health and Hygiene	• DLI 4. • Sewer pilot.	Reduction of sewage overflows, safer disposal of fecal sludge, and improved sanitation access.
Job Creation (short term)	• DLIs 1, 3, and 4. • NRW reduction (leak detection and repair; pressure management; and loss control measures). • Metering. • Sewer pilot.	Potential generation of temporary employment opportunities during construction activities. Given that most works are carried out by BWA staff, the scale of this benefit may be limited, and the creation of permanent jobs is not anticipated.
Better Jobs through Capacity Building	• DLIs 1, 2, 3, 4, and 5 • Technical and leadership training programs for staff. • Equitable policies and improved work environment.	Strengthened staff capacities and technical skills, enabling personnel to perform higher-level functions, assume greater responsibilities, and access better professional opportunities. Promotion of a female friendly, safe work environment (lactation rooms, separate bathrooms), staff trained on respectful, harassment-free behavior.
Climate and Disaster Resilience	• DLIs 1, 4, and 5. • NRW reduction (leak detection and repair; pressure management; loss control measures). • Metering. • Sewer pilot.	Strengthened resilience of water and sanitation services to climate variability and extreme events. Enhanced system efficiency and reliability. Populations are less exposed to service disruptions during extreme climate- and disaster-related events.
Improved Customer Service and Transparency	• DLIs 1, 2, 3, and 5. • Metering and data flow to billing systems. • Operational modernization and digitalization of commercial services (Customer Information System / Commercial Management System),	More transparent billing, reduced disputes, and stronger consumer confidence in water charges.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)
Environmental and Social Systems Report – Annex 2

Potential Social Benefit	DLIs and Activities Likely to Enable It	Nature of Benefit
Gender Empowerment and Inclusion	• DLI 3. • Promotion of gender inclusion in technical and managerial positions within BWA through capacity building and institutional policies.	Enhanced professional development and career advancement opportunities for women within BWA; greater gender balance in technical and managerial positions; improved organizational culture, diversity, and institutional performance.
Tourism and Local Economic Benefits from Improved Water Security	• DLIs 1, 4, and 5. • NRW reduction; sewer pilot (pollution control); service reliability.	Indirect, longer-term gains in tourism competitiveness and related jobs from reliable potable water and improved wastewater management.



ANNEX 3 – POTENTIAL ENVIRONMENTAL AND SOCIAL RISKS, IMPACTS, AND MITIGATION MEASURES

Potential Environmental Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Vegetation Loss (minor)	• DLIs 1, 4, and 5. • Sewer pilot.	Trenching and pipe-laying may require trimming or removal of roadside vegetation, ornamental plants, or small trees located along rights-of-way. This could affect local aesthetics, shade provision, and in some cases contribute to minor soil exposure or erosion.	• Conduct pre-construction surveys to identify and mark vegetation along rights-of-way. • Adjust trench alignment or excavation methods to avoid vegetation removal where feasible. • Limit trimming to the minimum necessary and avoid full removal unless absolutely required. • Restore disturbed areas after works, including replanting ornamental species or replacing removed trees. • Preserve and reuse topsoil during excavation to promote regrowth and reduce erosion. • Obtain permits and ensure compensatory replanting at a minimum ratio of two new trees for each protected or mature tree removed, prioritizing native or ecologically appropriate species, and maintain replacements until an 80–90 percent survival rate is achieved within 24 months. • Inform nearby residents and businesses about vegetation impacts and reinstatement measures.
Minor Soil Disturbance and Erosion	• DLIs 1, 4, and 5. • Sewer pilot.	Small-scale excavation for short pipe sections may cause limited soil disturbance or minor displacement of surface materials. Given the shallow depth and urban or previously modified settings, the potential for erosion or sediment runoff is low and localized, mainly during rainfall if trenches remain open. Proper sequencing and prompt reinstatement are sufficient to control these effects.	• Plan excavation during dry weather where possible. • Keep excavated soil stockpiled neatly away from drains or water channels. • Backfill and compact trenches promptly after pipe installation. • Use simple erosion-control measures (e.g., temporary barriers or silt fences) in sensitive areas. • Re-vegetate or restore disturbed verges and surfaces soon after completion. • Inspect reinstated areas after rainfall to address any localized erosion or settlement.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)
Environmental and Social Systems Report – Annex 3

Potential Environmental Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Dust, Noise, and Vibration (localized)	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Trenching, pipe laying, and movement of machinery and materials can generate dust emissions, elevated noise levels, and ground vibrations.	• Lightly spray exposed soil or excavation areas during dry or windy conditions to reduce dust. • Cover or dampen stockpiles and truck loads of excavated material during transport. • Limit work to daytime hours in residential or sensitive areas to minimize disturbance. • Use well-maintained equipment fitted with silencers or mufflers to reduce noise and vibration. • Avoid prolonged idling of machinery and sequence works to shorten exposure periods. • Notify nearby residents or institutions in advance of any short periods of elevated noise or vibration.
Minor Water Pollution Risks (sediment, debris, wastewater from pipe flushing)	• DLIs 1, 4, and 5. • Sewer pilot.	Improper disposal of spoil or flushing water may temporarily affect roadside drains. Impacts are short-term, localized, and unlikely to affect broader ecosystems.	• Collect and properly dispose of wastewater generated from pipe flushing in designated drains or disposal points approved by authorities. • Install silt fences, sediment traps, or barriers near drainage inlets to prevent runoff of spoil or debris. • Avoid direct discharge of flushing water, sediment, or waste into storm drains, waterways, or coastal areas. • Schedule flushing and dewatering activities during low-flow or dry weather periods to minimize downstream impacts. • Train contractors and crews on proper wastewater handling, disposal practices, and emergency spill response.
Small-Scale Solid, Construction, and Hazardous Waste Generation (old pipes, excavation spoil, packaging, broken meters, lubricants, asbestos fragments)	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Works will generate limited amounts of construction and solid waste, including spoil, old pipes, packaging, and broken meters, as well as small quantities of hazardous materials (oils, lubricants, possible asbestos).	• Segregate construction, solid, and hazardous waste at the worksite. Collect and transport waste to approved SSA or municipal disposal facilities. • Store hazardous waste (oils, asbestos fragments, lubricants) safely in sealed, labelled containers. • Train contractors on safe handling of hazardous and construction waste. • Reuse or recycle materials (e.g., pipes, packaging) where feasible. • Keep records of waste volumes and disposal routes. • Ensure appropriate final disposal of all waste streams through certified facilities or authorized export when required.



Potential Environmental Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Production Well Integrity and Water Quality Risks from Monitoring Installation	• DLIs 1 and 5. • Metering • Installation of well monitoring equipment.	If not properly designed/installed, instrumentation can compromise sanitary seals, introduce ingress pathways, create cable abrasion points, obstruct flow, or disturb the well column, leading to transient turbidity, biofouling risk, or interference with pumps/liners.	• Prefer the use of dedicated observation wells to install monitoring devices and avoid interference with production wells. • Maintain or upgrade sanitary seals and use waterproof cable connectors when instruments are installed in production wells. • Disinfect wells before and after installation and ensure that all materials and equipment are suitable for potable water use. • Secure and route cables properly to prevent damage, tangling, or interference with the pump and maintenance tools. • Conduct performance and water quality checks before returning the well to service to confirm safe operation. • Keep updated records of the installation, inspect seals and cables regularly, and plan for maintenance or safe replacement when needed. • Schedule works during periods of low demand and ensure standby supply arrangements to minimize service interruptions.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)
Environmental and Social Systems Report – Annex 3

Potential Social Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Temporary Access Disturbances (sidewalks, driveways, shop entrances blocked)	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Minor and short-term access disturbances may occur during trenching or excavation along sidewalks or near property entrances. Impacts are temporary, lasting from a few hours to a few days per site. In dense urban areas, disturbances are more likely due to restricted space and high pedestrian or commercial activity.	• Provide safe walkways, ramps, or bridging planks to maintain access to homes, shops, and community facilities. • Inform residents, businesses, and community institutions at least 48–72 hours in advance about timing, duration, and access arrangements. • Schedule works outside peak business hours in commercial areas where feasible, to reduce disruption to shops and vendors. • Install clear, accessible signage indicating detours, business access points, and expected duration of works. • Ensure affected individuals and businesses have access to a responsive grievance mechanism to report access issues or damages. • Include contractual clauses requiring contractors to minimize access disruption and reinstate sidewalks/driveways immediately after works are completed.
Minor and Short-term Business Livelihood Disruption (loss of customers, disturbance of vendors)	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Caused by reduced access, noise, dust, and visual obstruction near small shops and roadside vendors. Temporary, linked to construction period. Probability is medium to high in commercial strips where businesses depend on foot traffic.	• Hold short briefings with shop owners and vendors before works start to agree on phasing and hours of work. • Use signage, fencing with transparent panels, or temporary banners to indicate that businesses remain open during construction. • For roadside or informal vendors directly in trenching areas, provide temporary designated vending spots nearby to maintain customer flow. • Prioritize restoring sidewalks, driveways, and shop entrances in business districts to minimize revenue losses. • Include business disruption indicators (e.g., number of complaints, number of businesses temporarily inaccessible) in monitoring plans to adjust mitigation in real time.
Minor and Short-term Traffic and Mobility Impacts (lane closures, congestion, pedestrian safety issues)	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Caused by partial road closures, detours, and equipment occupying lanes.	• Require contractors/BWA to prepare Traffic Management Plans (TMPs) for each work site, approved by the Ministry of Transport and the Police, including detours, signage, and coordination with public transit routes. • Conduct works during off-peak hours, weekends, or nighttime in congested areas (Bridgetown, coastal roads) to reduce disruption. • Provide well-marked, barrier-protected pedestrian pathways, safe crossings, and access ramps around worksites, especially near schools, hospitals, and commercial strips.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)
Environmental and Social Systems Report – Annex 3

Potential Social Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Cultural Heritage Risks	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Excavation works may occasionally encounter subsurface cultural materials or take place near culturally sensitive sites such as cemeteries or churchyards. These risks are unlikely given that interventions will occur mainly within already disturbed or previously developed areas, and excavation depths are limited.	• Check alignments against the Physical Development Plan, known heritage inventories, churchyards/cemeteries, and designated conservation areas; consult the competent heritage authority early to confirm constraints and permit needs under applicable law. • Chance-find procedure (mandatory): Immediate stop-work in the discovery area; secure a 30–50 m cordon; notify BWA site lead and heritage authority within 24 hours; prohibit removal/handling; resume only upon written clearance and any required mitigation (recordation, salvage, redesign). Chance finds procedure to be included in contracts with contractors and supervising firms.
Occupational Health & Safety Risks (excavation hazards, underground utilities, traffic exposure, machinery, and heat stress)	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Work activities involve small-scale excavation, utility connections, and equipment operation along public roads or sidewalks, which may expose workers to traffic hazards, moving machinery, and high temperatures during outdoor work. Potential impacts include acute injuries or accidents, as well as temporary effects such as heat stress or dehydration. Probability is high without strict OSH standards, continuous supervision, provision of Personal Protective Equipment (PPE), and climate-adapted safety measures (hydration, shade, rest cycles).	• Require pre-excavation surveys (ground-penetrating radar, utility maps) and coordination with utility providers to prevent strikes on underground infrastructure. • Mandate shoring, sloping, or trench boxes for excavations >1.2m; ensure secured access/egress ladders and daily trench inspections by a competent person. • Implement Traffic Management Plans with barriers, reflective signage, night lighting, and flaggers; coordinate with the police/transport ministry for detours. • Adopt climate-adapted measures: shaded rest areas, hydration schedules, acclimatization for new workers, and work-rest cycles aligned with peak heat periods. • Emergency preparedness: Require contractors to maintain first aid kits, trained first-aid responders on site, and site-specific emergency response plans (including utility strikes, collapses, or heat-related illness). • Worker orientation & toolbox talks: Conduct mandatory induction for all workers on excavation hazards, machinery safety, PPE use, traffic risks, and heat illness prevention. Daily toolbox talks to reinforce key safety points. • Continuous supervision & reporting: Designate OSH officers on contractor teams, with daily inspection logs shared with BWA supervisors; establish mandatory incident/near-miss reporting.



Potential Social Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Community Health and Safety Risks	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Key hazards include unsecured trenches, excavation pits, construction machinery, and inadequate fencing or signage, which can result in accidents and injuries, particularly for children and pedestrians in dense neighborhoods. Interaction between workers and community members may also create risks of Sexual Exploitation and Abuse and Sexual Harassment (SEA/SH). Dust emissions from excavation and vehicle movement can reduce local air quality and visibility, while construction-related noise and vibration may disturb residents and sensitive receptors, such as schools, healthcare facilities, older persons, and individuals with disabilities or conditions such as autism who may be highly sensitive to sensory disturbances.	• Worksite safety and access control: Install secure fencing, barricades, and warning signage around open trenches, pits, and machinery; provide safe pedestrian walkways, crossings, and lighting in dense neighborhoods. • Traffic management: Coordinate with traffic police and local authorities to prepare and implement traffic management plans, including detours, flaggers, and restricted vehicle movement near schools, clinics, and markets. • Dust, noise, and vibration control: Use dust suppression (e.g., water spraying), noise barriers or scheduling of works during less sensitive hours, and equipment maintenance to minimize community disturbance. • Vector and sanitation management: Prevent stagnant water in trenches and spoil piles; ensure timely backfilling or covering to avoid mosquito breeding and public health risks. • Waste handling: Collect and dispose of spoil, old pipes, and packaging promptly through Sanitation Service Authority (SSA)–approved systems; avoid informal dumping near residences or drains. • Community engagement: Provide advance notice of works (scope, duration, impacts), clear signage, and direct communication channels (hotline, SMS, community liaison) to inform residents and businesses. • SEA/SH: Finalize BWA’s draft SH Policy Require contractor/BWA Codes of Conduct, worker training, and awareness measures; establish confidential grievance pathways and referral mechanisms for survivors. • Sensitive groups inclusion: Adapt communication and safety measures for children, older persons, persons with disabilities, and individuals with conditions like autism (e.g., quieter hours near schools/clinics, clear visual signage).



Potential Social Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Digital Divide / Exclusion of Vulnerable Users (difficulty accessing new digital tools or services)	• DLIs 2, 3, and 5. • Operational modernization and digitalization of commercial services (Customer Information System / Commercial Management System)	The reliance on digital platforms for customer service functions may create barriers for certain groups, including older persons, low-income households, or individuals with limited digital literacy or internet access, potentially restricting their ability to fully access program benefits and services.	• Maintain hybrid service delivery: Ensure that physical service counters, phone hotlines, and paper-based options remain available alongside digital systems, particularly for bill payment and grievance redress. • Inclusive design of digital platforms: Require that online portals and mobile applications follow international accessibility standards (e.g., WCAG), with compatibility for screen readers, large-print modes, and simplified navigation. • Targeted outreach and training: Provide community workshops, simple guides, and awareness campaigns to help older persons, low-income households, and persons with disabilities access and use digital services. • Mobile-friendly solutions: Prioritize smartphone-based services (low-data use) rather than computer-only portals, recognizing that low-income households may rely exclusively on mobile access. • Equity monitoring: Track usage data (disaggregated by gender, age, location, and disability where feasible) to identify groups underrepresented in digital access and adapt strategies accordingly. • Fallback mechanisms during outages: Maintain clear procedures for manual handling of complaints, billing, and service requests when digital systems are down.
Concerns from Private Users and Operators	• DLIs 2, 3, 4, and 5. • Operationalizing the Water and Wells Management Committee (WWMC). • Wells census and database. • Metering.	As the WWMC and Wells Management Committee and the Water Resources and Environmental Management Unit become operational, some private users may express concern about new data requests, registration, or monitoring procedures related to well abstraction. The overall likelihood of conflict is low, provided that communication and guidance are clear.	• Early engagement: Communicate objectives and procedures clearly to private users during implementation. • Transparency: Publish user-friendly guidance materials on data submission and monitoring requirements. • Capacity support: Offer information sessions and technical assistance for users unfamiliar with reporting processes. • Feedback channels: Maintain accessible channels for questions, concerns, and feedback during rollout.



Potential Social Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Disruption and Elimination of Informal or Unmarked Connections	• DLIs 1, 4, and 5. • NRW reduction measures. • Metering.	Caused by removal of illegal/unrecorded household or business connections during works. Impacts are temporary but can translate into permanent disconnection if not regularized.	• Pre-works survey: Require contractors and BWA field staff to identify and document informal/unmarked connections before trenching or replacement works. • Notification and communication: Provide advance notice to households and businesses with suspected informal connections, explaining potential impacts and available options. • Regularization pathways: Establish a simplified, affordable mechanism for users to formalize connections (e.g., temporary permits, fee waivers, or phased payments). • Temporary service continuity: Where feasible, provide interim measures to ensure households are not left without water during regularization. • Coordination with GRM: Allow affected users to raise concerns through the grievance redress mechanism, ensuring accessibility for vulnerable groups. • Monitoring: Require reporting of all disconnections and their resolution status as part of contractor supervision and BWA oversight.
Service disruptions due to underground Utility works	• DLIs 1, 4, and 5. • Leak repair. • Metering. • Sewer pilot.	Risk of accidental damage to unmarked or poorly mapped underground infrastructure, including water, sewer, telecommunications, or electrical lines.	• Consultation with utility providers (telecom, electricity, sewerage, water) before works to obtain the most updated maps and records. • Pre-excavation surveys: Conduct ground-penetrating radar (GPR) or other detection methods in high-risk or poorly documented areas to identify hidden infrastructure. • Controlled excavation methods: Apply safe digging techniques (manual excavation, trial pits) in areas where utilities are suspected, avoiding heavy machinery until confirmation. • Emergency response protocols: Develop procedures for immediate containment, repair coordination, and public safety in case of a strike (e.g., gas leaks, power outages, flooding). • Communication and notification to affected communities and businesses, with rapid updates if service disruptions occur. • Train contractors and BWA crews on utility strike prevention, reporting, and emergency response measures. • Ensure contracts include clear provisions for liability, compensation, and rapid repair costs in case of accidental strikes.



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Barbados Water Security and Sector Performance Program (P508674)
Environmental and Social Systems Report – Annex 3

Potential Social Risk / Impact	DLIs and Activities Likely to Cause It	Impact/Risk Description	Mitigation Measures
Potential opposition or disruption among BWA staff due to organizational reforms	• DLIs 1, 2, and 3. • Organizational restructuring and introduction of new management and digital systems.	The introduction of external professionals, new performance-based management systems, and changes in roles or workflows may generate uncertainty, resistance, or temporary disruptions among BWA staff. Such dynamics could affect staff morale, cooperation, or the pace of reform implementation.	• Early and transparent communication with BWA staff on reform objectives, scope, and expected benefits. • Active participation of staff representatives and unions in reform planning and monitoring. • Capacity-building and on-the-job training programs to strengthen ownership and facilitate knowledge transfer from external professionals. • Inclusion of change management and staff engagement strategies and institutional strengthening plan.



ANNEX 4 – NATIONAL LEGAL FRAMEWORK

Law / Regulation / Strategy	Key Provisions	Applicability to the Program
Environmental, Land Use, and Heritage Legislation		
Planning and Development Act 2019 (as amended 2020) and Environmental Impact Assessment (EIA) Regulations 2021 (S.I. 88)	Establish the Planning and Development Board and procedures for development control; require environmental review for projects with potentially significant impacts; set rules for EIA preparation, review, and public participation; include provisions for cultural heritage, conservation areas, and tree preservation.	For Program activities (NRW reduction measures, metering improvements, small sewer pilots), full EIAs are generally unlikely given their small scale and location within existing rights-of-way. However, an Initial Environmental Evaluation (IEE) or EIA may still be required if works are proposed near sensitive zones (e.g., coastal areas, heritage sites, conservation areas, or high groundwater zones).
Environment Protection Act, 1997	Provides the framework for environmental management and pollution control; empowers authorities to regulate emissions, waste, and natural resources.	Program activities generate construction waste and possible hazardous substances (e.g., oils, asbestos) and must comply with pollution prevention and waste-handling rules.
Trees (Preservation) Act, 1981	Authorizes the designation of individual trees or groups of trees for legal protection through preservation orders. Prohibits cutting, lopping, topping, or uprooting without prior consent from the relevant authority. Requires to get a permit from the Chief Town Planner to “kill” a tree (applies to trees with circumference ≥ 1 m measured ≥ 0.5 m above ground). Any permit can carry conditions/restrictions, and the Chief Town Planner may also issue a notice requiring planting or replanting specifying the size, species, and location, plus maintenance/protection. The Act does not prescribe a numerical ratio (e.g., 1:1 or 2:1). The Act also allows requiring a monetary deposit as security to ensure the planting/landscaping conditions are carried out.	For Program works, if trenching necessitates removing roadside or urban trees (including those under a TPO), BWA/contractors must obtain the relevant permit(s) and comply with any conditions imposed; replanting or other mitigation applies only when specified.
Soil Conservation (Scotland District) Act (Cap. 396)	Establishes soil conservation controls in the Scotland District, a highly erosion-prone area in northeast Barbados. Regulates clearing, excavation, and development activities, requiring permits and conservation practices.	If Program works extend into the Scotland District, permits and specific erosion control measures are mandatory. If works remain outside the District, no obligations are triggered.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)

Environmental and Social Systems Report – Annex 4

Law / Regulation / Strategy	Key Provisions	Applicability to the Program
Coastal Zone Management Act, 1998 (Cap. 394)	Provides for integrated management of coastal resources and establishes the Coastal Zone Management Unit (CZMU). Regulates coastal works, erosion control, and conservation of marine habitats. Requires permits for activities affecting coastal waters, beaches, or reefs.	If Program works (e.g., sewer pilots) occur in or near coastal areas, permits and mitigation measures may be required to avoid erosion, sediment runoff, or marine pollution. Otherwise, inland activities may not trigger obligations.
Physical Development Plan (Amendment), 2023	Statutory national land-use plan under the Planning and Development Act. Guides zoning, land allocation, and development priorities, including protected areas and infrastructure corridors.	Provides spatial planning framework that must be observed in siting Program works, especially where they intersect with designated water protection zones, conservation areas, or urban corridors.
National Conservation Commission Act (Cap. 393)	Establishes the Commission responsible for management and maintenance of public parks, beaches, open spaces, and conservation areas. Authorizes rules for protection of natural resources in these areas.	If Program works intersect with parks, beaches, or public open spaces, coordination with the Commission is required to ensure restoration and protection.
Third-Party Utility Interference		
Electric Light & Power Act, Cap. 278 (electric lines/cables)	Works crossing or in proximity to licensed electricity infrastructure must follow the licensee's safety standards and obtain the licensee's written consent. Crossings/encroachments are typically authorized via permit conditions tied to planning approval or through a bilateral wayleave/easement with the licensee.	Before works overhead/underground power lines, BWA/contractors must coordinate with the electricity licensee, secure written consent/wayleave, comply with clearance and protection requirements, and implement the licensee-approved method statements and safety controls.
National Petroleum Corporation Act, Cap. 280 (gas lines)	Any works near NPC gas pipelines require NPC's prior consent and compliance with NPC safety requirements (e.g., minimum clearances/cover, protection measures, supervised crossings, and approved construction methods). Authorization may be issued via permit/wayleave with specified conditions.	For crossings or works within/near NPC rights-of-way, BWA/contractors must notify NPC, obtain written consent, follow NPC's technical and safety conditions (including standby/supervision if required), and incorporate these requirements in site-specific method statements and contractor EHS plans.



Law / Regulation / Strategy	Key Provisions	Applicability to the Program
Road Traffic Act, Cap. 295 (as amended, incl. Act 2018-31)	Governs works affecting public roads and traffic. Requires authorization to occupy/excavate carriageways, shoulders, or sidewalks; mandates temporary traffic management plans per directions of the competent road authority (e.g., signage, barriers, lighting, speed control, flaggers, safe pedestrian detours, access to premises). Obligates prompt reinstatement of road and footpath surfaces and allows enforcement/stop-work for non-compliance.	Before works in/along public roads, BWA/contractors must secure the necessary road-opening/occupation approval, prepare and implement a site-specific Traffic Management Plan, coordinate with the road authority (and police/emergency services as required), maintain safe access for pedestrians/businesses, and restore the roadway/sidewalks to the required standard.
Cultural Heritage Protection		
Preservation of Antiquities and Relics Act, 2011	Provides for the identification, protection, and preservation of antiquities, relics, and cultural property of historical significance. Prohibits excavation, removal, or disturbance of relics without authorization from the Minister or designated authority. Requires reporting and safeguarding of any finds.	If trenching, excavation, or other Program works uncover relics or antiquities, BWA and contractors are legally obligated to halt works, notify authorities, and comply with prescribed protection measures. Ensures cultural heritage safeguards are integrated into construction activities.
Water and Sanitation		
Barbados Water Authority Act, 1980 (Cap. 274A)	Establishes the Barbados Water Authority (BWA) as the statutory body responsible for managing, developing, and supplying water resources and sewerage services; empowers BWA to regulate abstraction, issue licenses, and manage water and wastewater infrastructure.	The Program is implemented through BWA. The Act provides the legal mandate for activities such as NRW reduction, metering, and sewer pilots, as well as institutional authority for planning, regulation, and enforcement.
Utilities Regulation Act, 2002 (Cap. 282), as reinforced by the Barbados Water Authority (Amendment) Act, 2010 (No. 4)	Establishes the Fair Trading Commission (FTC) as the independent regulator for utilities. The 2010 Amendment clarified that BWA tariffs and charges fall under FTC oversight, inserting section 29A in the BWA Act. Provides mechanisms for addressing complaints and ensuring transparency in utility operations.	Places the BWA under the regulatory oversight of the FTC, which approves tariffs and monitors service quality and consumer protection.
Barbados Water Authority (Water Services) Regulations, 1982 (Cap. 274A)	Set obligations for BWA and consumers regarding water distribution. Cover supply duties, connection costs, metering standards, billing and payment rules, service discontinuation, and consumer responsibilities (e.g., avoiding waste, illegal connections).	Regulations ensure compliance with standards for meters, billing cycles, and continuity of supply, shaping how improvements affect customers.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)

Environmental and Social Systems Report – Annex 4

Law / Regulation / Strategy	Key Provisions	Applicability to the Program
Barbados Water Authority (Amendment) Act, 2023	Expands definitions (e.g., groundwater, water resources). Revises BWA functions, powers, and governance. Establishes the Water and Wells Management Committee. Introduces provisions on water zones, public supply areas, and links to the Planning and Development Act for water protection zones.	Strengthens BWA's mandate for water resource management and protection, especially groundwater and supply zones. Program activities must align with these provisions, particularly where works occur in designated water zones.
Barbados Water Authority (Water and Sewerage Rates) (Amendment) Regulations, 2019	Establishes tariffs for water supply and sewerage services provided by BWA, including different rates by category of use.	Any tariff adjustments remain governed by these regulations and overseen by the FTC.
Sewerage Regulations, 1980 (S.I. No. 151)	Regulates how domestic and industrial premises must connect to and use the public sewer system; prohibits discharges of industrial wastewater without permission; requires submission of construction plans and approval for sewerage works; controls types of discharges (e.g. grease, noxious substances), and ensures private sewers comply with public system standards.	Pilot sewer works must be designed, submitted, and approved under BWA's standards; industrial or non-domestic discharges must receive permission; and private sewer connections upgraded must meet system compliance.
Health Services (Building) Regulations, 1969	Prohibits discharge of sullage or waste matter into streets or public places without approval of the Medical Officer of Health. Establishes sanitary standards for waste disposal in buildings and public health oversight.	Program sewer pilot activities must comply with these rules, ensuring that wastewater and sullage are properly managed through approved systems. Supports alignment with public health requirements and minimizes risks of unsanitary discharges during works.
Health Services Act, 1969 (Cap. 44)	Establishes the legal framework for public health in Barbados. Empowers the Minister and Chief Medical Officer to regulate matters affecting health, including sanitation, waste disposal, drainage, and prevention of communicable diseases. Provides authority to make subsidiary regulations (e.g., Health Services (Building) Regulations).	Program activities that involve wastewater management, sewer connections, or trenching must align with public health standards under this Act. It provides the overarching authority for sanitary controls that directly intersect with sewer pilots and safe disposal of waste during works.
Health Services (Nuisances) Regulations, 1969	Defines nuisances injurious to health (e.g., accumulation of waste, foul water, defective drains, insanitary premises). Empowers health authorities to abate nuisances and enforce sanitary standards.	Program works must avoid creating nuisances such as stagnant water, blocked drains, or poor waste disposal during construction. Ensures alignment with public health obligations and reduces risk of community grievances.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)

Environmental and Social Systems Report – Annex 4

Law / Regulation / Strategy	Key Provisions	Applicability to the Program
Health Services (Disposal of Offensive Matter) Regulations, 1969	Regulates safe handling, transport, and disposal of offensive matter (e.g., refuse, waste, animal remains). Requires permits or approvals from health authorities and sets conditions to prevent risks to public health and the environment.	During Program works (e.g., sewer pilots, trenching), excavated waste, sludge, or other offensive matter must be disposed of in accordance with these regulations. Compliance helps minimize health risks and community complaints.
Health Services (Collection and Disposal of Refuse) Regulations, 1975	Provides rules for collection, storage, and safe disposal of refuse to protect public health. Requires approval of disposal methods and prohibits indiscriminate dumping. Empowers health authorities to enforce compliance.	Program activities that generate solid waste (e.g., excavated materials, packaging, old pipes) must follow approved collection and disposal methods. Supports proper waste management and reduces risks of illegal dumping or health nuisances during works.
Sanitation Service Authority Act, 1974	Establishes the Sanitation Service Authority (SSA) to manage waste collection, treatment, and disposal, including landfill operations.	Excavated spoil, old pipes, and packaging from Program works must be disposed of through SSA-approved systems.
Water Reuse Act, 2023 (Act 2023-10)	Creates a permit system to produce, supply, or sell reclaimed (non-potable) water. Establishes inspection, enforcement, suspension/revocation, offences, and administrative penalties. Establishes a Water Reuse Committee to advise on permits and compliance. Defines allowable uses of reclaimed water (e.g., irrigation, equipment cooling, groundwater recharge). EPD conducts inspections and environmental monitoring.	Relevant if any Program activity produces, supplies, stores, transports, or uses reclaimed wastewater for a beneficial purpose. This could include using treated effluent for construction activities such as dust suppression, site irrigation, or equipment washing, or if the sewer pilot involves treatment and reuse of wastewater for irrigation, industrial, or other non-potable uses.
Workplace Safety and Employment Rights		
Safety and Health at Work Act, 2005 (Cap. 356A), as amended in 2022, and subsidiary regulations (e.g., Workplace Sanitary Conveniences, PPE, and General Duties Regulations, 2007)	Barbados' main occupational safety and health framework. Establishes employer duties to provide safe working conditions, conduct risk assessments, train workers, and supply protective equipment. The 2022 Amendment expanded obligations, worker rights, and inspector powers, aligning with ILO C155 and C187. Subsidiary regulations set practical requirements: adequate sanitary facilities, mandatory PPE use, and general duties of employers and employees to ensure workplace safety.	Applies to all Program works carried out by BWA and contractors. Covers Program-specific risks such as traffic exposure, handling hazardous materials (e.g., oils, asbestos), operation of heavy equipment, and heat stress. Requires preventive safety measures, training, PPE provision, sanitary facilities, and compliance with inspector directives.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)

Environmental and Social Systems Report – Annex 4

Law / Regulation / Strategy	Key Provisions	Applicability to the Program
Workplace (Noise) Regulations, 2007	Subsidiary regulation under the Safety and Health at Work Act. Establishes permissible noise exposure levels, requires monitoring and controls, and obliges employers to provide hearing protection and training.	Construction works will generate noise. Contractors must comply with exposure limits, monitoring, and mitigation (e.g., PPE, scheduling, barriers).
Employment Rights Act, 2012-19	Establishes fundamental labor rights, including written employment terms, protection against unfair dismissal, redundancy pay, and conditions of service.	Ensures fair labor standards for BWA staff and contractors. Aligns with ESSA Core Principle 3 (labor and working conditions), though not specific to construction or water works.
Accidents and Occupational Diseases (Notification) Act, 1983	Requires employers to notify authorities of workplace accidents and occupational diseases, enabling monitoring and enforcement of worker protection.	Directly applicable: Contractors and BWA must report incidents during Program works involving trenching, equipment operation, or hazardous materials.
National Insurance and Social Security Act (Cap. 47)	Provides the framework for Barbados' national insurance scheme, covering pensions, sickness, maternity, unemployment, and employment injury benefits.	Ensures that workers employed under the Program are covered for social protection and employment injury benefits. Compliance falls on employers/contractors.
Land acquisition and tenure		
Limitation and Prescription Act (Cap. 232)	Establishes the conditions under which private rights over land and water may be acquired by long-term use ("prescription"). A right of way, easement, or use of a watercourse can be legally claimed after 20 years of continuous and uninterrupted use, and becomes indefeasible after 40 years. Defines what counts as interruption of use, exceptions (e.g., persons under disability), and evidentiary standards.	The Program's interventions are short-term maintenance, upgrades, or installation works, not claims over private land rights or establishing easements. Unless a project crosses land where prescriptions or easements are disputed (e.g. laying pipes across third-party land where there is no formal grant), this law won't typically affect normal Program works. However, it's good to be aware of, in case any works involve crossing or using private lands, pathways or rights of way, to avoid inadvertent legal conflicts.
Land Acquisition Act, Cap. 228	Governs compulsory acquisition of private land by the Crown for public purposes. Requires notice, inquiry, and compensation to affected landowners. Compensation is based on market value and determined by the courts if contested.	Relevant only if Program works require land outside existing public rights-of-way, which is excluded. Provides the legal framework for compulsory acquisition, though most works are expected to remain within established corridors.
Land Registration Act, Cap. 229 (2002)	Establishes a land title registration system, regulates land transfers, and provides legal certainty for ownership and interests in land. Aims to reduce disputes and fraud.	Ensures clarity of ownership for any lands where Program works may occur. Helps identify legitimate landowners and avoid disputes during easement or temporary access arrangements.
Disaster Risk Management		



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)

Environmental and Social Systems Report – Annex 4

Law / Regulation / Strategy	Key Provisions	Applicability to the Program
Catastrophe Fund Act, 2020	Establishes a fund to provide financial aid to eligible persons and qualifying businesses adversely affected by catastrophes, including natural disasters and other major shocks, enabling the State to mitigate or remedy adverse effects.	It provides a national financial safety net that could be mobilized in the event that Program activities or infrastructure are affected by extreme weather events.
Emergency Management Act, 2007 (Cap. 160), as amended in 2020	Establishes the Department of Emergency Management (DEM) and the national disaster management system. Provides legal authority for preparedness, mitigation, response, and recovery. The 2020 amendment strengthened coordination and response powers during emergencies.	BWA and contractors must comply with DEM directives during hurricanes, floods, or other disasters. Includes obligations to secure worksites, protect infrastructure, and follow official emergency orders.
Gender and vulnerable groups		
Sexual Offences Act (Cap. 154)	Establishes criminal offences for sexual violence, exploitation, and abuse. Provides the legal framework for prosecution of SEA/SH cases	Establishes criminal offences for sexual violence, exploitation, and abuse. Provides the legal framework for prosecution of SEA/SH cases
Domestic Violence (Amendment) Act, 2016-2	Expands protections against domestic violence to include physical, sexual, emotional, financial, and child abuse. Provides for protection orders and support services for survivors.	Strengthens the social protection context in Barbados and supports integration of SEA/SH prevention and survivor referral mechanisms in Program safeguards.
Sexual Harassment in Employment (Prevention) Act, 2017-21	Prohibits sexual harassment in workplaces. Defines employer obligations, establishes complaint procedures, and empowers the Employment Rights Tribunal to hear cases.	Requires workplace policies and grievance channels for SH, aligning with World Bank Core Principle 5 on protecting vulnerable groups.
Employment (Prevention of Discrimination) Act, 2020	Prohibits discrimination in employment based on sex, sexual orientation, disability, HIV status, religion, and other protected categories.	Relevant to Program labor management: Contractors and BWA must ensure non-discriminatory hiring and employment practices.
Barbados 2023 Human Rights Report (U.S. State Department)	Notes that sexual harassment is prohibited under law, but enforcement gaps persist and no high-profile cases were reported in 2023.	Highlights enforcement challenges, underscoring the need for Program-level safeguards (training, awareness, grievance redress) to go beyond formal compliance.
Discrimination (Persons with Disabilities) Act, 2020	Prohibits discrimination on the basis of disability in employment, education, and access to services. Requires reasonable accommodation to ensure equal participation. Establishes mechanisms for complaints and enforcement.	Ensures that persons with disabilities are not excluded from employment opportunities under the Program, and that service delivery (e.g., metering, billing, grievance mechanisms) remains accessible.
National Strategies		



Law / Regulation / Strategy	Key Provisions	Applicability to the Program
Integrated Solid Waste Management Program (ISWMP)	A national program (not legislation) supported by international partners to modernize Barbados' solid waste management system. Focus areas include improved landfill operations, recycling, hazardous waste handling, and institutional strengthening.	The Program may generate construction waste (old pipes, excavation spoil, packaging), which should align with ISWMP objectives for safe disposal and sustainable waste practices (not legally binding).
Comprehensive Disaster Management (CDM) Policy, 2022	A national policy framework aligning with the regional CDEMA strategy. Not legally binding but provides strategic direction for integrating disaster risk reduction and climate resilience in the water sector. Program design and operations should be consistent with its principles.	Not legally binding, but relevant as strategic guidance. Program activities should integrate CDM principles by ensuring climate- and disaster-resilient design, incorporating emergency preparedness (e.g., securing worksites before storms), and aligning with national and regional resilience strategies.



ANNEX 5 – INTERNATIONAL LEGAL FRAMEWORK

Law / Regulation	Key Provisions	Applicability to the Program
Convention for the Protection and Development of the Marine Environment of the Wider Caribbean Region (Cartagena Convention), 1983	Regional treaty ratified by Barbados; provides a legal framework for the protection and sustainable development of the marine and coastal environment in the Wider Caribbean; includes protocols on land-based sources of marine pollution, specially protected areas and wildlife (SPAW), and oil spill response.	The Program's activities are land-based and small in scale. Direct impacts on marine ecosystems are unlikely, but works in coastal areas or with drainage/wastewater discharges could indirectly affect coastal waters. The Cartagena Convention reinforces good practices for wastewater handling, sediment control, and stormwater management to prevent marine pollution.
Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal (1989)	International treaty regulating cross-border movement and disposal of hazardous waste. Requires prior informed consent of receiving countries, environmentally sound management of wastes, and restrictions on export to non-parties. Barbados acceded in 1994.	The Program will generate small quantities of hazardous waste (e.g., oils, hydrocarbons, possibly asbestos). As Barbados lacks full local treatment capacity; therefore, these must be securely stored, disposed of in controlled landfill cells, or exported under Basel procedures.
ILO C155 – Occupational Safety and Health Convention, 1981	International labor standards require a national policy on occupational safety and health (OSH). Obligates employers to ensure safe working environments, prevent accidents, and protect workers' health. Establishes principles for OSH systems, training, and worker participation. Barbados ratified in 2019.	The Program involves field works (e.g., pipe replacement, metering, and sewer pilots) that carry OSH risks such as exposure to traffic, handling of pressurized water systems, use of heavy equipment, and heat stress, among others. Compliance with OSH obligations is required for BWA and contractors.
ILO C187 – Promotional Framework for Occupational Safety and Health Convention, 2006	Establishes a framework for continuous improvement of occupational safety and health (OSH). Calls for a national OSH system with institutions, programs, data collection, and a culture of prevention. Promotes integration of OSH into national development policies. Barbados ratified in 2019.	Reinforces the requirement to adopt a preventive approach to OSH in Program activities (e.g., metering, sewer pilots). Supports development of training, monitoring, and safety culture to reduce risks such as traffic exposure, hazardous waste handling, and heat stress.
Montreal Protocol on Substances that Deplete the Ozone Layer, 1987	International treaty (ratified by Barbados in 1992) to phase out the production and consumption of ozone-depleting substances (ODS), such as CFCs, halons, and HCFCs. Includes obligations for control, reporting, and gradual substitution with ozone-friendly alternatives.	Procurement or use of certain refrigeration, air-conditioning, or fire suppression equipment during construction/operation must comply with Barbados' commitments under the Protocol.



THE WORLD BANK

Barbados Water Security and Sector Performance Program (P508674)

Environmental and Social Systems Report – Annex 5

Law / Regulation	Key Provisions	Applicability to the Program
Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW), 1979	Requires State Parties to eliminate discrimination against women in political, economic, social, cultural, and civil spheres. Obligates adoption of measures to ensure equality in employment, education, and public life. Calls for protection against exploitation and harmful practices.	Reinforces the need for gender-sensitive Program design and safeguards. Program contractors and BWA should ensure women are not excluded from jobs, training, or benefits, and that women in communities are consulted and protected from disproportionate risks during works.
Inter-American Convention on the Prevention, Punishment and Eradication of Violence Against Women (“Belém do Pará”), 1994	Regional treaty recognizing women’s right to live free from violence, including physical, sexual, and psychological violence in both public and private spheres. Obligates States to prevent, investigate, and sanction violence against women.	Strengthens the case for SEA/SH into Program activities. Supports adoption of Codes of Conduct for contractors, awareness training, and safe grievance redress mechanisms.
ILO Equal Remuneration Convention (No. 100, 1951)	Requires equal pay for men and women for work of equal value, without discrimination. Obligates states and employers to ensure wage equity across sectors and occupations.	BWA and contractors must ensure that women employed in Program activities (e.g., pipe laying, metering, administration) receive equal remuneration to men in equivalent roles.
ILO Discrimination (Employment and Occupation) Convention (No. 111, 1958)	Prohibits discrimination in employment and occupation based on sex, race, religion, political opinion, or other grounds. Requires states to promote equality of opportunity and treatment.	Contractors and BWA must adopt non-discriminatory hiring and employment practices for Program works, ensuring equal access for women and vulnerable groups.
ILO Violence and Harassment Convention (No. 190, 2019)	Framework to prevent, address, and remedy violence and harassment in the world of work. Covers physical, psychological, and sexual harassment, including gender-based violence. Requires prevention measures, complaints mechanisms, and victim protection.	Program contractors must implement Codes of Conduct, worker training, grievance mechanisms, and monitoring to prevent SEA/SH risks at worksites and in surrounding communities.
Ratification of ILO Recommendation No. 206 (2019)	International commitment to address violence and harassment in the world of work, complementing ILO Convention No. 190. Encourages integration of prevention measures into national systems.	It supports the adoption of preventive measures against workplace harassment, reinforcing the Program’s labor management requirements.



ANNEX 6 – NATIONAL SYSTEM CAPACITY TO RESPOND TO IDENTIFIED ENVIRONMENTAL AND SOCIAL RISKS

Potential Environmental Impact / Risk	System instruments to assess and manage the risk/impact	Addressing Identified Gaps
Vegetation Loss (minor)	Planning & Development Act (2019, as amended 2020) + EIA Regulations (2021): Enable screening and, where triggered, conditions of approval (e.g., tree protection, reinstatement/landscaping, erosion control) near sensitive sites. Trees (Preservation) Act (1981): Protects designated trees; requires consent for cutting/lopping/topping/uprooting; can require mitigation/replanting. Physical Development Plan (Amendment) 2023: Provides spatial guidance (conservation/green corridors, water protection areas) to flag locations where vegetation removal should be minimized and restoration specified. Soil Conservation (Scotland District) Act (Cap. 396): Requires permits and conservation practices for works in the erosion-prone District (relevant if alignments enter this area). Coastal Zone Management Act (1998): Where works are near coastal habitats, permits/conditions can require sediment control and protection of coastal vegetation. National Conservation Commission Act (Cap. 393): Coordination/consent for works within public parks/open spaces and requirements for site restoration. BWA arrangements: SOPs for works exist (not available for review); Customer Service Department and Compliance Unit provide first-level grievance intake (with escalation to FTC for consumer matters).	While the national framework effectively regulates vegetation management, these safeguards remain largely reactive and compliance-based. The PAP will ensure BWA SOPs are updated to reflect the national framework and contain specific requirements to identify vegetation/landscaping impacts in routine right-of-way works. SOPs will include requirements for protection, replanting, and basic erosion/sediment controls are unclear. Consultation/GRM: Existing customer channels appear oriented to billing/service issues; no evidence of a mechanism configured for advance works notifications and for receiving/resolving construction-phase grievances. Updating the grievance system to include construction issues specifically will be included as an action in the PAP.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE H — RETIREMENT ANNUITY AND GUARANTEED CONTINUOUS EMPLOYMENT PLANS (Continued)

the plan. The plan includes a feature in which participants could elect to receive an annual Company contribution of 2% of their base pay in exchange for forfeiting certain benefits under the pension plan.

The changes in the pension plans’ benefit obligations were as follows:

	2002	2001
Obligation at January 1	\$475,418	\$445,634
Service cost	15,018	11,768
Interest cost	33,264	31,783
Participant contributions	773	667
Plan amendments	577	189
Actuarial loss	26,915	10,062
Benefit payments	(25,447)	(23,148)
Currency translation	5,181	(1,537)
Obligation at December 31	\$531,699	\$475,418

The changes in the fair values of the pension plans’ assets were as follows:

	2002	2001
Plan assets at January 1	\$428,460	\$452,697
Actual return on plan assets	(45,893)	(12,057)
Employer contributions	21,400	12,079
Participant contributions	786	668
Benefit payments	(25,447)	(23,148)
Currency translation	4,774	(1,779)
Plan assets at December 31	\$384,080	\$428,460

The funded status of the pension plans was as follows:

	2002	2001
Plan assets (less than) projected benefit obligations	\$(147,619)	\$(46,958)
Unrecognized net loss	178,682	67,594
Unrecognized prior service cost	7,711	8,531
Unrecognized transition assets, net	(476)	(861)
Net amount recognized in the balance sheets	\$ 38,298	\$ 28,306

The amounts recognized in the consolidated balance sheets was composed of:

	2002	2001
Prepaid pension cost	\$ 3,345	\$ 38,784
Accrued pension liability	(102,881)	(13,779)
Intangible asset	7,620	1,566
Accumulated other comprehensive income	130,214	1,735
Net amount recognized in the balance sheets	\$ 38,298	\$ 28,306

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS — Continued

NOTE H — RETIREMENT ANNUITY AND GUARANTEED CONTINUOUS EMPLOYMENT PLANS (Continued)

A minimum pension liability adjustment was recorded during the fourth quarter as a liability with a corresponding decrease to shareholders' equity. At December 31, 2002, the Company recorded an after-tax charge to shareholders' equity of \$79,697.

The projected benefit obligation, accumulated benefit obligation, and fair value of plan assets for the U.S. pension plans with accumulated benefit obligations in excess of plan assets were \$482,943, \$439,604 and \$346,207, respectively; as of December 31, 2002 and \$13,100, \$12,863 and \$0, respectively as of December 31, 2001. The projected benefit obligation, accumulated benefit obligation, and fair value of plan assets for the non-U.S. pension plans with accumulated benefit obligations in excess of plan assets were \$18,455, \$17,186 and \$11,874, respectively; as of December 31, 2002 and \$15,994, \$14,842 and \$11,953, respectively as of December 31, 2001.

A summary of the components of total pension expense was as follows:

	2002	2001	2000
Service cost — benefits earned during the year	\$ 13,393	\$ 11,768	\$ 12,321
Interest cost on projected benefit obligation	33,264	31,783	31,077
Expected return on plan assets	(37,461)	(40,117)	(40,733)
Amortization of transition asset	(437)	(426)	(436)
Amortization of prior service cost	1,377	1,339	1,177
Amortization of net loss (gain)	340	(21)	(209)
Net pension cost of defined benefit plans	10,476	4,326	3,197
Defined contribution plans	4,387	3,629	2,040
Total pension expense	\$ 14,863	\$ 7,955	\$ 5,237

Weighted-average assumptions used in accounting for the defined benefit plans as of December 31, 2002 and 2001, were as follows:

	2002	2001
Discount rate	6.7%	7.2%
Rate of increase in compensation	4.9%	4.9%
Expected return on plan assets	9.1%	9.1%

The Company anticipates lowering the expected rate of return on plan assets to approximately 8.8% for 2003. U.S. plan assets consist of fixed income and equity securities. Non-U.S. plan assets are invested in non-U.S. insurance contracts and non-U.S. equity and fixed income securities. The Company does not have, and does not provide for, any postretirement or postemployment benefits other than pensions.

The Cleveland, Ohio, area operations have a Guaranteed Continuous Employment Plan covering substantially all employees which, in general, provides that the Company will provide work for at least 75% of every standard work week (presently 40 hours). This plan does not guarantee employment when the Company's ability to continue normal operations is seriously restricted by events beyond the control of the Company. The Company has reserved the right to terminate this plan effective at the end of a calendar year by giving notice of such termination not less than six months prior to the end of such year.

LINCOLN ELECTRIC HOLDINGS, INC. AND SUBSIDIARIES
CONSOLIDATED BALANCE SHEETS

	December 31	
	2002	2001
	(In thousands of dollars, except share data)	
LIABILITIES AND SHAREHOLDERS' EQUITY		
CURRENT LIABILITIES		
Notes payable to banks	\$ 945	\$ 13,163
Trade accounts payable	63,894	67,569
Accrued employee compensation and benefits	38,663	26,456
Accrued expenses	12,722	13,519
Taxes, including income taxes	34,699	34,216
Dividend payable	6,746	6,355
Other current liabilities	20,514	22,788
Current portion of long-term debt	13,433	13,434
TOTAL CURRENT LIABILITIES	191,616	197,500
Long-term debt, less current portion	174,146	24,181
Accrued pensions	92,066	17,849
Deferred income taxes	1,070	31,751
Other long-term liabilities	13,218	11,332
SHAREHOLDERS' EQUITY		
Preferred Shares, without par value — at stated capital amount:		
Authorized — 5,000,000 shares in 2002 and 2001; Issued and Outstanding — none	—	—
Common Shares, without par value — at stated capital amount:		
Authorized — 120,000,000 shares in 2002 and 2001; Issued — 49,282,306 shares in 2002 and 2001; Outstanding — 42,087,115 shares in 2002 and 42,369,145 shares in 2001	4,928	4,928
Additional paid-in capital	106,237	105,380
Retained earnings	597,495	594,701
Accumulated other comprehensive income (loss)	(132,350)	(66,726)
Treasury shares, at cost — 7,195,191 shares in 2002 and 6,913,161 shares in 2001	(147,157)	(139,585)
TOTAL SHAREHOLDERS' EQUITY	429,153	498,698
TOTAL LIABILITIES AND SHAREHOLDERS' EQUITY	\$ 901,269	\$ 781,311

See notes to these consolidated financial statements.

NOTE 1: SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES (Continued)

Arrow applies the provisions of SFAS No. 114 to all impaired commercial and commercial real estate loans over \$250, and to all loans restructured subsequent to the adoption of SFAS No. 114. Allowances for loan losses for the remaining loans are recognized in accordance with SFAS No. 5. Under the provisions of SFAS No. 114, Arrow determines impairment for collateralized loans based on fair value of the collateral less estimated costs to sell. For other loans, impairment is determined by comparing the recorded investment in the loan to the present value of the expected cash flows, discounted at the loan's effective interest rate. Arrow determines the interest income recognition method on a loan-by-loan basis. Based upon the borrowers' payment histories and cash flow projections, interest recognition methods include full accrual or cash basis.

Other Real Estate Owned and Repossessed Assets - Real estate acquired by foreclosure and assets acquired by repossession are recorded at the lower of the recorded investment in the loan or the fair value of the property less estimated costs to sell. Subsequent declines in fair value, after transfer to other real estate owned and repossessed assets, are recognized through a valuation allowance. Such declines in fair value along with related operating expenses to administer such properties or assets are charged directly to operating expense.

Premises and Equipment - Premises and equipment are stated at cost, less accumulated depreciation and amortization. Depreciation and amortization included in operating expenses are computed largely on the straight-line method. The provision is based on the estimated useful lives of the assets and, in the case of leasehold improvements, amortization is computed over the terms of the respective leases or their estimated useful lives, whichever is shorter. Gains or losses on disposition are reflected in earnings.

Income Taxes - Arrow accounts for income taxes under the asset and liability method required by SFAS No. 109. Deferred tax assets and liabilities are recognized for the future tax consequences attributable to differences between the financial statement carrying amounts of existing assets and liabilities and their respective tax bases. Deferred tax assets and liabilities are measured using enacted tax rates expected to apply to taxable income in the years in which those temporary differences are expected to be recovered or settled. The effect on deferred tax assets and liabilities of a change in tax rates is recognized in income tax expense in the period that includes the enactment date. Arrow's policy is that deferred tax assets are reduced by a valuation allowance if, based on the weight of available evidence, it is more likely than not that some or all of the deferred tax assets will not be realized.

Goodwill and Other Intangible Assets - Arrow adopted SFAS No. 147, "Acquisitions of Certain Financial Institutions," during the quarter ended September 30, 2002. SFAS No. 147 affects the accounting for an unidentifiable intangible asset acquired in the acquisition of a bank or thrift (including acquisitions of branches), where the fair value of the liabilities assumed exceeded the fair value of the assets acquired. Under SFAS No. 147, if such a transaction met the criteria for a business combination, the carrying amount of the unidentifiable intangible asset is reclassified to goodwill (reclassified goodwill) as of the date SFAS No. 142 was applied in its entirety, which for Arrow was January 1, 2002. The carrying amounts of any recognized intangible assets that meet the recognition criteria of SFAS No. 141 that have been included in the amount reported as an unidentifiable intangible asset, and for which separate accounting records have been maintained, should be accounted for apart from the unidentifiable intangible asset and should not be reclassified to goodwill. The reclassified goodwill should be accounted for and reported prospectively as goodwill under SFAS No. 142, for which amortization is not required, but which must be evaluated annually for impairment. Annually, we test for any impairment of goodwill and other intangible assets by comparing the carrying amount of those assets to the fair value of each reporting unit's intangible assets, applying rates derived from recent actual transactions.

In April 2005, Arrow completed the cash purchase of three branches from HSBC Bank USA, N.A. Arrow recorded the following intangible assets as a result of the acquisition: goodwill (\$3,690) and core deposit intangible asset (\$2,247). The value of the core deposit intangible asset is being amortized over ten years.

In November 2004, Arrow acquired all of the outstanding shares of common stock of CFG in a tax-free exchange for Arrow's common stock (62,805 shares, as restated for stock dividends). As adjusted for subsequent contingency payments, Arrow recorded the following intangible assets as a result of the acquisition (none of which are deductible for income tax purposes): goodwill (\$1,460), covenant (\$117) and expirations (\$686). The value of the covenant is being amortized over five years and the value of the expirations is being amortized over twenty years. The agreement provides for annual contingent future payments of Company stock, based upon earnings, over a five-year period. Management has concluded that, under criteria established by SFAS No. 141, these payments will be recorded as additional goodwill at the time of payment. The amount of additional goodwill recorded in 2005 was \$91 (3,227 shares).

Critical Analysis on Government Contracts: An Introduction

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Government Contracts are nothing but legal agreements between two governments or between government and a private party. The government may be either central government or state government outlining the terms for provision of goods or services or with reference to immovable property also. Article 298 of Indian Constitution is dealing with Government Contracts. It must be in the format prescribed by Article 299 and in case the agreement is not in the prescribed format, it is not enforceable in the court of law. It stated that the agreement should be entered by the officer of the Government on behalf of Union or State Government and in the name of President of India or Governor of the State. Article 298 of the Indian Constitution of India confers the power to the Union as well as the State to carry out trade, or acquisitions, hold and dispose of property, and further make contracts for any purpose.

Article 299 prescribes the following conditions:

- a) The concerned authority should enter the contract as an agent of the president of the union or by the governor of the State.
- b) The contract must be in writing.
- c) It must be entered on behalf of Union of India or on behalf of any State Government.
- d) Agreements must be expressed in writing and executed by a person duly authorized by the President or the Governor on their behalf.
- e) President or the Governor cannot be personally held liable for contract, it does not grant immunity to the government from the legal provisions of the contract.

It was held that a Government Contract has to comply with the provisions of Article 299 in addition to the requirements of Indian Contract Act, 1872. The contractual liability of the Government is the same as that of any individual under the ordinary law of the contract. (State of Bihar v. Majeed AIR 1954 SC 245).

When an agreement is not signed in the name of the Governor of a State and the name of the Governor is not mentioned in the deed, it was held that it was not valid and binding on the Government. (Mrs. Aliakutty Paul v. The State of Kerala and Ors AIR 1995 Ker 291).

Reasonableness and Fairness:

Every action of the Government must be informed with reason and should be free from arbitrariness. This is the very essence of the rule of law and its bare minimal requirement. (Roman Dayaram Shetty v. International Airport Authority of India (1979) 3 SCC)

Public Interest:

When an award of contract may not be causing any loss to the public exchequer manifestly, it may still be liable to quash for being unfair, unreasonable, discriminatory and violative of the guarantee contained in Article 14. (Shri Sachindanand Pandey v. State of West Bengal AIR 1987 SC 1109).

Equality, Non-Arbitrariness:

The principle of reasonableness, which legally as well as philosophically, is an essential element of equality or non-arbitrariness pervades Article 14 like a brooding omni-presence and the procedure contemplated by Article 21 must answer the test of reasonableness in order to be in conformity with Article 14. (Menaka Gandhi v. Union of India AIR 1978 SC 597).

Contractual Liability:

Article 299 (2) provides immunity to the President or Governor or any Officer entered into agreement with third party in case of breach of contract by the Government. At the same time it did not provide immunity to the Government. The Government liability is practically the same as that of a private person, subject of course, to any contract to the contrary. (State of Bihar v. Abdul Majid AIR 1954 SC 245)

Government Contract and Section 70 of Indian Contract Act, 1872:

Even though the contract is not in the prescribed format, in case the contract is executed by the third party and the government derives any benefit, the Government is liable to compensate the third party under quasi contracts. The quasi contracts are called implied contracts and these contracts are based on the principles of justice, equity and good conscience and no person should become rich at the cost of others.

Mandatory nature of Article 299:

It is a settled law that Article 299 is a mandatory provision, and not directory one. It was confirmed in number of judgments of the Supreme Court, more specifically specified in State